

A

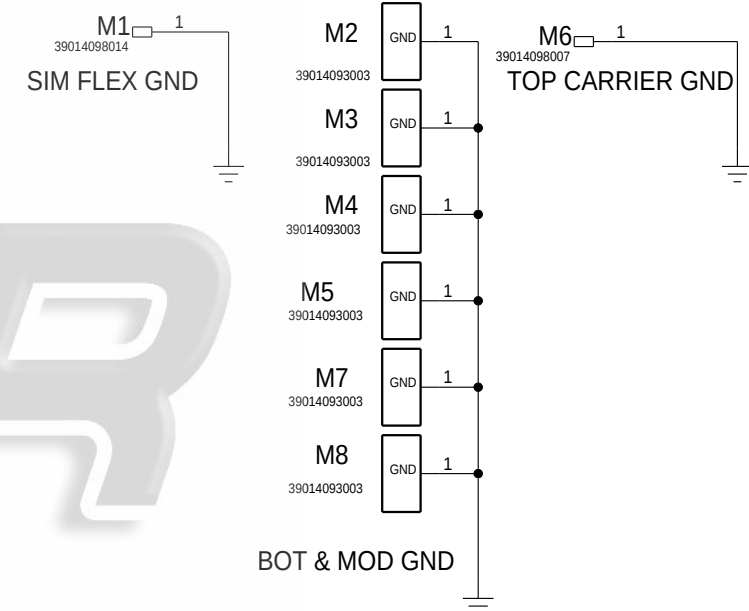
A

HOUDINI ME/SHIELDS

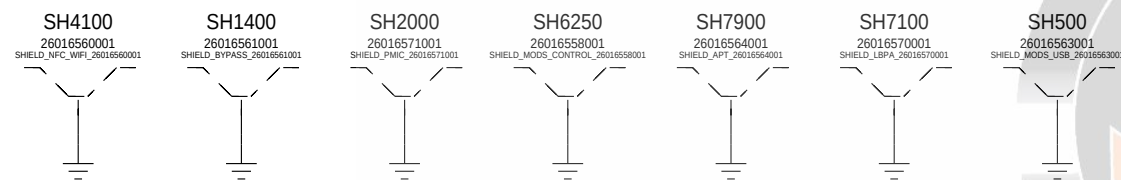
REFERENCE DESIGNATOR RANGES BY BLOCK

500 - 599: USER INPUT
600 - 799: POWER
1000-2999: MSM8953 + PM8953
3000-3199: WTR3925
4000-4499: CONNECTIVITY
4500-4999: AUDIO FE
5000-5499: DISPLAY
5500-5999: CAMERA
6000-6999: XMCS
7000-7999: RFFE
8000 - 8099: ANTENNA

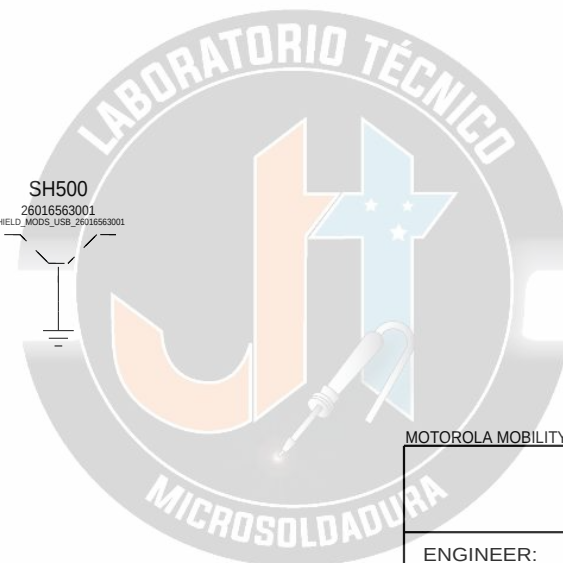
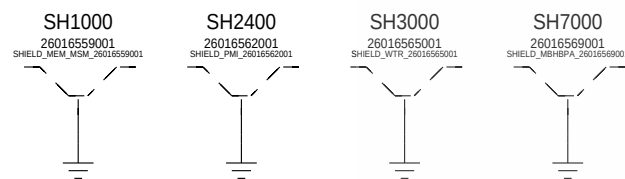
GROUND CONTACTS



TOP SIDE PCB SHIELD



BOTTOM SIDE PCB SHIELD



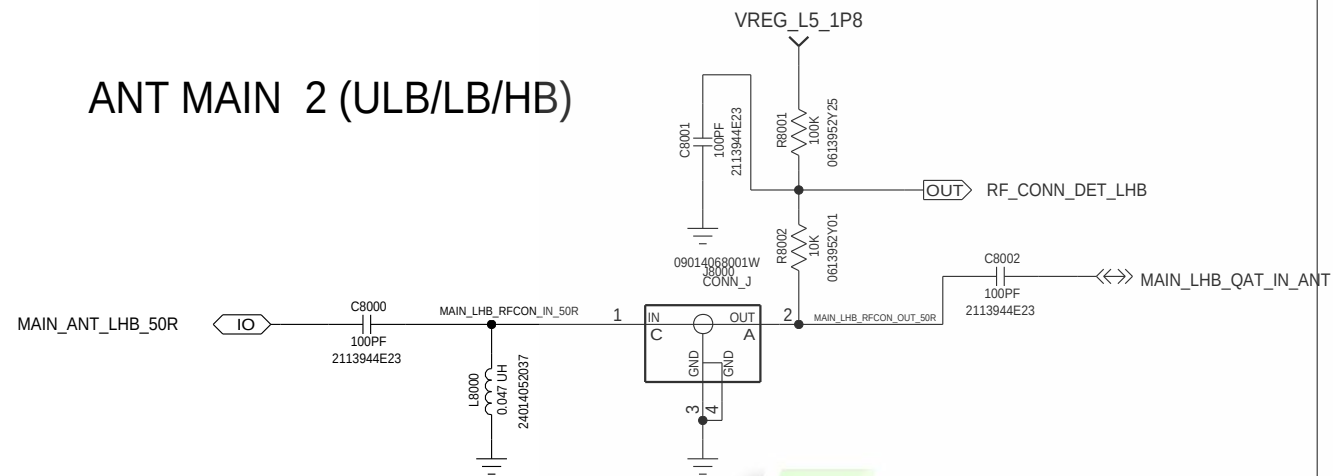
MOTOROLA MOBILITY CONFIDENTIAL RESTRICTED-MDB TRACKED TRACKEDANDBLOCKED

APPROVALS		 MOTOROLA MOBILE DEVICES		
ENGINEER:		BLOCK TITLE: houdini_09		
LAST MODIFIED BY: Thu Jan 05 14:29:33 2017				
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		B houdini_09.cpm		01
		CAD FILE NAME:		PAGE 2 OF 64

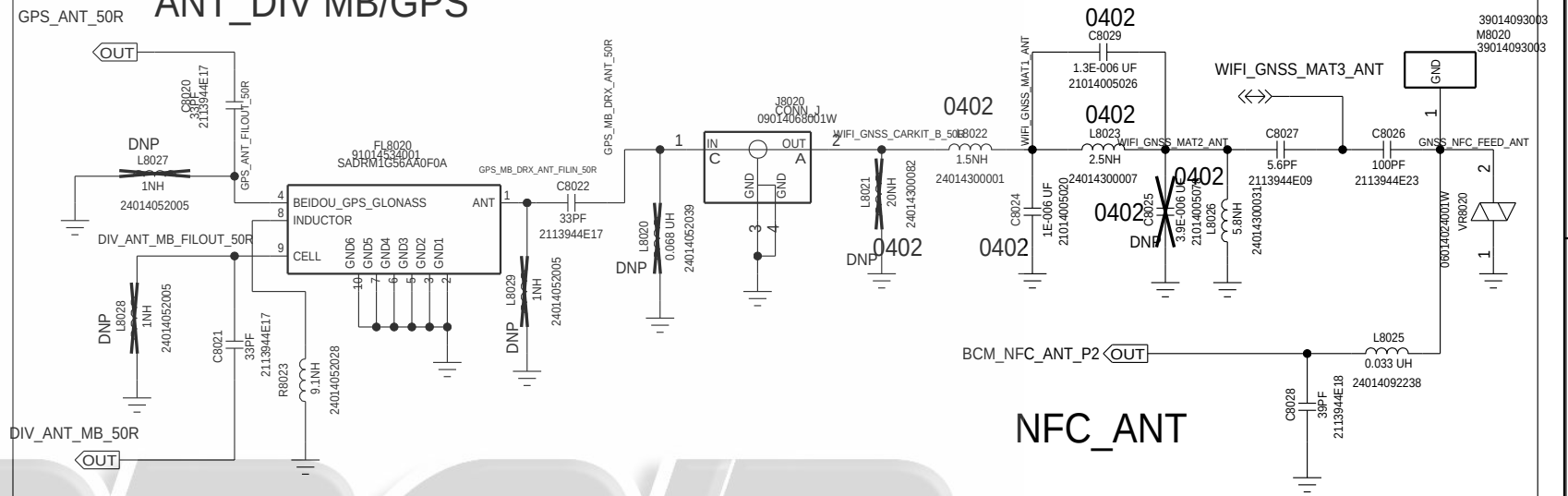
8000:antennas

REF 8000-8049

ANT MAIN 2 (ULB/LB/HB)

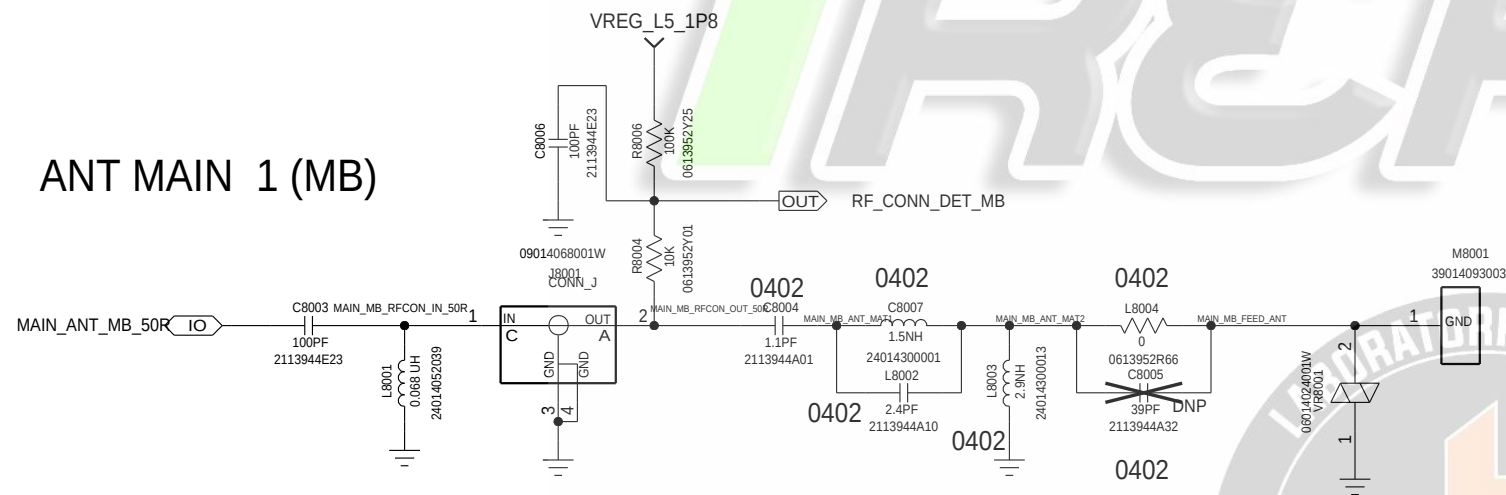


ANT_DIV MB/GPS

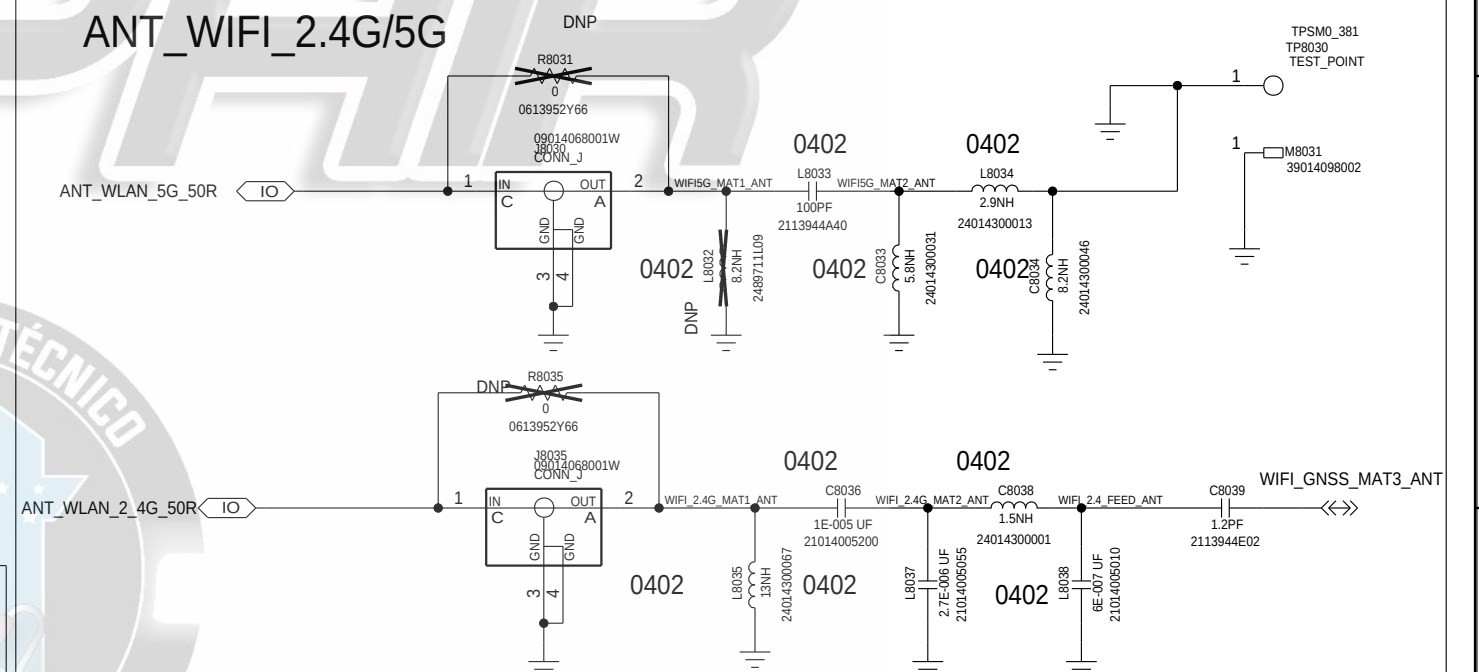


NFC_ANT

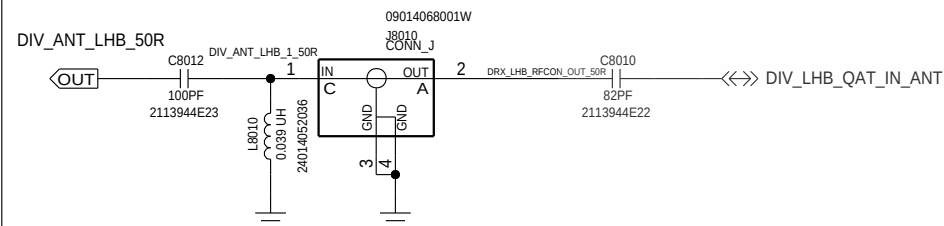
ANT MAIN 1 (MB)



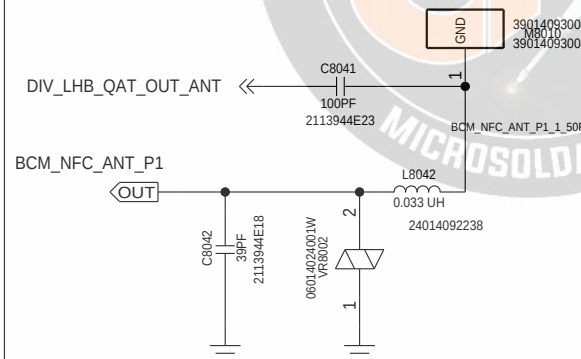
ANT_WIFI_2.4G/5G



ANT 3 DIV ULB/LB/HB



DIV & NFC ANT



APPROVALS

ENGINEER:

LAST MODIFIED BY:

Fri Jan 20 11:26:58 2017

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MOTOROLA
MOBILE DEVICES

BLOCK TITLE:

antennas

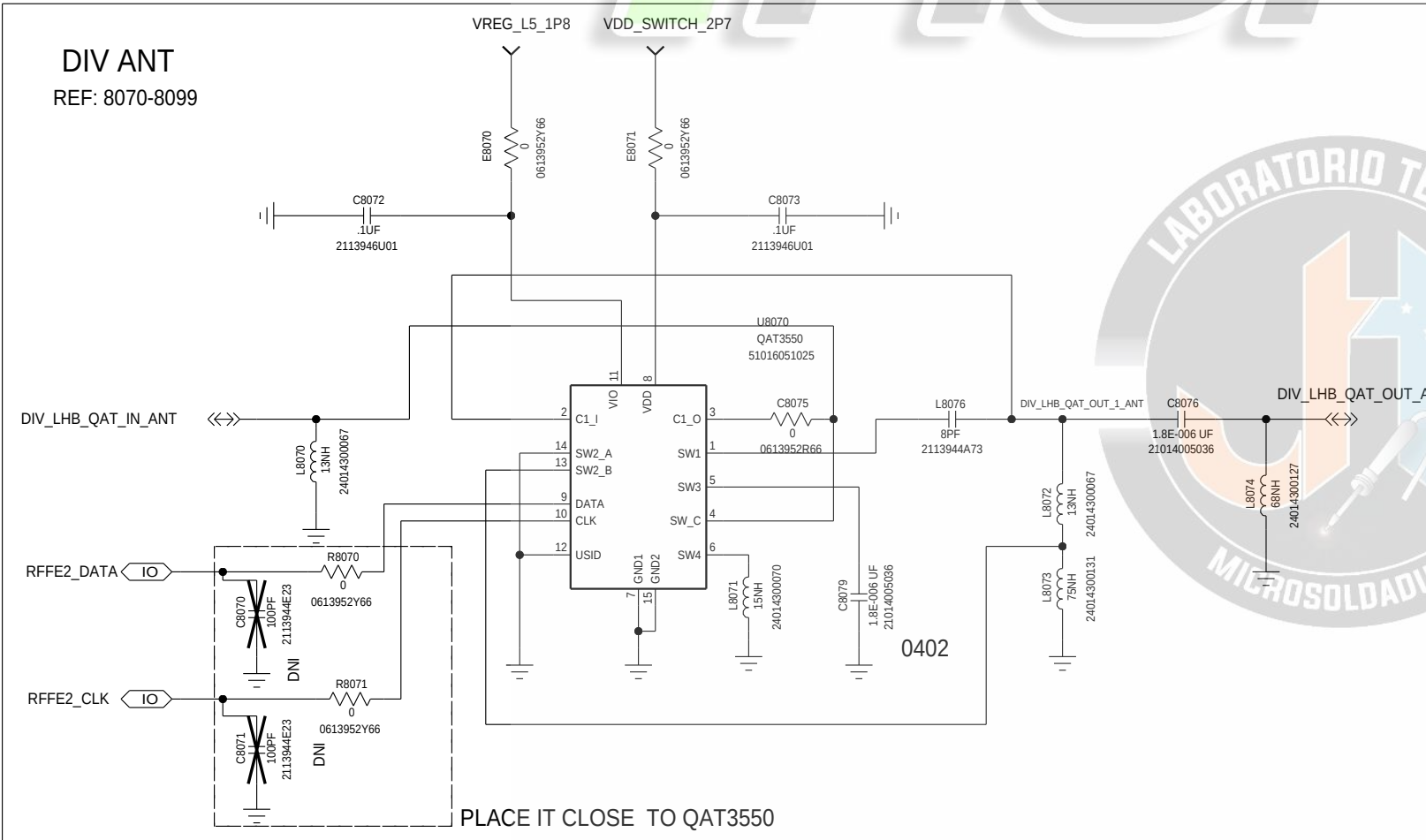
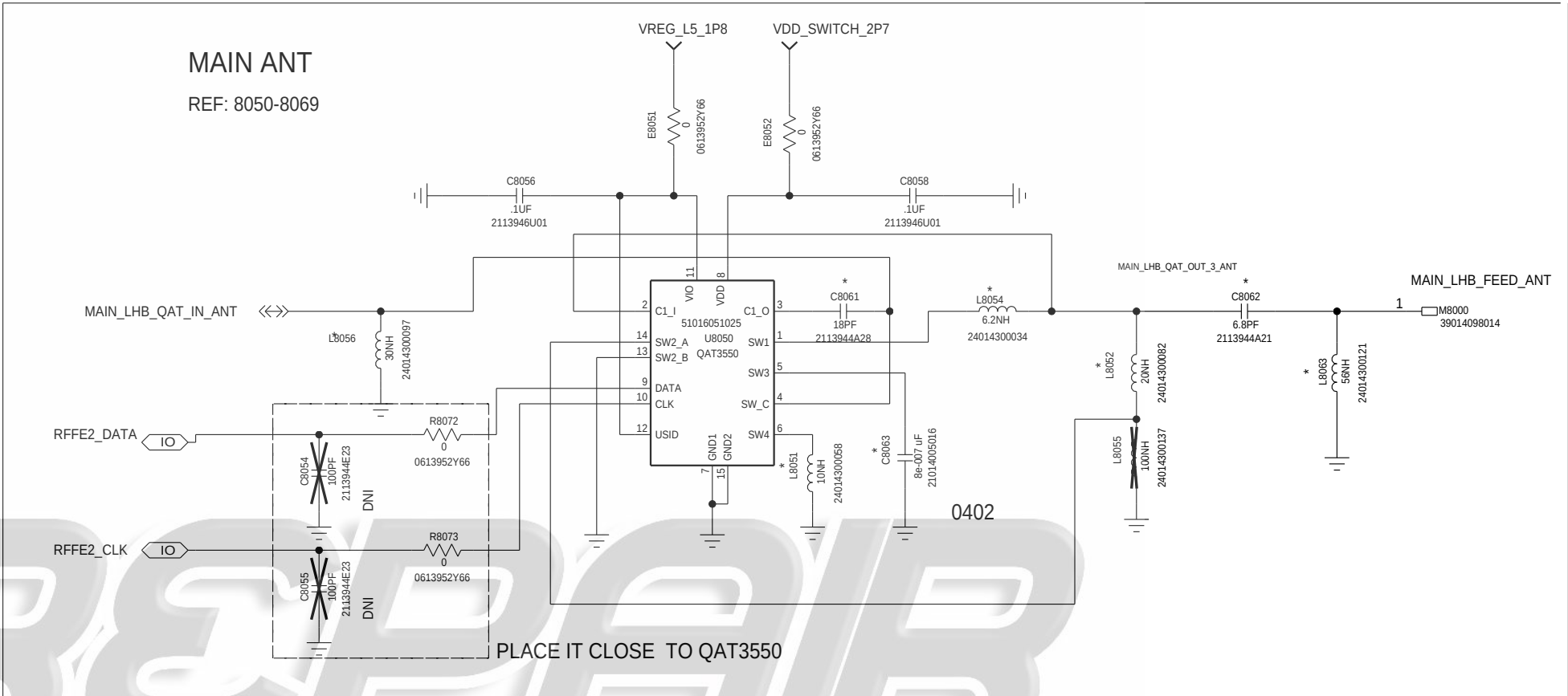
PROJECT NAME
houdini_09.cpm

CAD FILE NAME:

REV
01

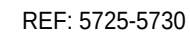
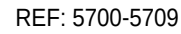
PAGE 3 OF 64

8050:ANT_TUNER_QAT3550
REF 8050-8099



APPROVALS		MOTOROLA MOBILE DEVICES	
ENGINEER:		BLOCK TITLE: antennas	
LAST MODIFIED BY: Thu Dec 22 11:36:14 2016		PROJECT NAME houdini_09.cpm	
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		REV 01	

REF: 5700-5799

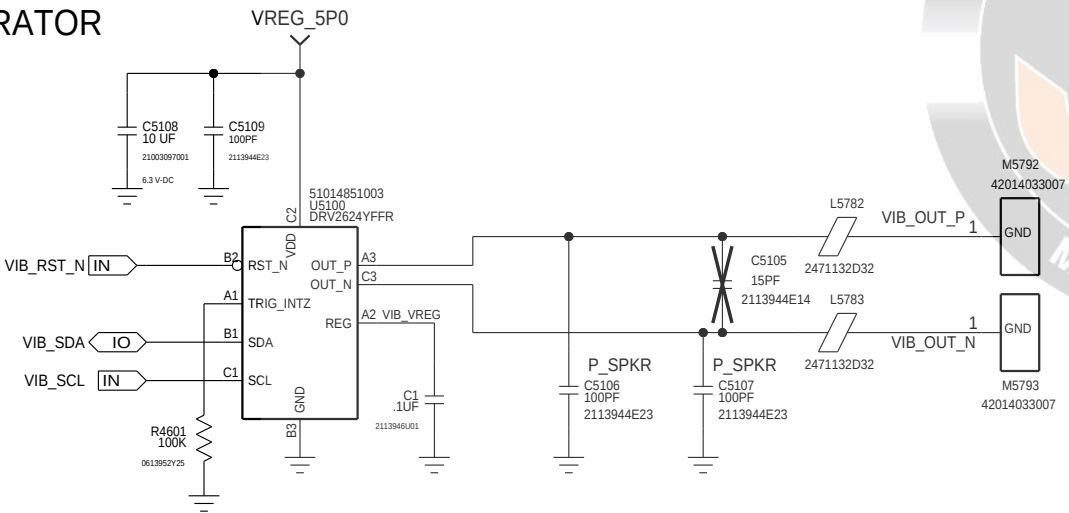
PAGE 5 OF 64

AFE: SPEAKERS

LOUDSPEAKER/EAR PIECE
REF:5790-5799



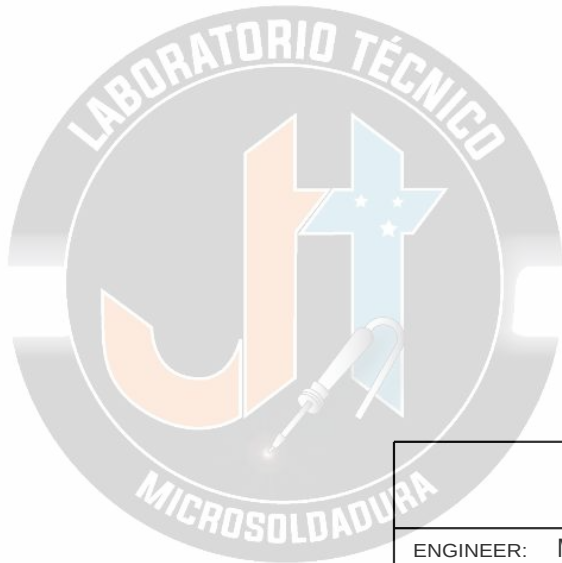
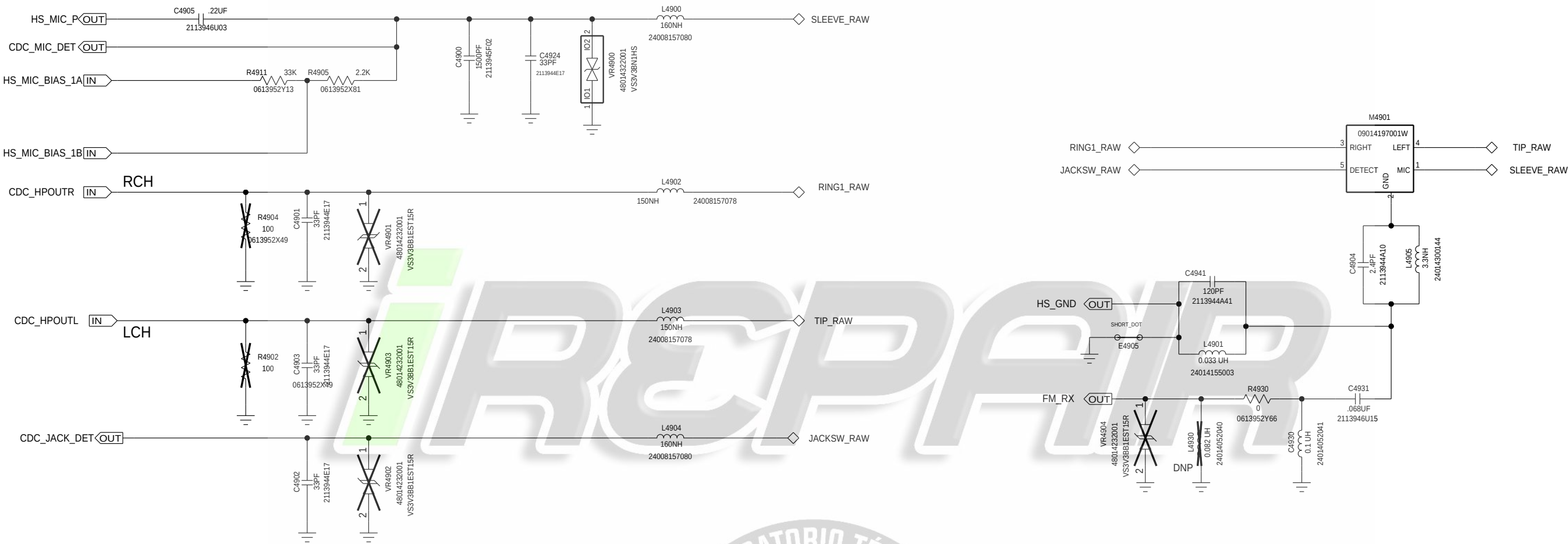
LINEAR VIBRATOR
REF 5100-5119



APPROVALS		MOTOROLA MOBILE DEVICES	
ENGINEER: DENNIS BUDNICK		BLOCK TITLE: audio_fe	
LAST MODIFIED BY: HUNG PHAM Thu Nov 17 13:08:10 2016		PROJECT NAME houdini_09.cpm	
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		PAGE 6 OF 64	

AFE: 3.5MM HEADSET

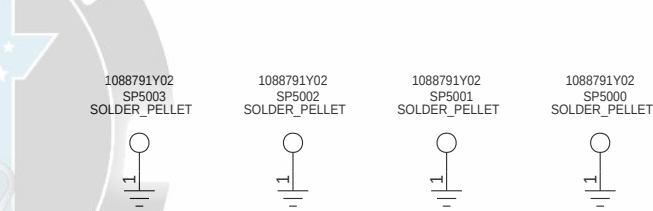
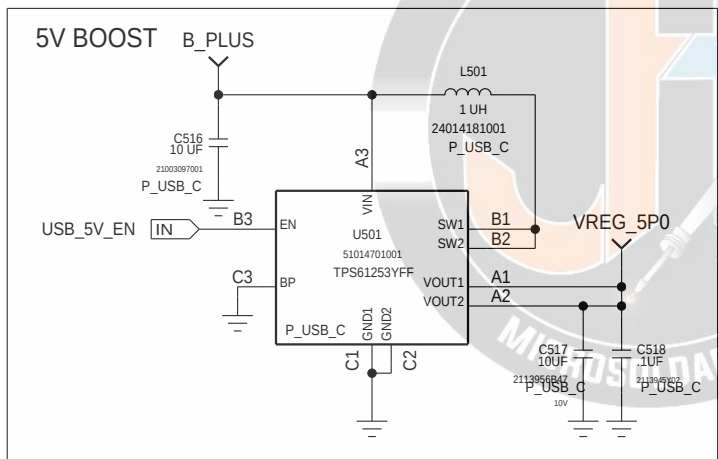
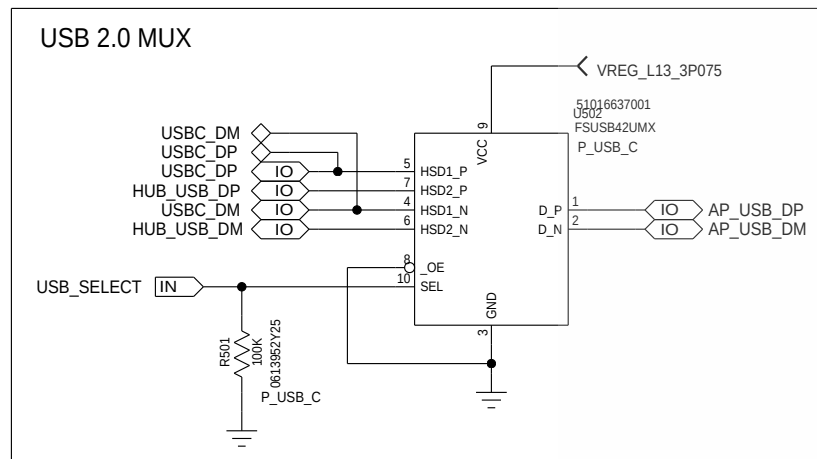
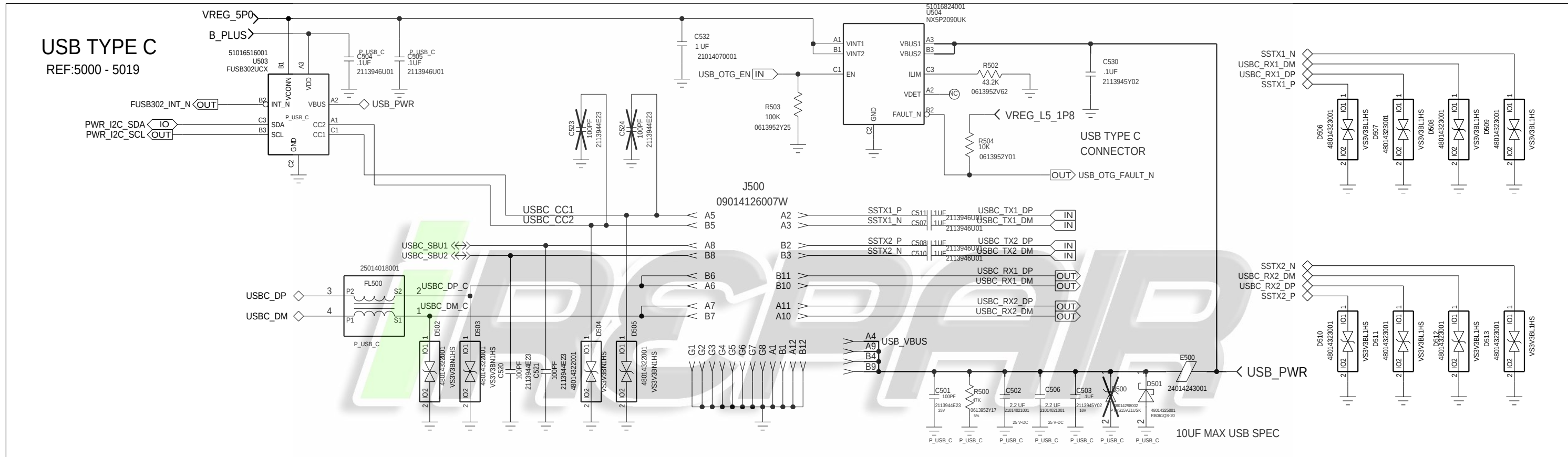
REF:4900-4949



APPROVALS		 MOTOROLA MOBILE DEVICES		
ENGINEER: MMI		BLOCK TITLE: audio_fe		
LAST MODIFIED BY: MMI Tue Nov 15 13:46:18 2016				
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		B houdini_09.cpm		01
		CAD FILE NAME:		PAGE 7 OF 64

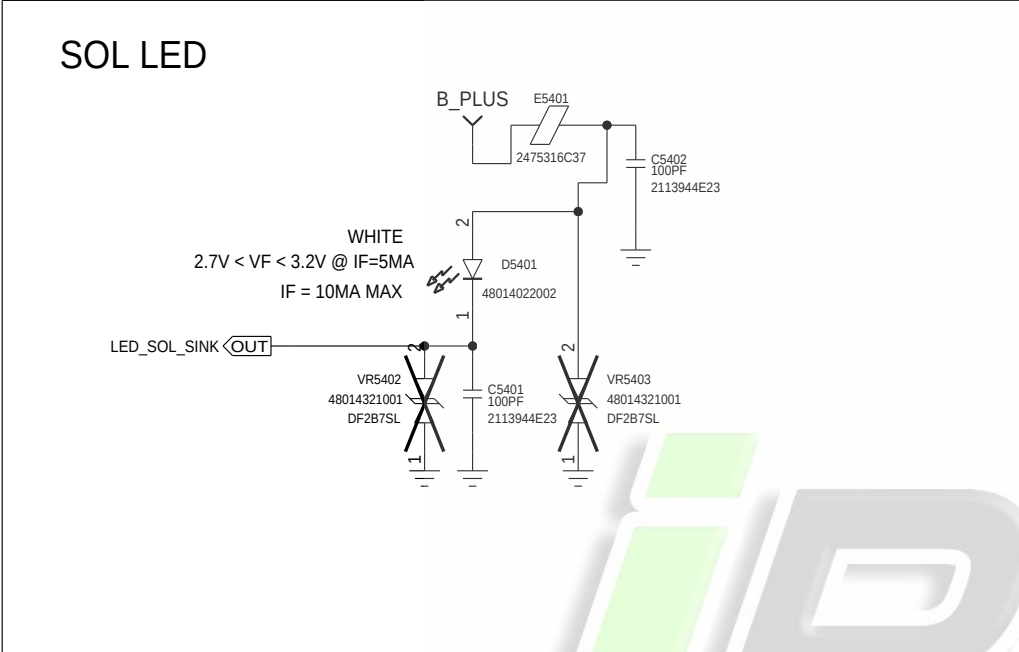
PWR: USB-C

REF:5000 - 5219



APPROVALS		MOTOROLA MOBILE DEVICES	
ENGINEER: RICH NG		BLOCK TITLE: power	
LAST MODIFIED BY: NATE CONNELL Tue Nov 15 13:46:14 2016		PROJECT NAME houdini_09.cpm	
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		PAGE 8 OF 64	

PWR: LIGHTING



REPAIR



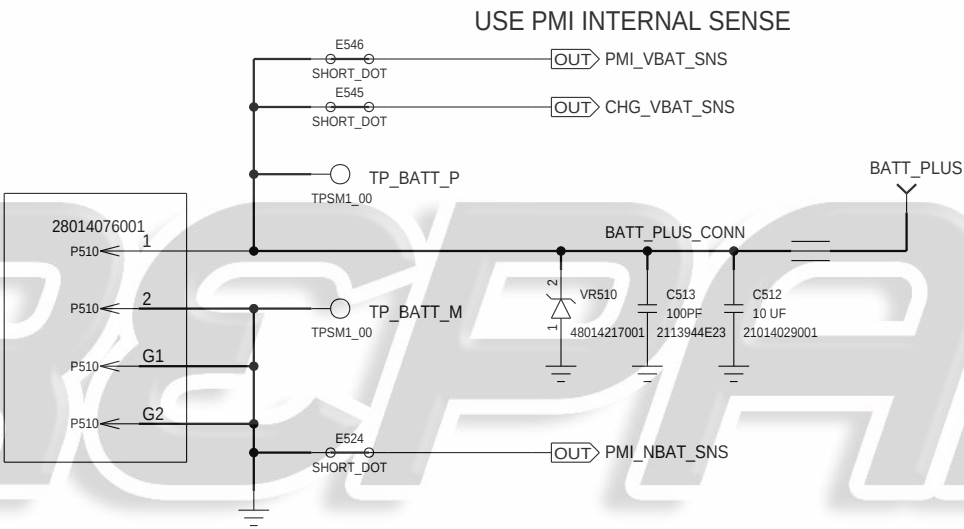
MOTOROLA MOBILITY CONFIDENTIAL RESTRICTED-MDB TRACKED TRACKEDANDBLOCKED

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LAST MODIFIED BY: TIM MILLER Tue Nov 15 13:46:15 2016		power	
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		REV 01	
		CAD FILE NAME:	
		PAGE 9 OF 64	

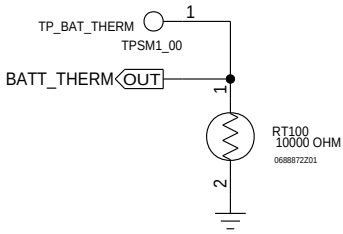
PWR: BATT

REF:510 - 550

BATTERY BLOCK

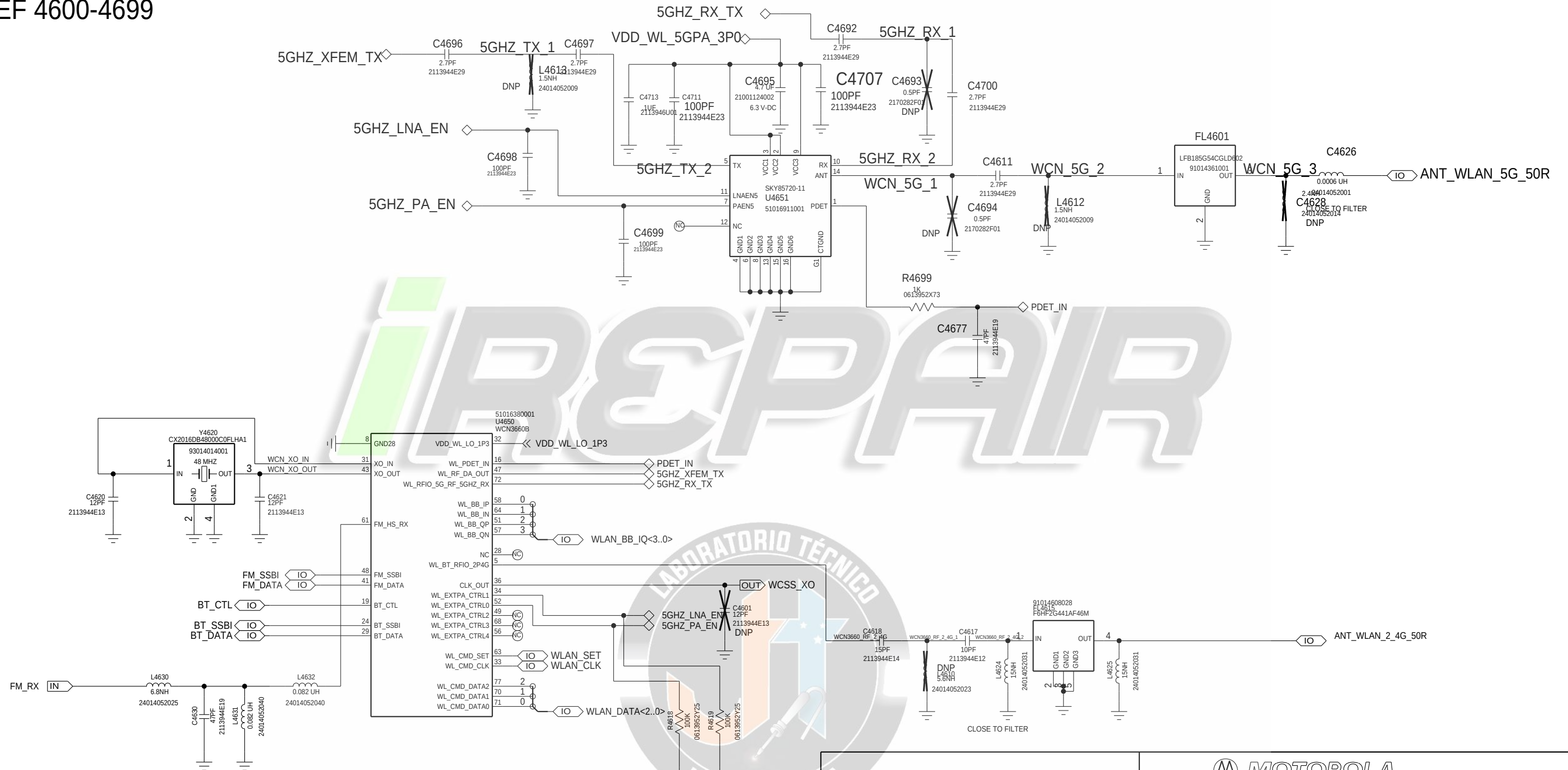


BATT THERMISTOR



APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: RICH NG		BLOCK TITLE:	
LAST MODIFIED BY: NATE CONNELL Tue Dec 13 16:58:42 2016		power	
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		CAD FILE NAME:	
		PAGE 10 OF 64	

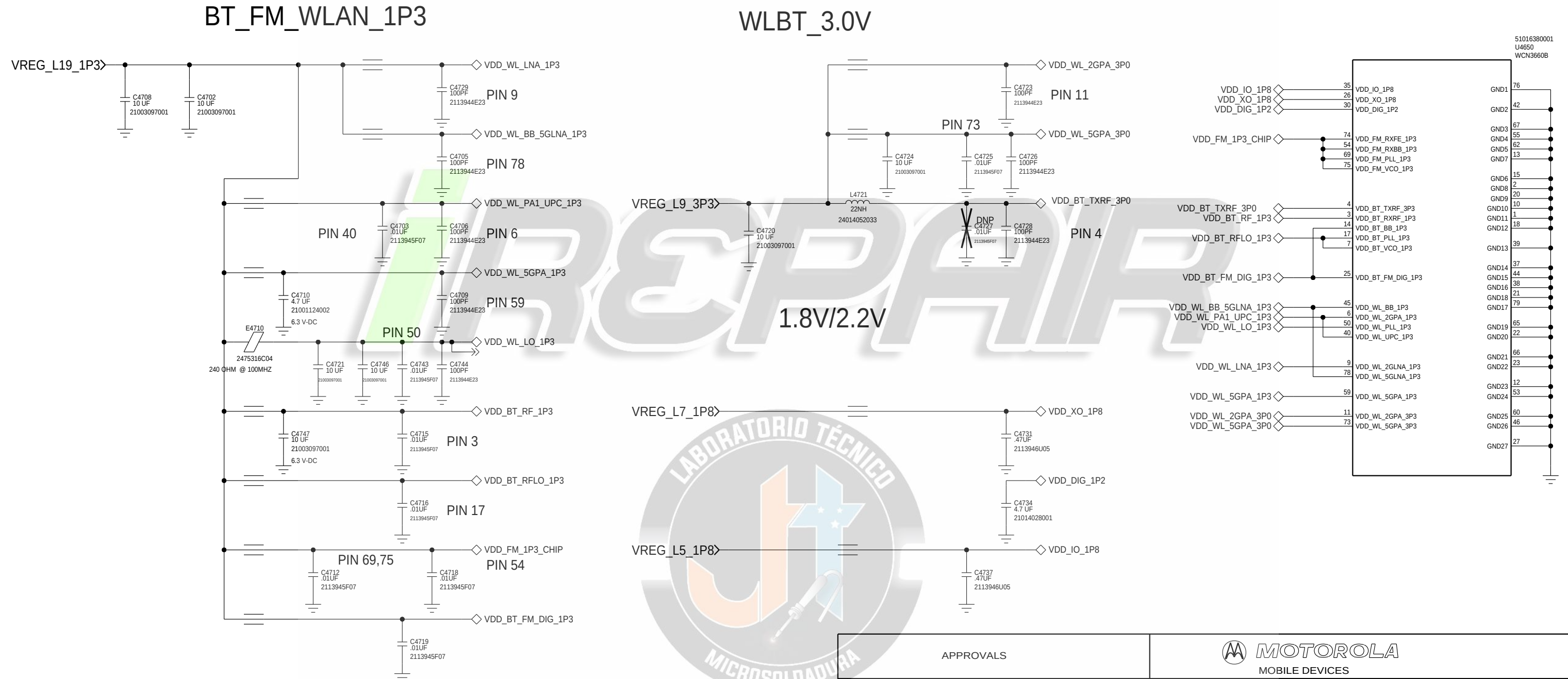
WCN3660B RF/CTRL
REF 4600-4699



BASED ON 80-WL006-44 REV C

APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER:	BLOCK TITLE: connectivity		
LAST MODIFIED BY: FROEHLING Fri Dec 16 14:34:31 2016			
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		CAD FILE NAME:	PAGE 11 OF 64

WCN3660B POWER
REF 4700-4799



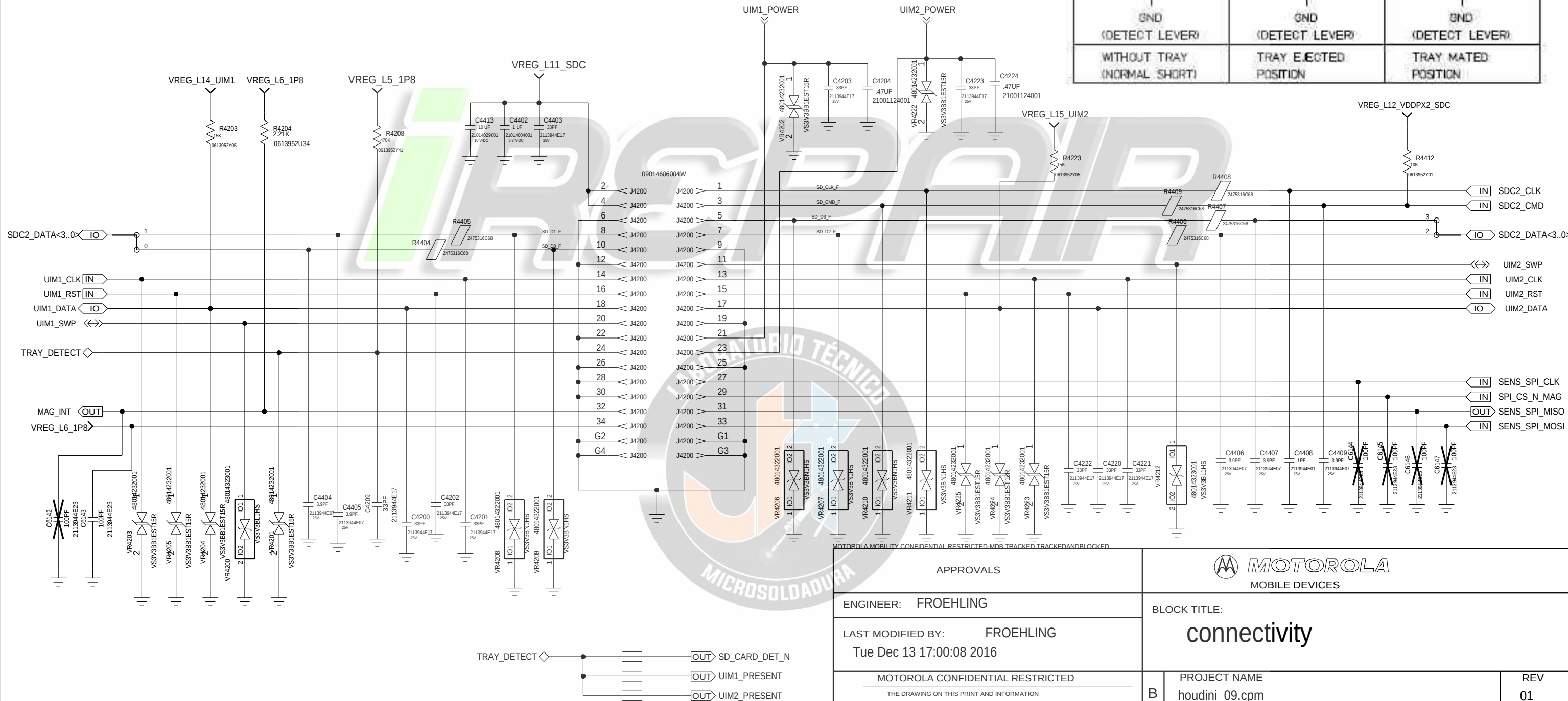
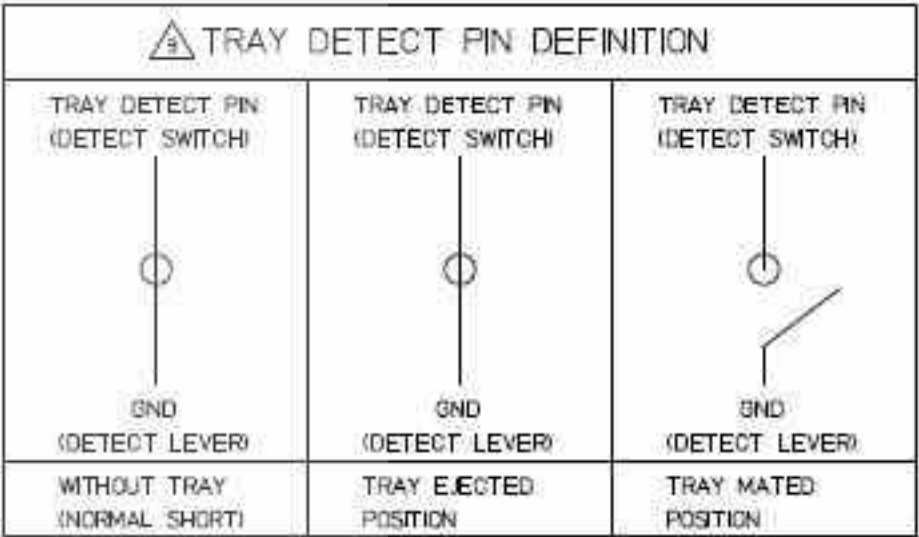
NOTE: RESERVE C4717 AND C4718 FOR 0.01UF BYPASSES

BASED ON 80-WL006-44 REV C

APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER:	BLOCK TITLE: connectivity		
LAST MODIFIED BY: FROEHLING Tue Nov 15 13:46:10 2016			
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		CAD FILE NAME:	PAGE 12 OF 64

CONN: UIM1/UIM2/SDCARD

REF:4200-4219



NOTE 1: USE 24.9K PULL-UP TO SUPPORT HOT SWAP WITH FREEZE-IO FEATURE.

NOTE 2: UIM_PWR BYPASS OF 0.33UF RECOMMENDED BY NFC IC SUPPLIER, PLACE NEAR CONNECTOR.

APPROVALS		MOTOROLA MOBILE DEVICES	
ENGINEER: FROEHLING		BLOCK TITLE:	
LAST MODIFIED BY: FROEHLING Tue Dec 13 17:00:08 2016		connectivity	
MOTOROLA CONFIDENTIAL RESTRICTED		PROJECT NAME	REV
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		CAD FILE NAME:	PAGE 13 OF 64

USR: FPC

REF: 6600 - 6699

8

7

6

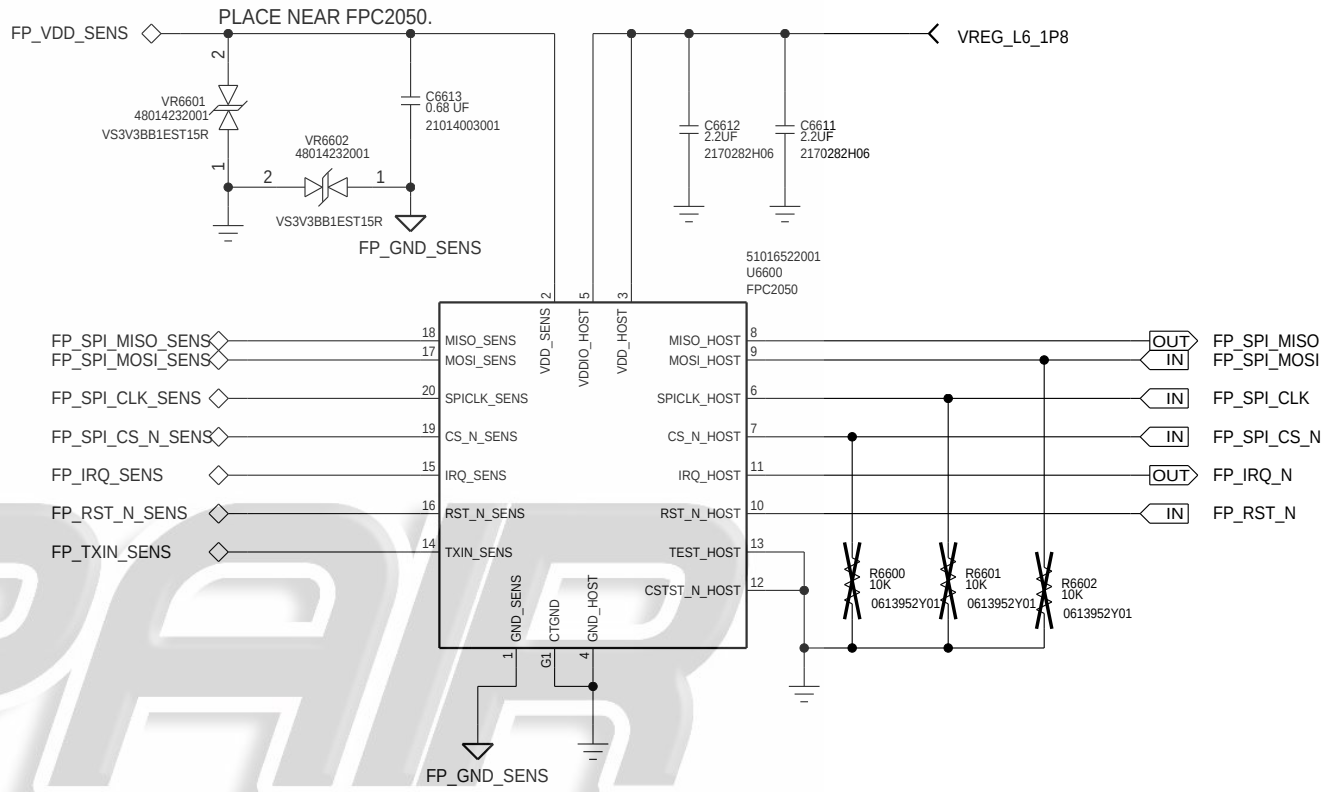
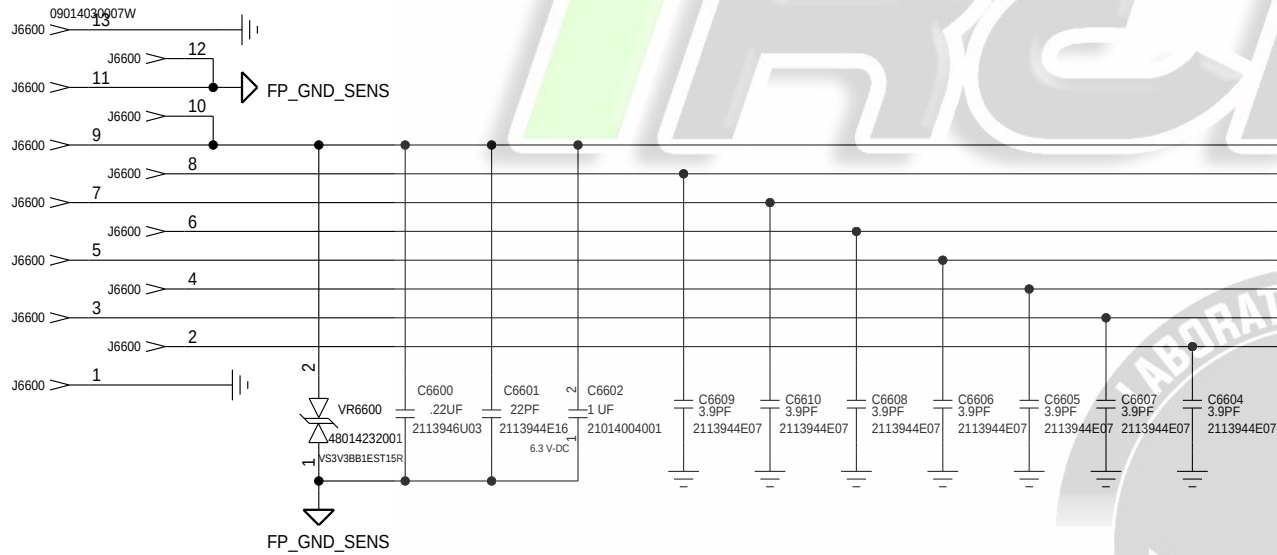
5

4

3

2

1



APPROVALS		 MOTOROLA MOBILE DEVICES	
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LAST MODIFIED BY: NATE CONNELL Tue Nov 15 13:46:12 2016		connectivity	
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		CAD FILE NAME:	PAGE 14 OF 64

8

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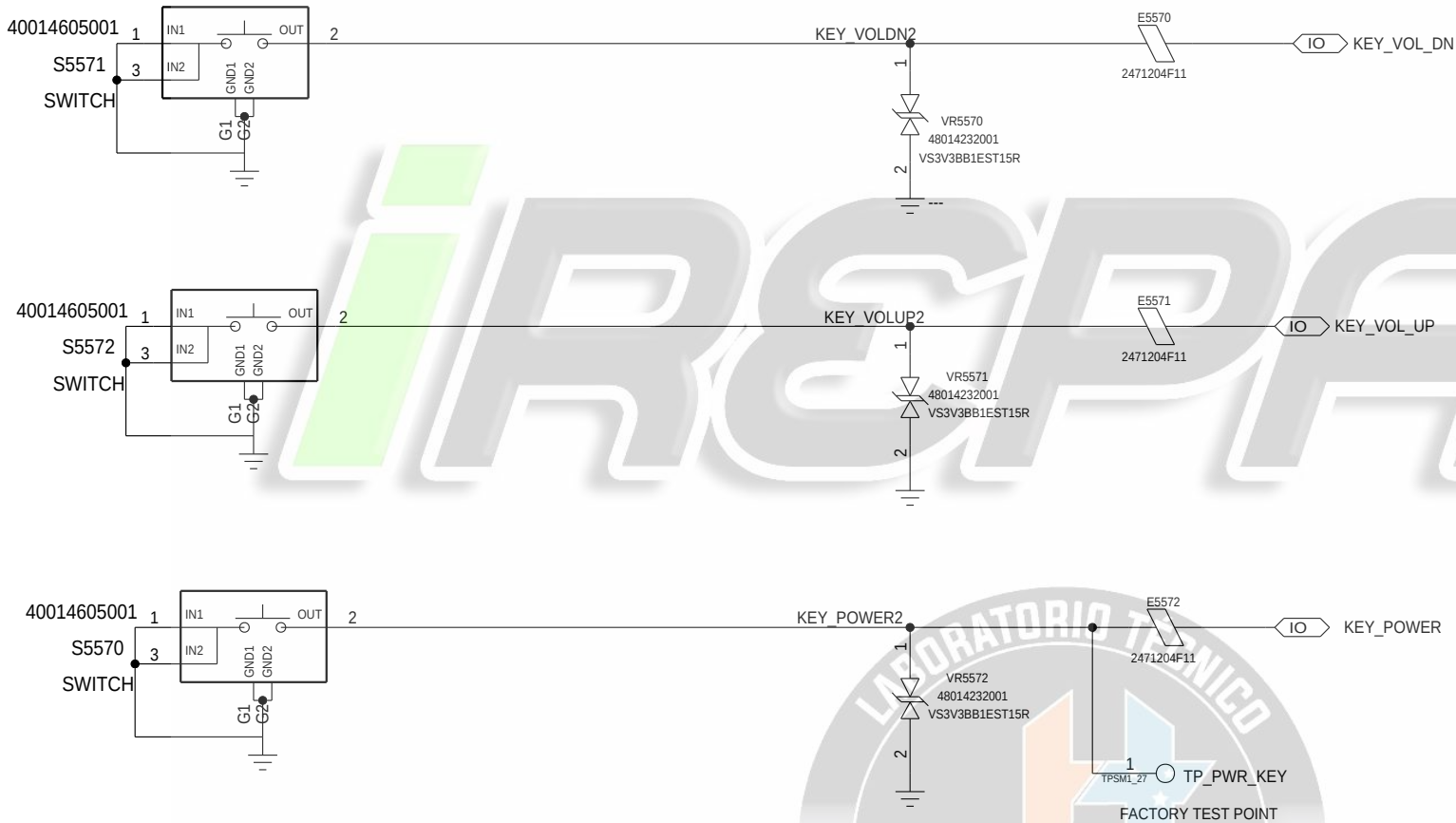
1

SIDE KEY

REF:5570-5579

MECHANICAL KEYS

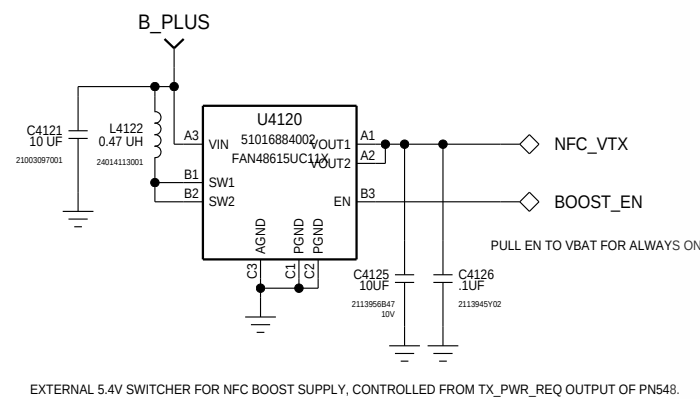
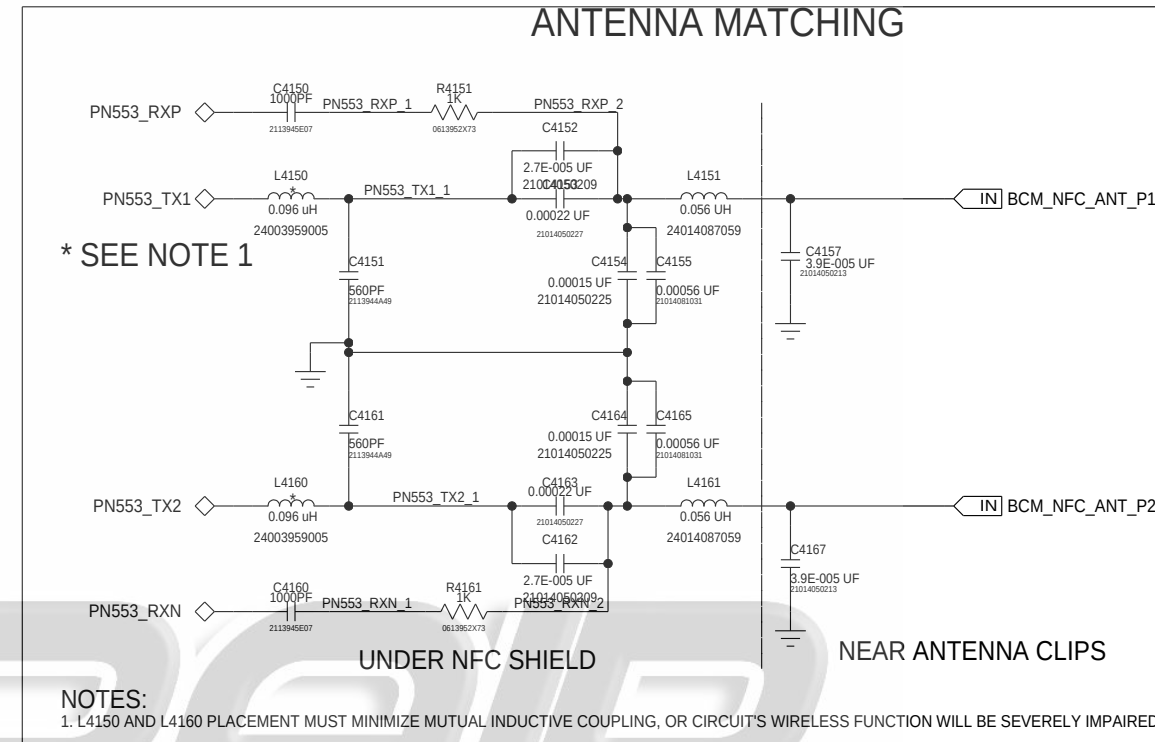
VOL_UP/DOWN/GND TPS NEED P0 BRINGUP ONLY



APPROVALS		MOTOROLA MOBILE DEVICES	
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LAST MODIFIED BY: NATE CONNELL Tue Nov 15 13:46:12 2016		connectivity	
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		CAD FILE NAME:	
		PAGE 15 OF 64	

REF: 4100-4199

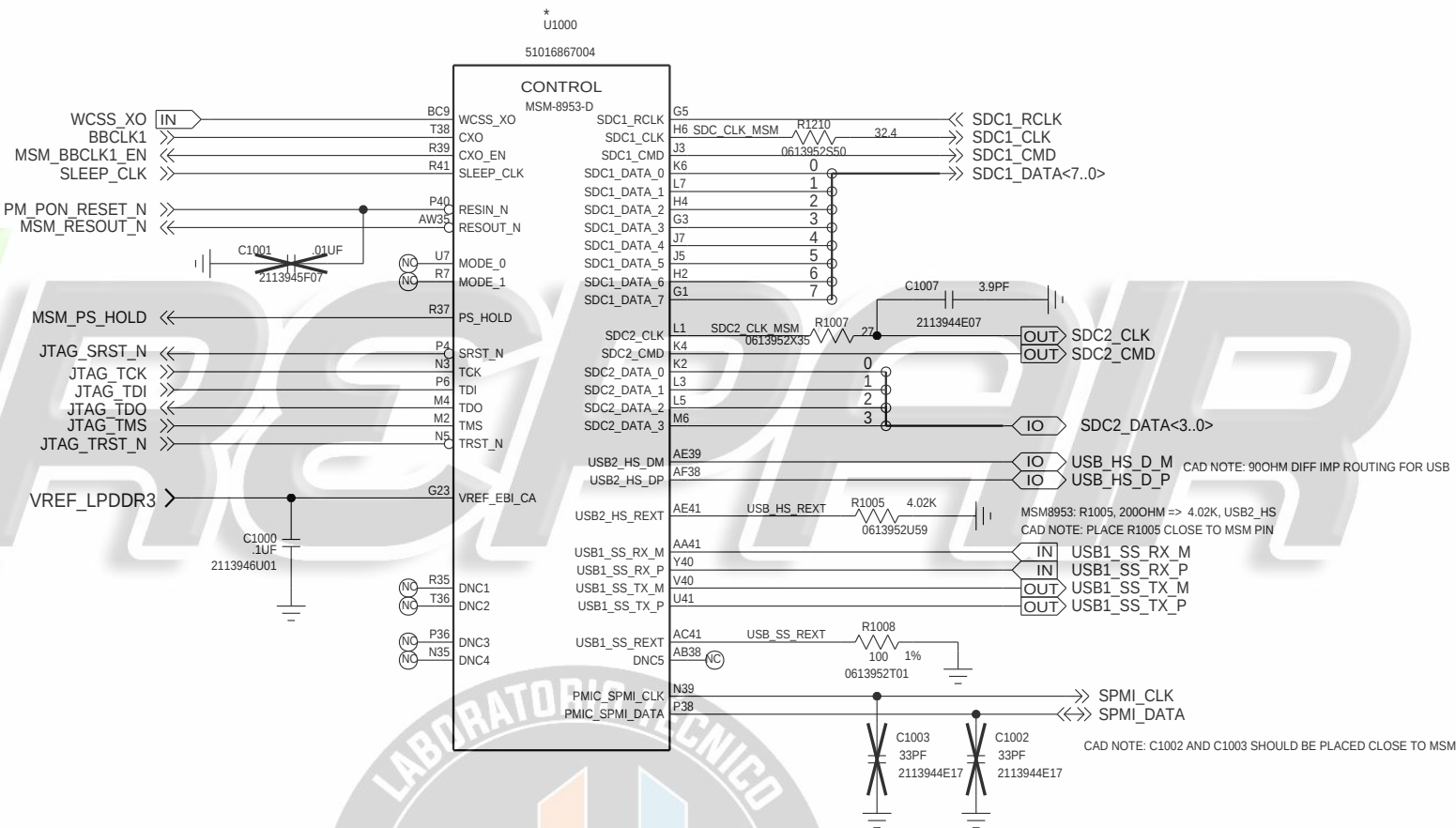
HIF2	HIF1	WRITE	READ
LOW	LOW	0X50	0X51
LOW	HIGH	0X52	0X53
HIGH	LOW	0X54	0X55
HIGH	HIGH	0X56	0X57



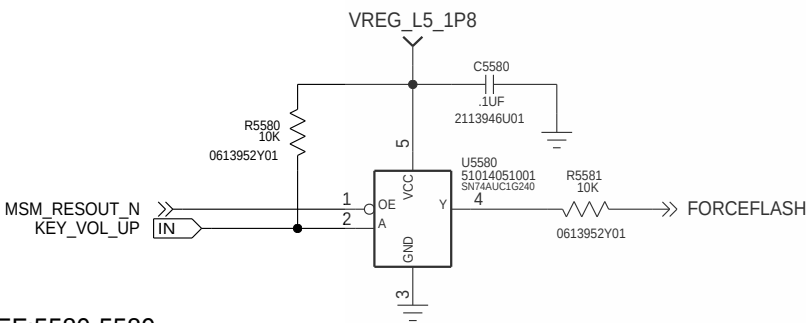
APPROVALS 		 MOTOROLA MOBILE DEVICES	
ENGINEER: MOTOROLA		BLOCK TITLE:	
LAST MODIFIED BY: MOTOROLA Mon Jan 09 09:39:59 2017		connectivity	
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		CAD FILE NAME:	
		PAGE 16 OF 64	

MSM: CTRL

REF 1000-1099



FORCE FLASH SUPPORT
WITH VOL UP KEY
(DNI FOR PRODUCTION)



APPROVALS

ENGINEER:

LAST MODIFIED BY: FAN021
Tue Nov 15 13:45:58 2016

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MOTOROLA
MOBILE DEVICES

BLOCK TITLE:

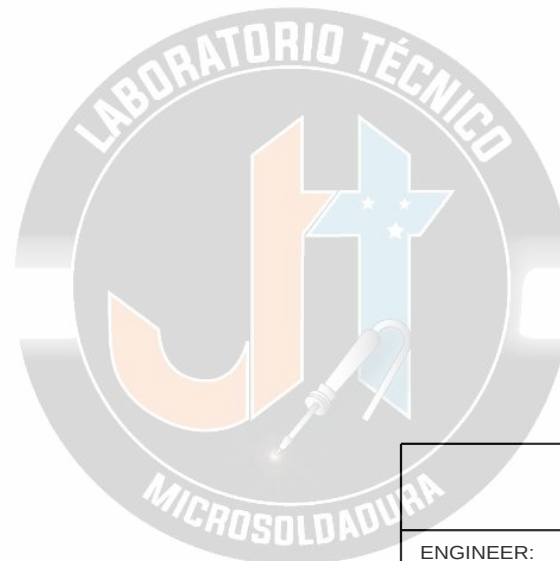
ap_pmic

PROJECT NAME
houdini_09.cpm

CAD FILE NAME:

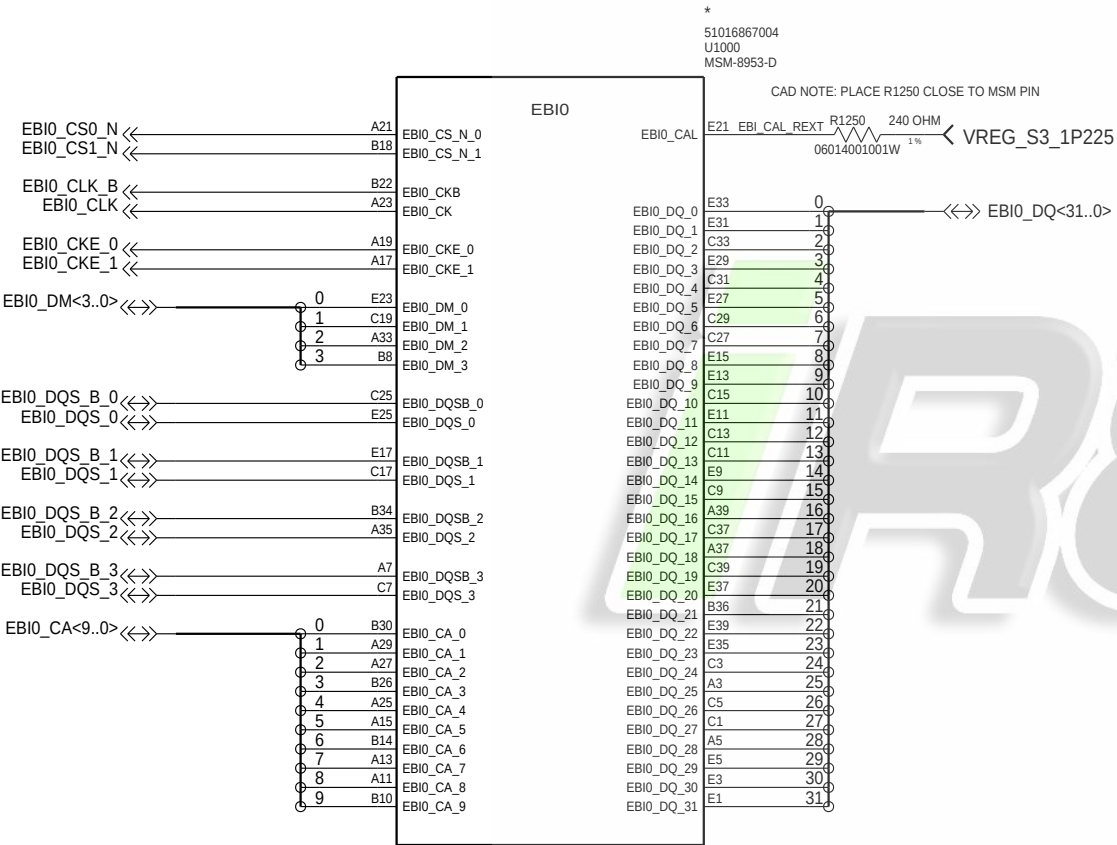
REV
01

REF 1000-1099

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MSM: MIPI/DSI/CSI/EBIO

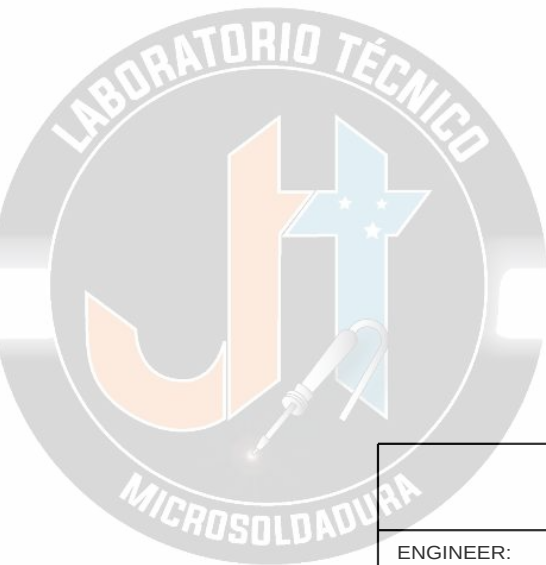
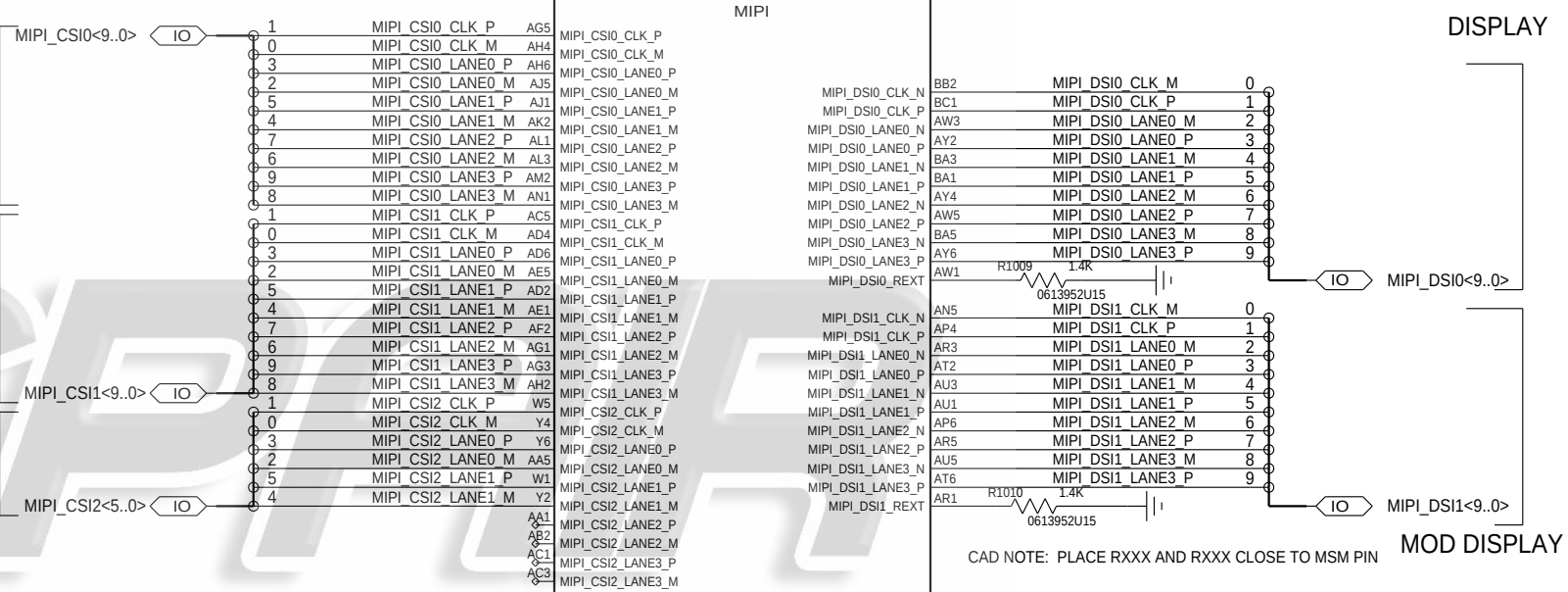
REF 1200-1299



MODS CAMERA

REAR CAMERA

FRONT CAMERA



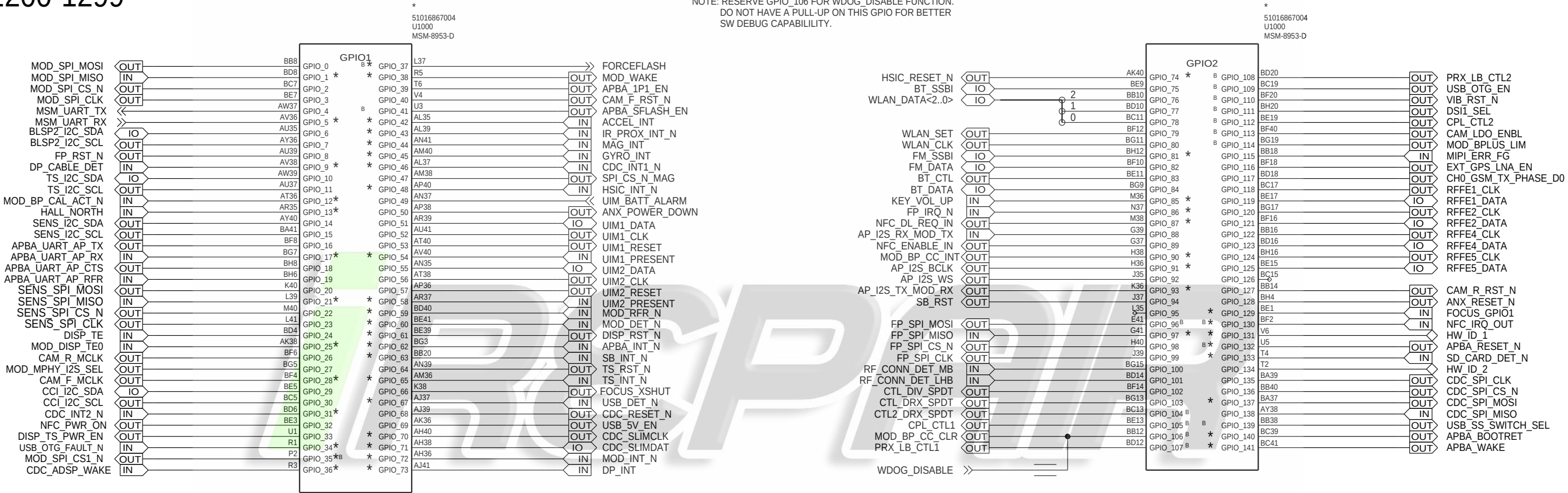
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ENGINEER:		BLOCK TITLE:	
LAST MODIFIED BY:		ap_pmic	
Tue Nov 15 13:45:59 2016			
MOTOROLA CONFIDENTIAL RESTRICTED		PROJECT NAME	
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		REV 01	
		PAGE 19 OF 64	

MSM: GPIO (0-141)
REF 1200-1299

NOTE: ASTERISK(*) INDICATE MODEM POWER MANAGEMENT (MPM) WAKE-UP PINS.

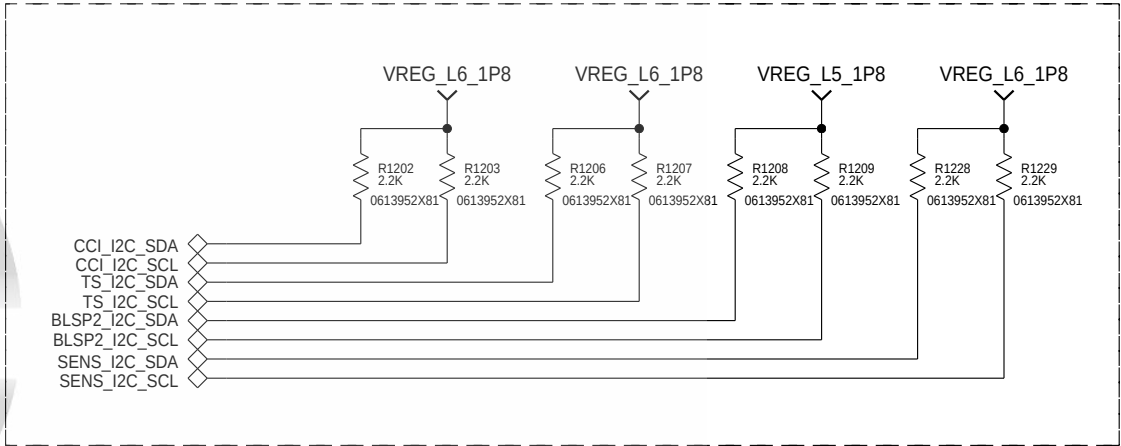
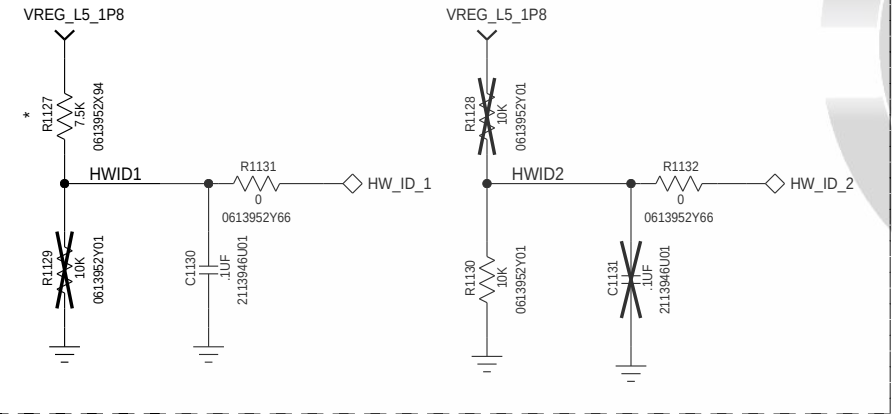
NOTE: DO NOT HAVE PULL-UPS ON THE FOLLOWING GPIOs UNLESS INTEND FOR BOOT OR SECURE BOOT RELATED CONFIGURATIONS:
GPIO_8, 9, 91, 107, 108, 109

NOTE: RESERVE GPIO_106 FOR WDOG_DISABLE FUNCTION.
DO NOT HAVE A PULL-UP ON THIS GPIO FOR BETTER SW DEBUG CAPABILITY.



MOT BOARD ID

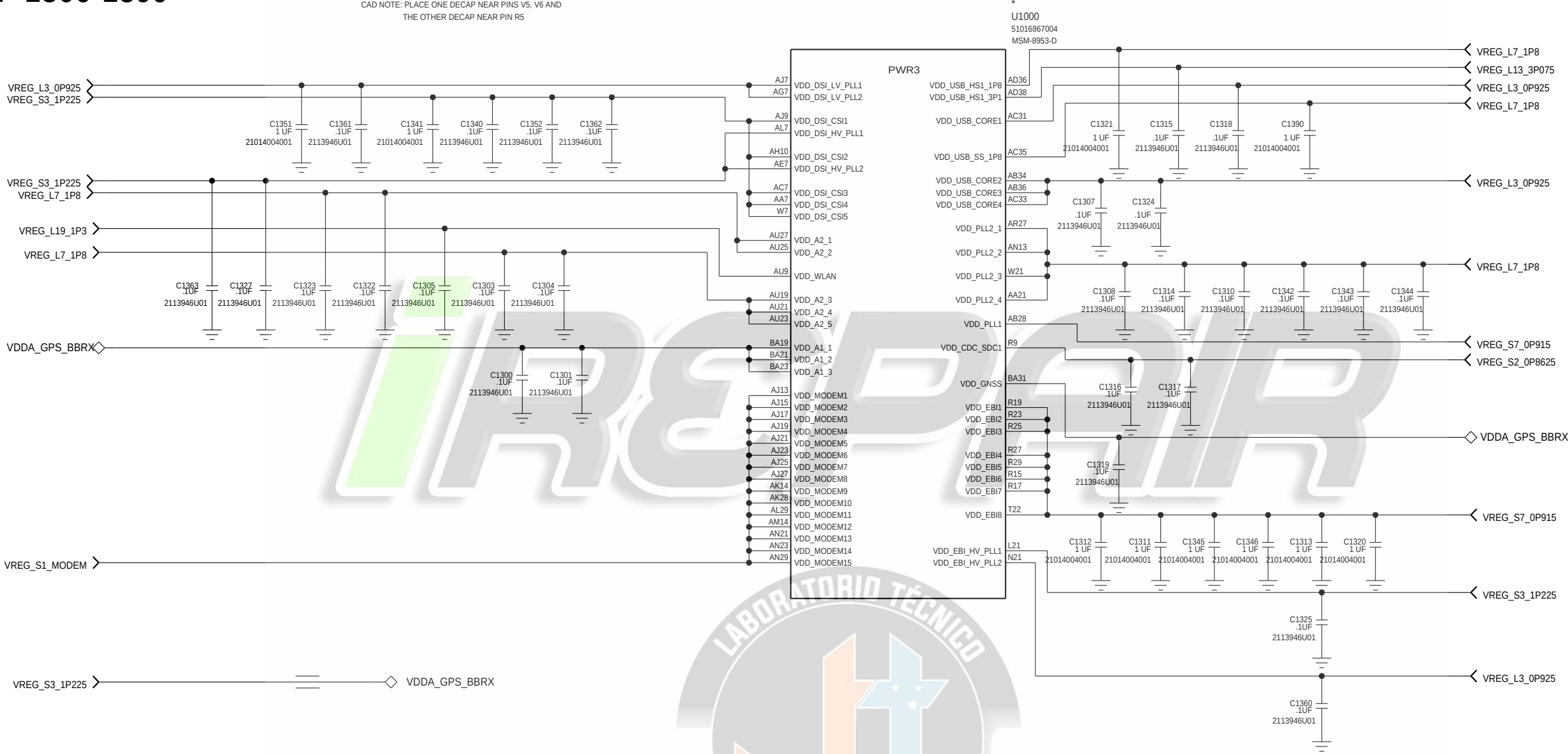
FOR RUNTIME REVISION INFORMATION, REFER TO:
<https://sites.google.com/a/motorola.com/runtime-product-ids>



APPROVALS		MOTOROLA MOBILE DEVICES	
ENGINEER:		BLOCK TITLE:	
LAST MODIFIED BY: MOTOROLA Tue Nov 15 13:46:00 2016		ap_pmic	
MOTOROLA CONFIDENTIAL RESTRICTED		PROJECT NAME	REV
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		CAD FILE NAME:	PAGE 20 OF 64

MSM: PWR1
REF 1300-1399

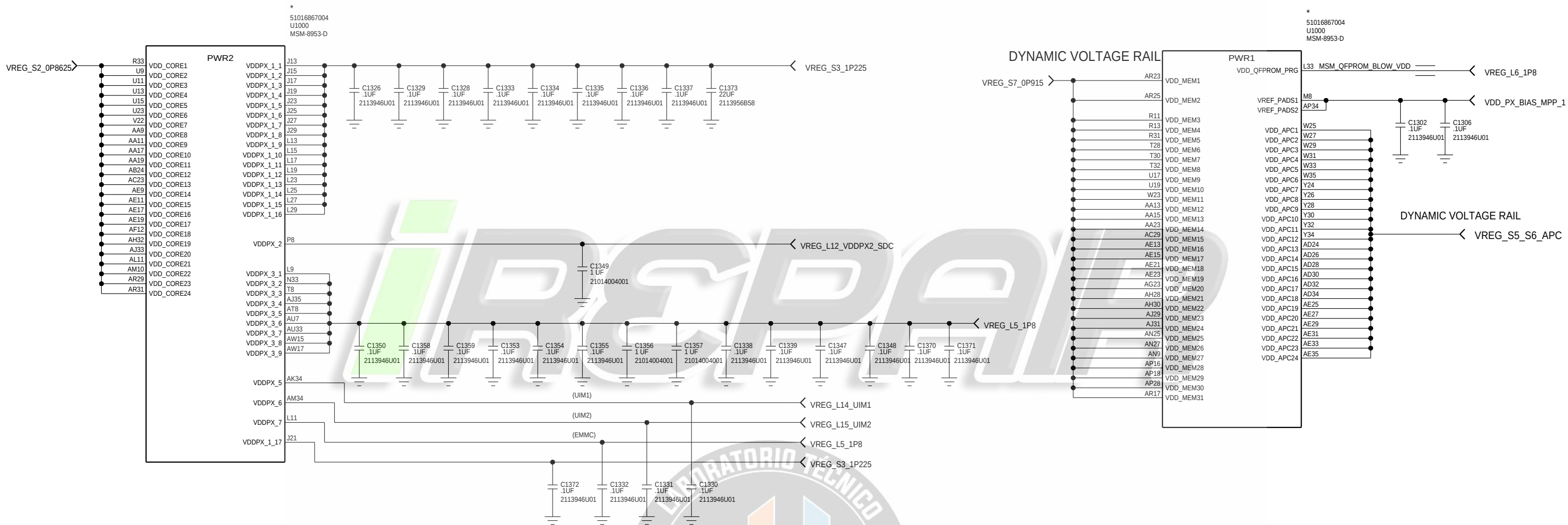
CAD NOTE: PLACE ONE DECAP NEAR PINS V5, V6 AND
THE OTHER DECAP NEAR PIN R5



APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER:		BLOCK TITLE:	
LAST MODIFIED BY: FAN021 Tue Nov 15 13:46:01 2016		ap_pmic	
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		CAD FILE NAME:	
		PAGE 21 OF 64	

MSM: PWR2

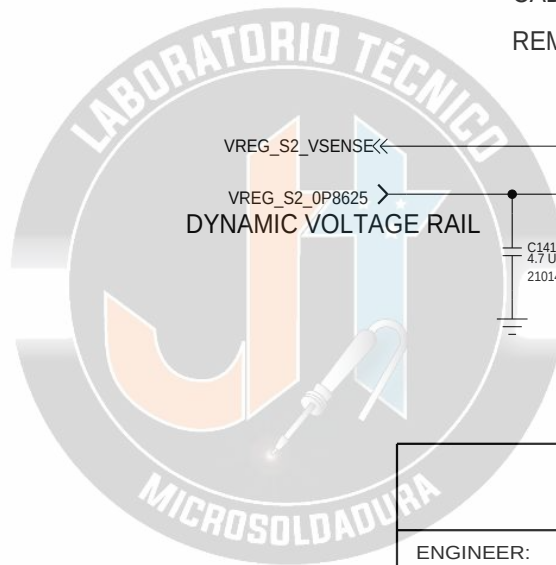
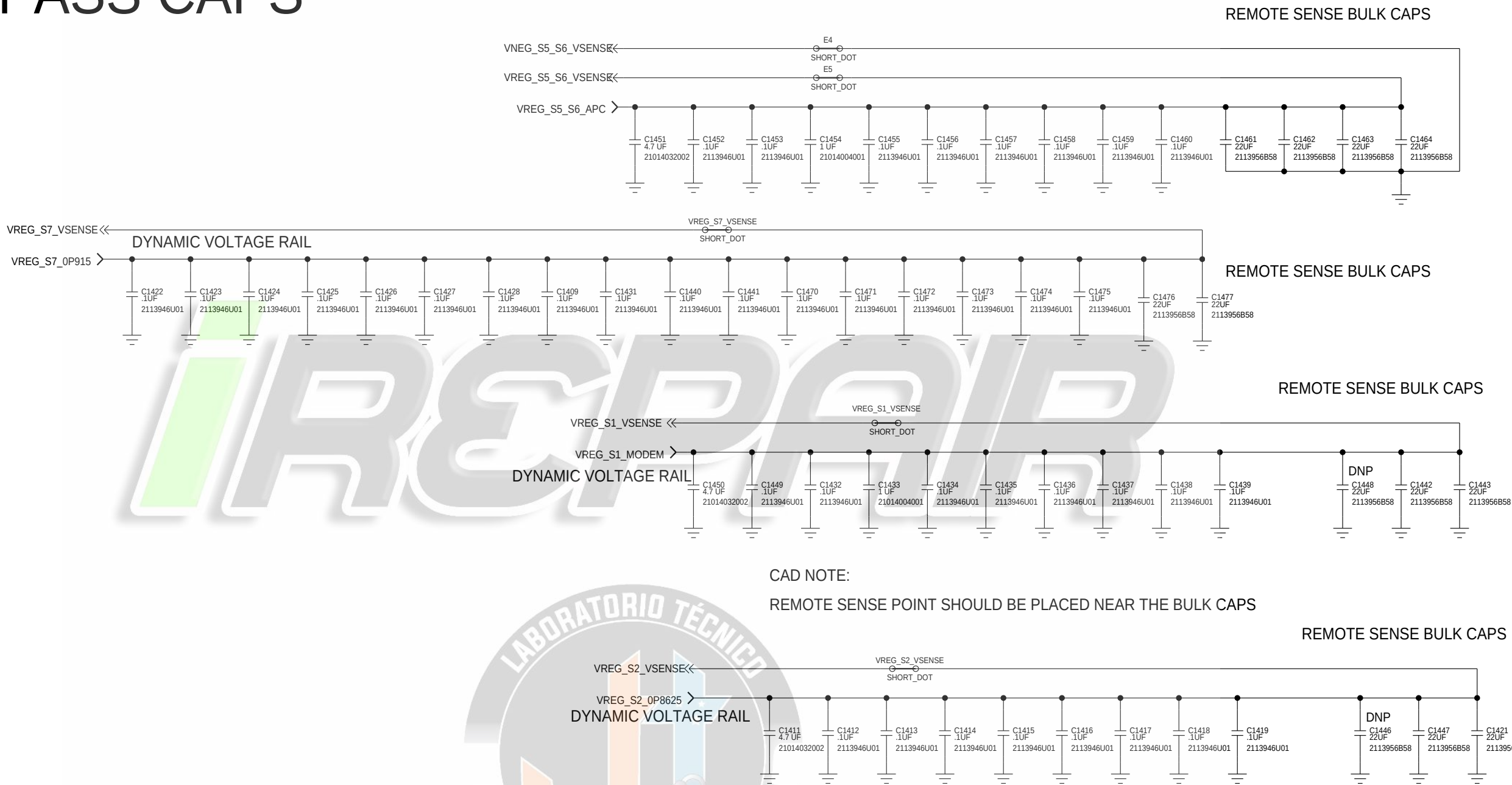
REF 1300-1399



APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER:		BLOCK TITLE:	
LAST MODIFIED BY: FAN021 Tue Nov 15 13:46:01 2016		ap_pmic	
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		CAD FILE NAME:	
		PAGE 22 OF 64	

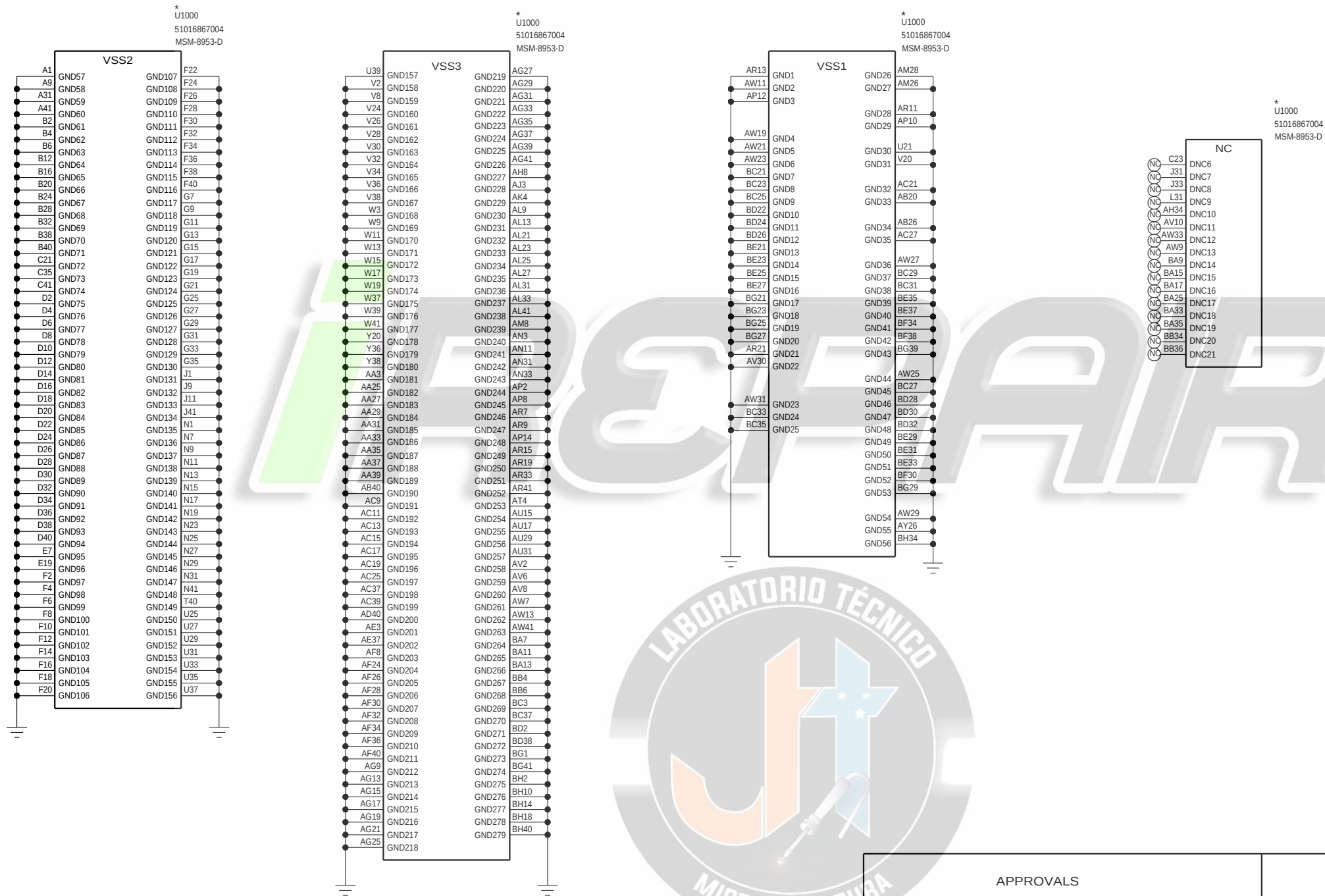
MSM: BYPASS CAPS

REF 1400-1499



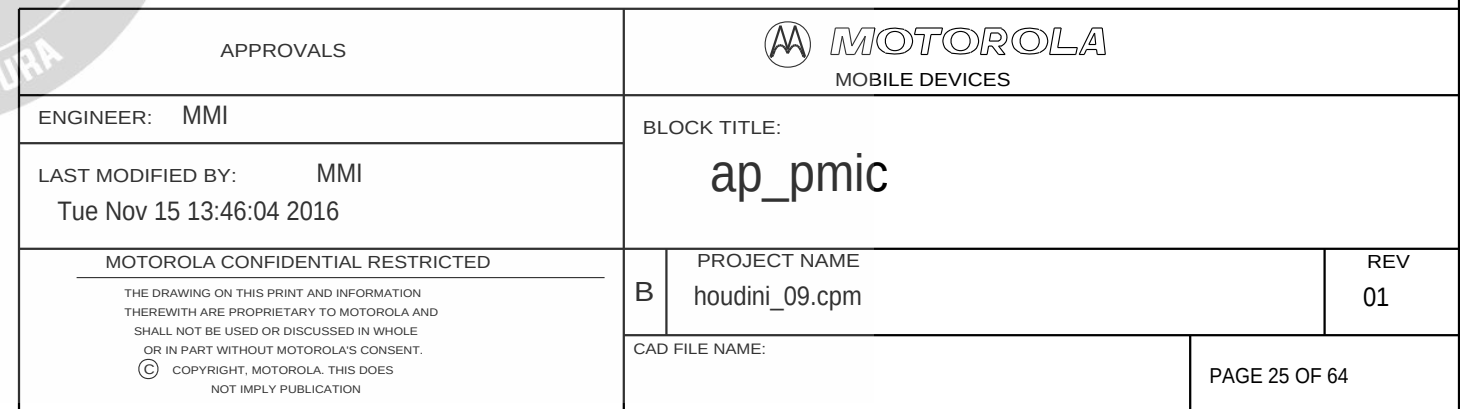
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CAD FILE NAME:			PAGE 23 OF 64

MSM: GROUND/DNC



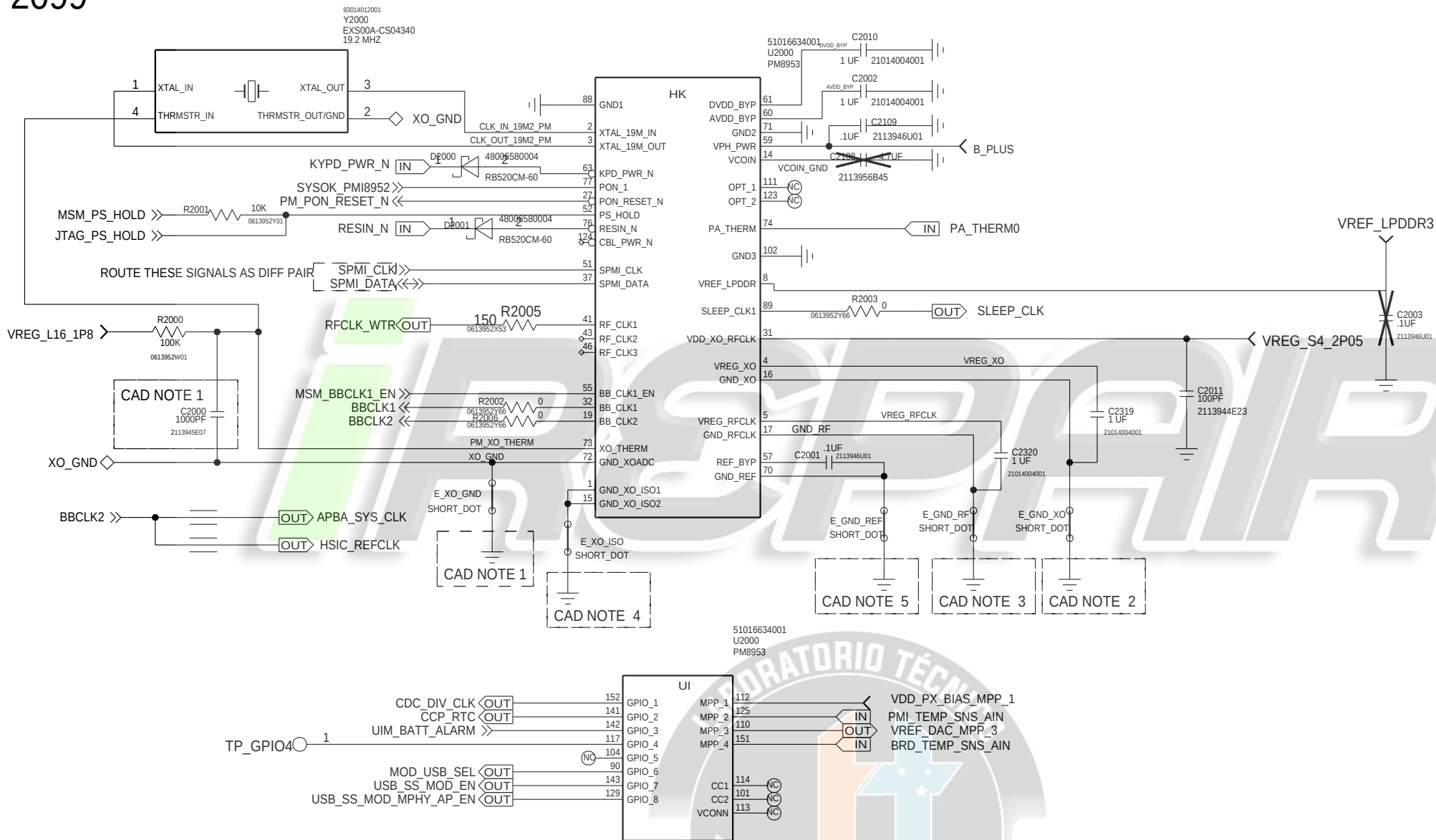
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		B	houdini_09.cpm
		REV 01	
		PAGE 24 OF 64	

REF: 1600-1699 MCP: 16GB EMMC, 2GB LPDDR3



PM8953: HK/GHRG/LED/GPIO

REF: 2000-2099

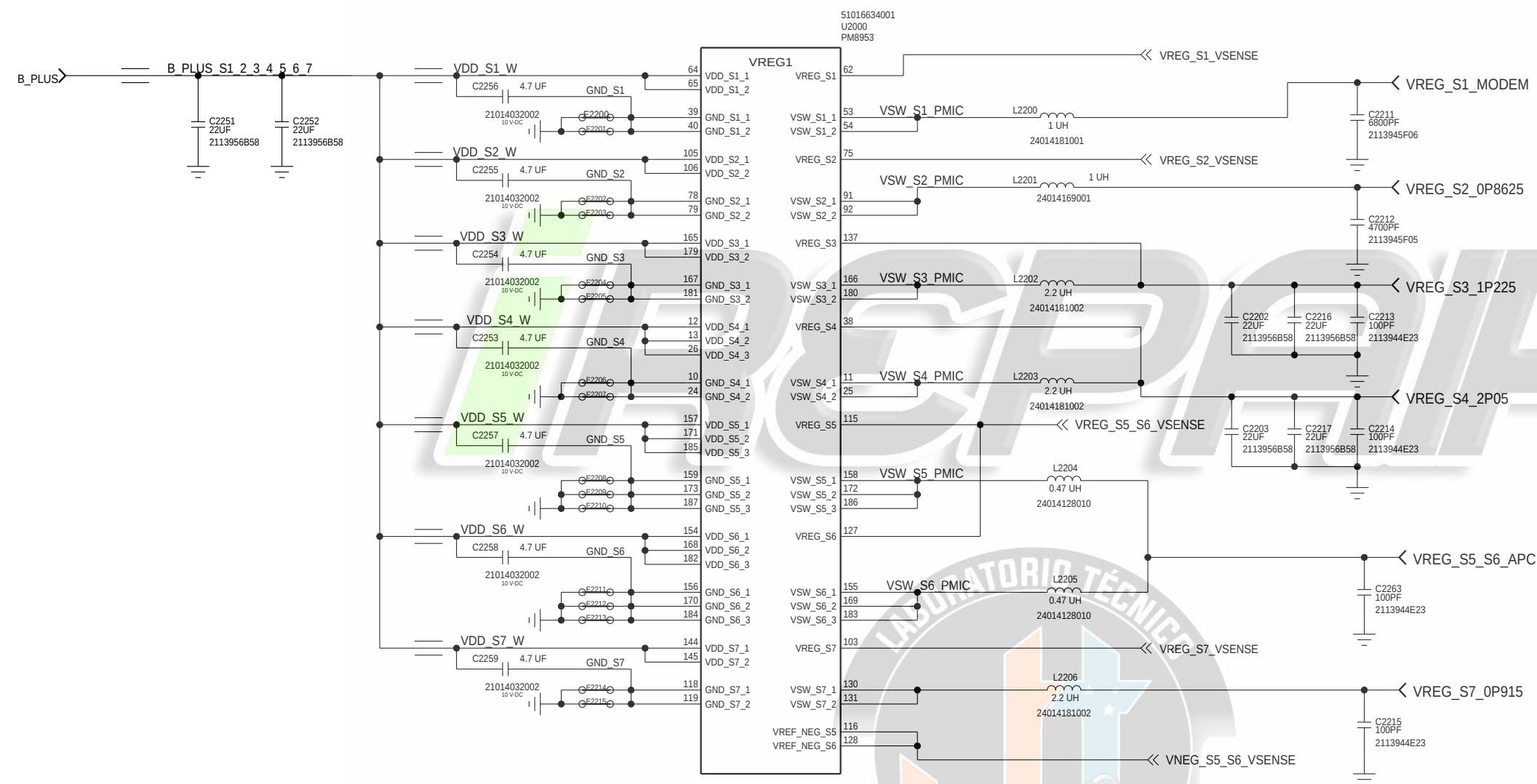


CAD NOTE 1 : PLACE C2000 CLOSER TO Y2000 AND CONNECT GND_XOADC TO CAP GND AND THEN CONNECT TO MAIN GND PLANE THROUGH DEDICATED VIA
CAD NOTE 2 : CONNECT GND_XO TO C2319 CAP GND AND THEN CONNECT TO MAIN GND USING DEDICATED VIA.
CAD NOTE 3 : CONNECT GND_RFCLK TO C2320 CAP GND AND THEN CONNECT TO MAIN GND PLANE USING DEDICATED VIA.
CAD NOTE 4 : CONNECT GND_XO_ISO TO MAIN GND PLANE USING DEDICATED VIA. DO NOT SHARE THIS GND WITH OTHER GROUNDS.
CAD NOTE 5 : CONNECT GND_REF TO CAPACITOR C2001 GND AND THEN CONNECT TO MAIN GND PLANE USING DEDICATED VIA. DON NOT SHARE THIS GND WITH OTHER GND.

APPROVALS		MOTOROLA MOBILE DEVICES	
ENGINEER: MMI		BLOCK TITLE:	
LAST MODIFIED BY: MMI Tue Nov 15 13:46:05 2016		ap_pmic	
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		CAD FILE NAME:	
		PAGE 26 OF 64	

PM8953: SMPS

REF: 2200-2299

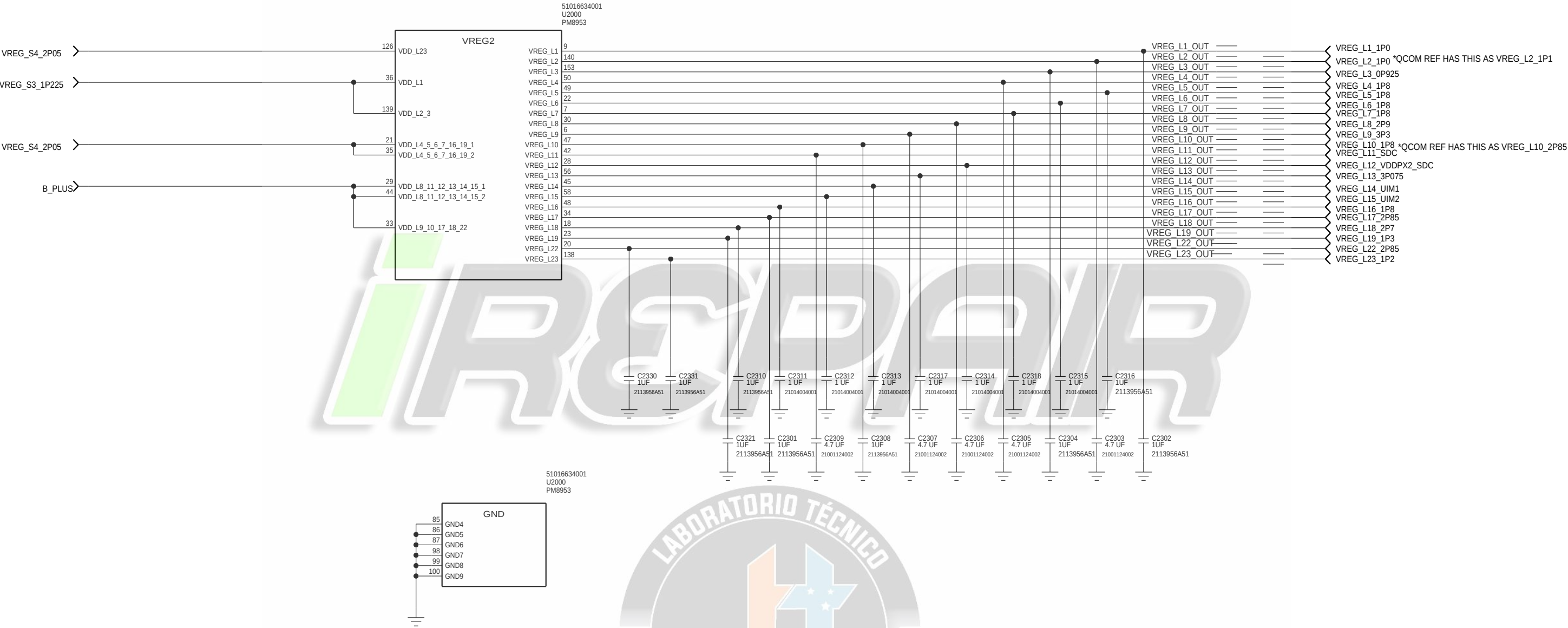


NOTE: PLACE INPUT CAPS CLOSE TO RESPECTIVE VIN PINS
EACH CAP GND SHOULD BE STAR-ROUTED

APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: MMI	BLOCK TITLE: ap_pmic		
LAST MODIFIED BY: MMI Tue Nov 15 13:46:05 2016			
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		CAD FILE NAME:	PAGE 27 OF 64

PM8953: VREGS

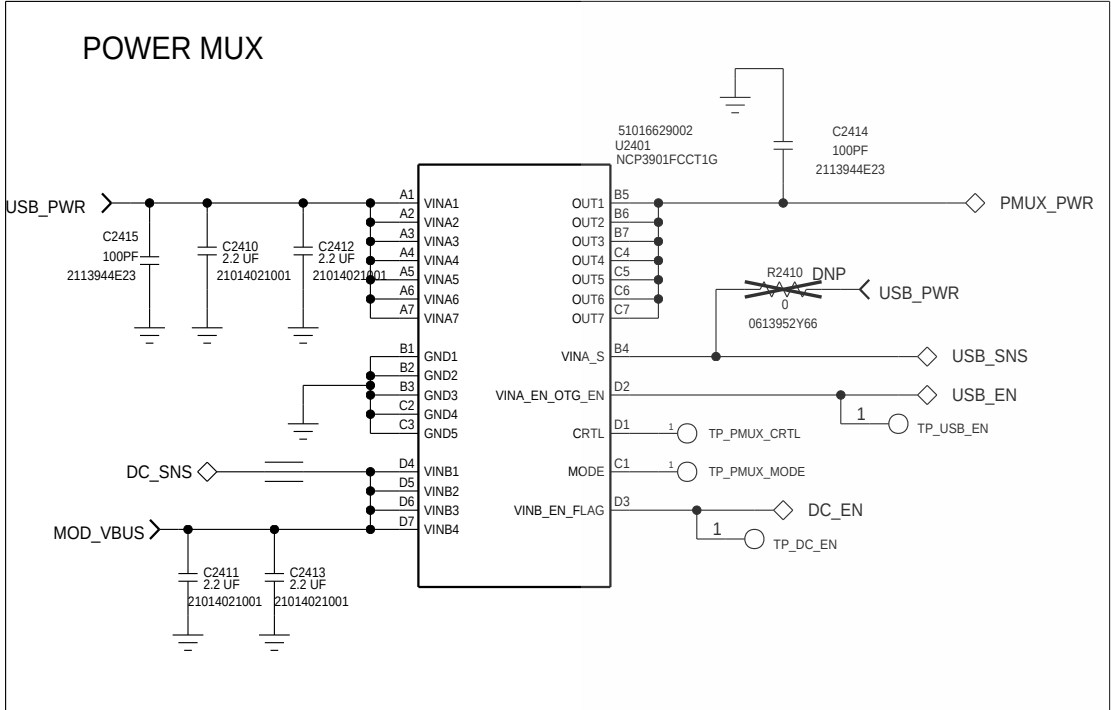
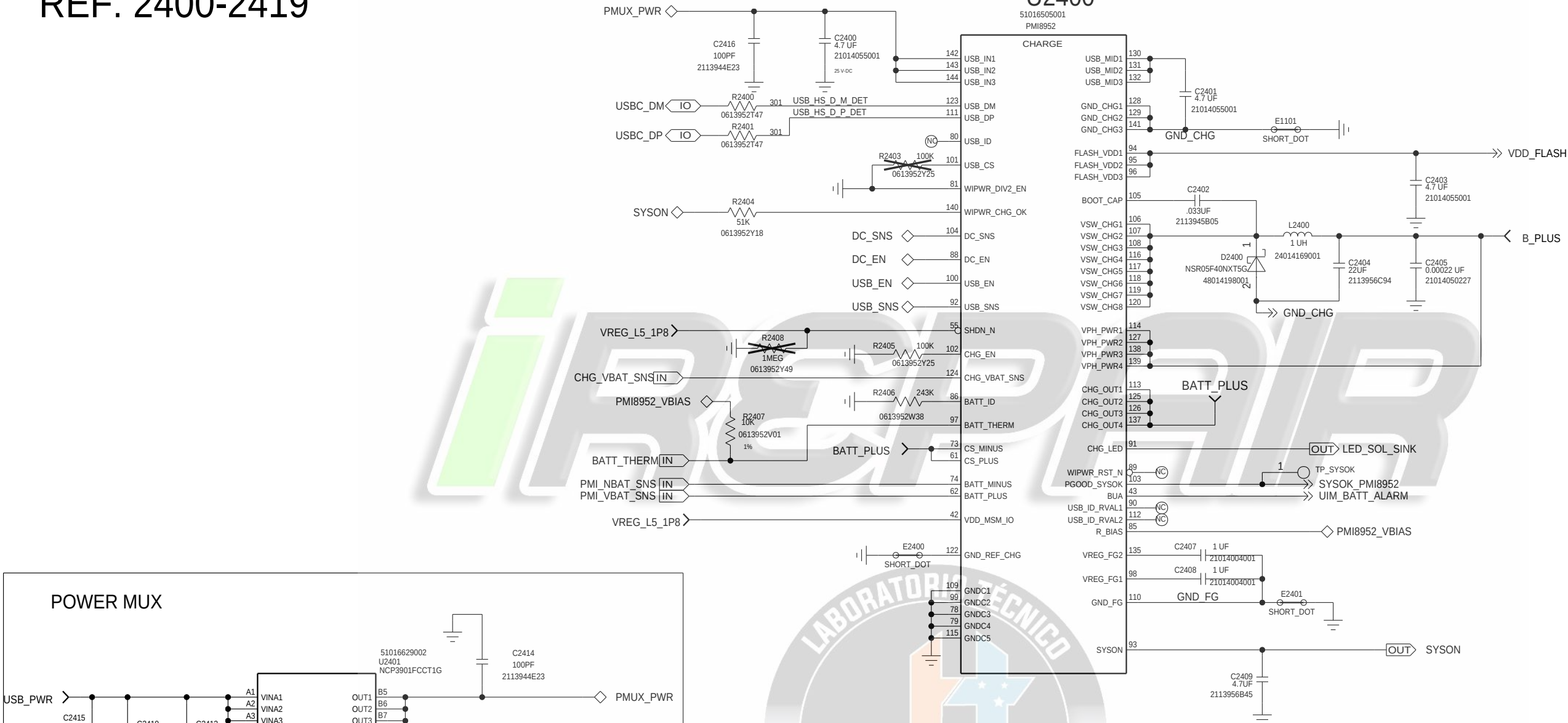
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		B houdini_09.cpm		01
		CAD FILE NAME:		PAGE 28 OF 64

PMI8952: CHARGING

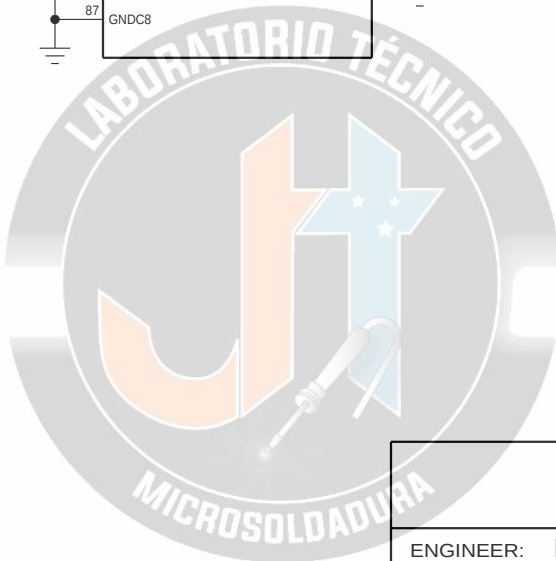
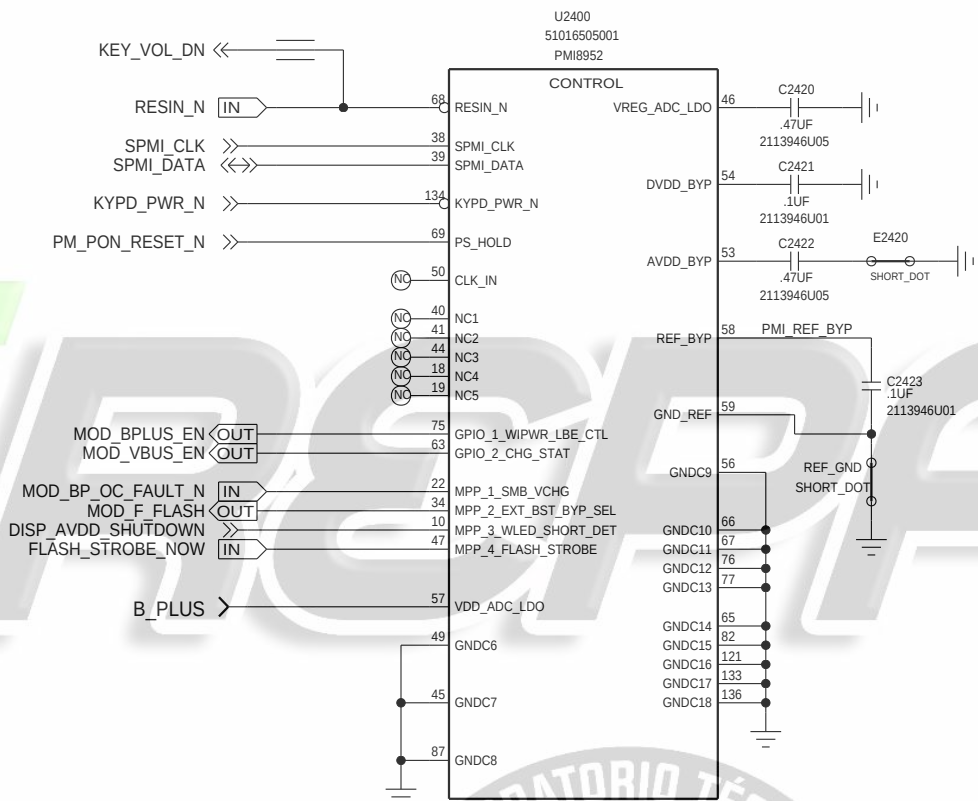
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		CAD FILE NAME:	
		PAGE 29 OF 64	

PMI8952: IO_REF

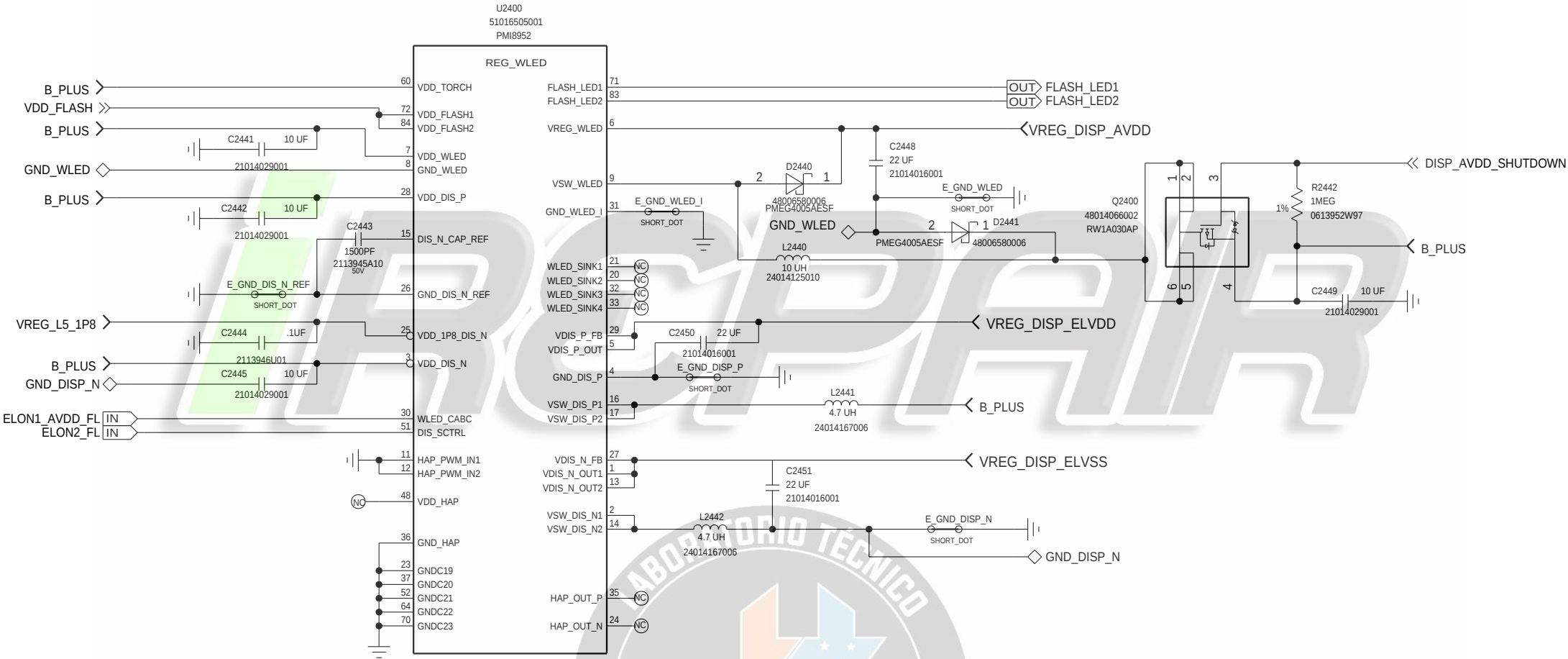
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		REV 01	
		CAD FILE NAME:	
		PAGE 30 OF 64	

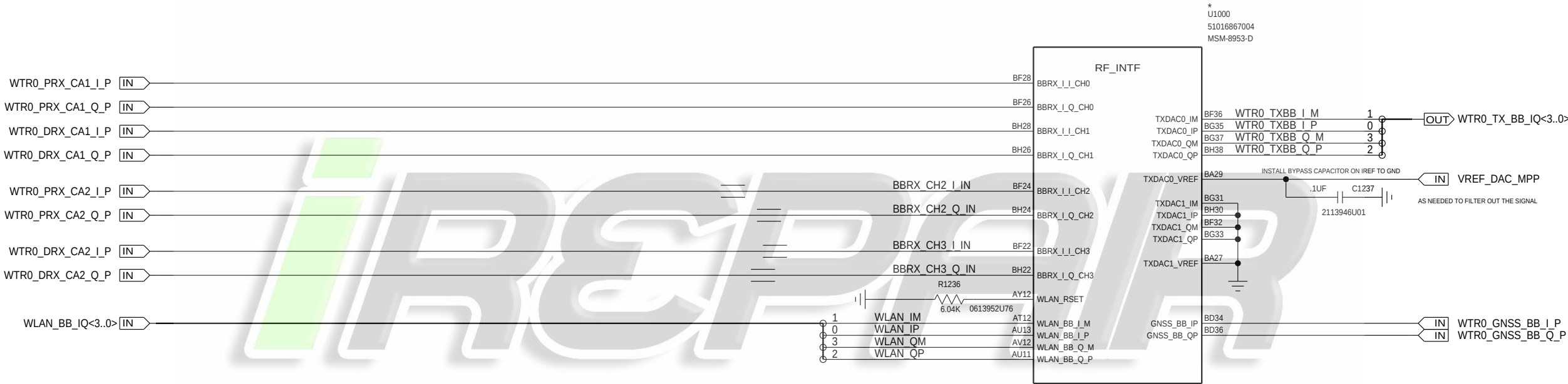
PMI8952: REG_WLED

REF: 2440-2459



APPROVALS		 MOTOROLA MOBILE DEVICES	
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		REV 01 PAGE 31 OF 64	

MSM: RF INTERFACE
REF 1200-1299

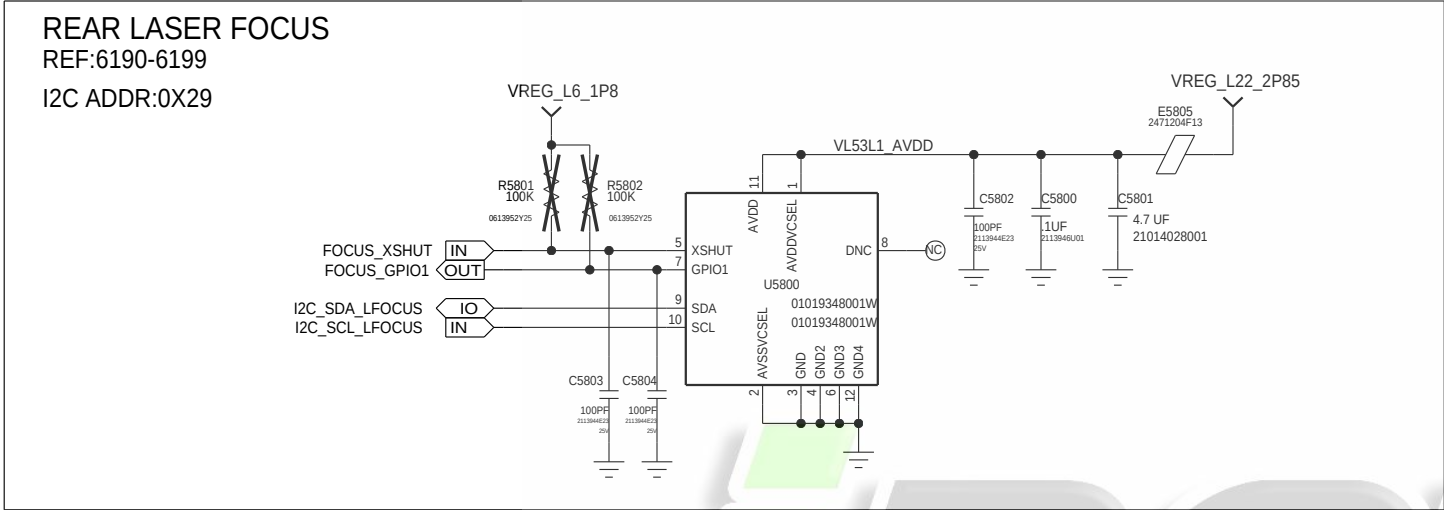


NEED TO APPLY LC FILTER FOR REDUCE B40 RX SPURS(QC 80-NP237-56)
LC FILTER COULD BE PLACED ANYWHERE BETWEEN WTR BBRX_IQ AND MSM BBRX_IQ PINS
BUT THE GND FOR SHUNT C SHOULD NOT BE SHARED WITH EITHER WTR OR MSM GND ON ANY LAYER.
A DIRECT VIA FROM SHUNT C TO MAIN GND IS RECOMMENDED

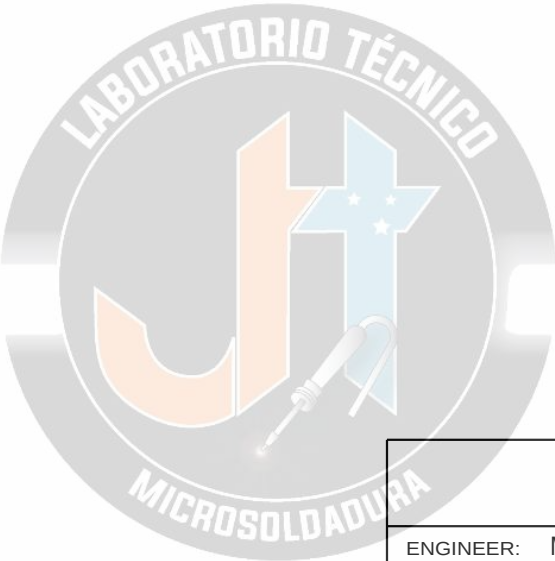
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		CAD FILE NAME:	
		PAGE 32 OF 64	

CAM: REAR LASER FOCUS

REF:5800-5899

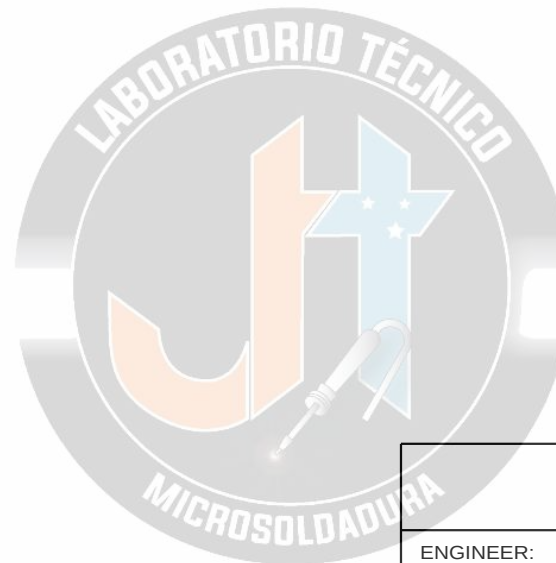
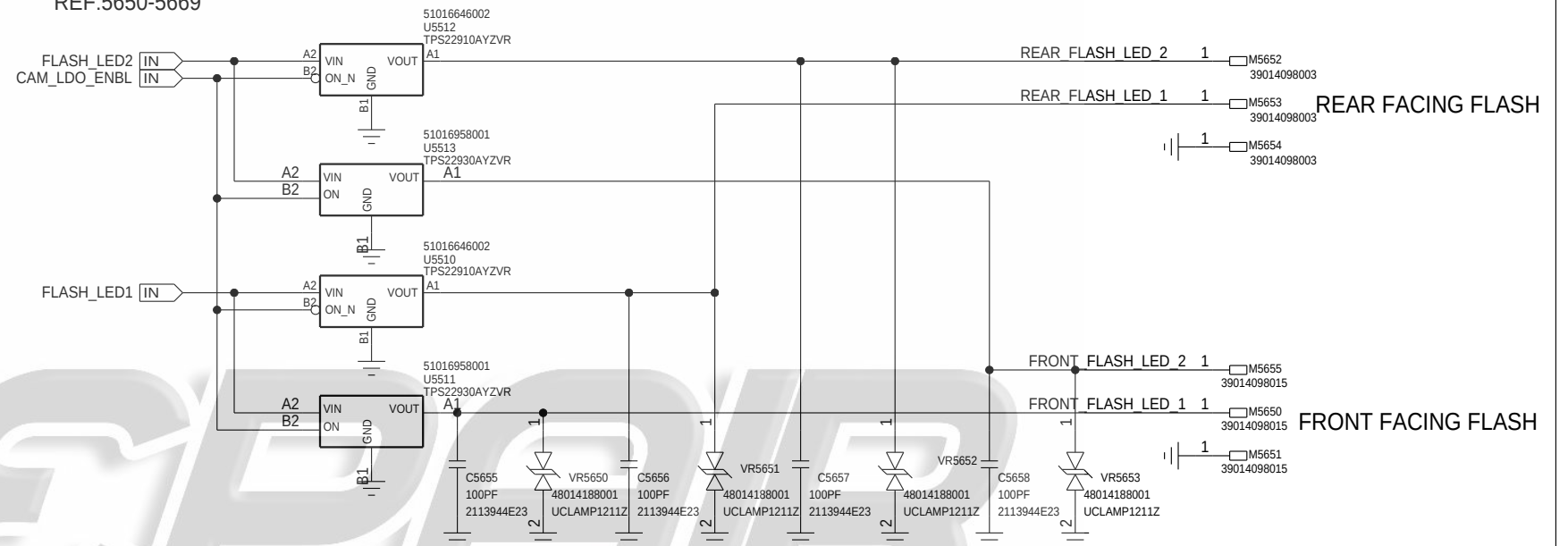
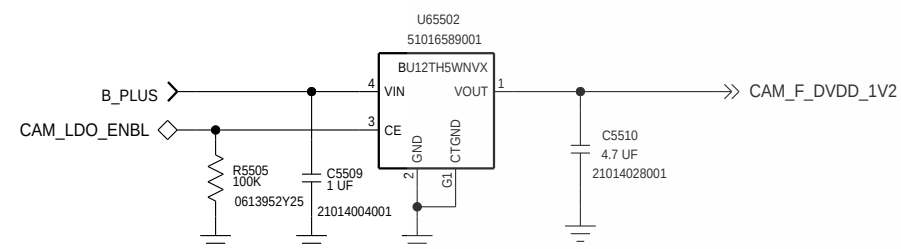
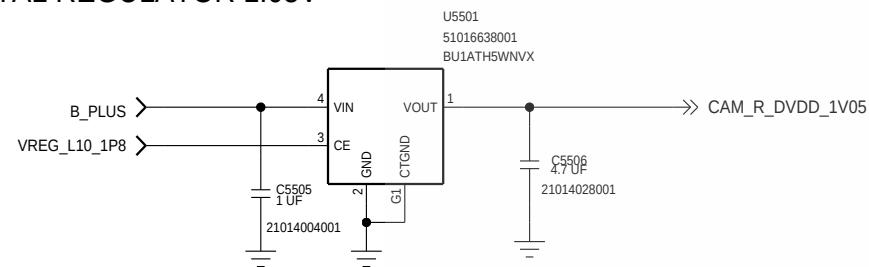
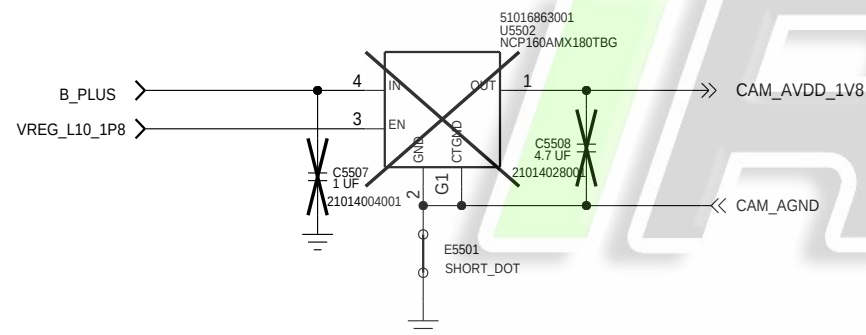
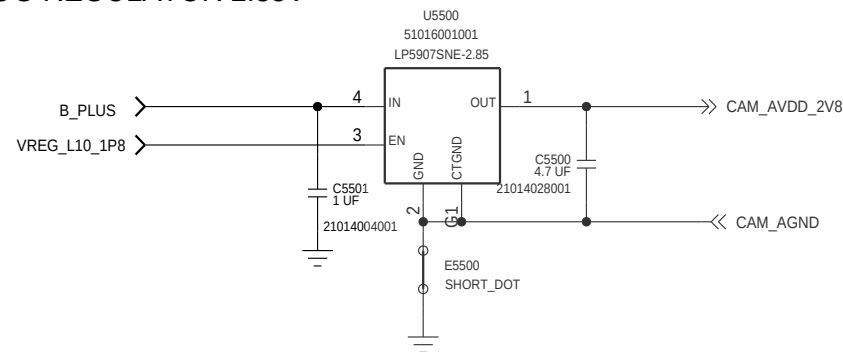


IREPAIR



APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: MOTOROLA		BLOCK TITLE:	
LAST MODIFIED BY: MOTOROLA Tue Nov 15 13:45:52 2016		multimedia	
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		CAD FILE NAME:	
		PAGE 33 OF 64	

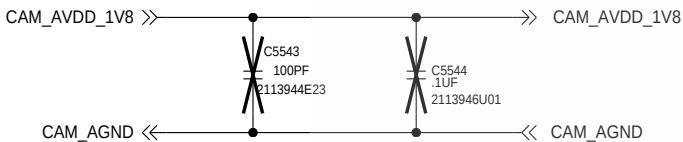
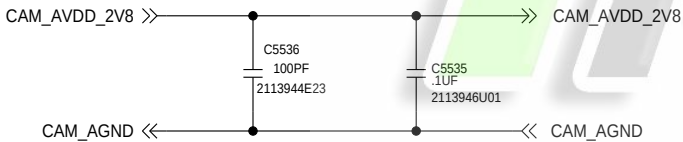
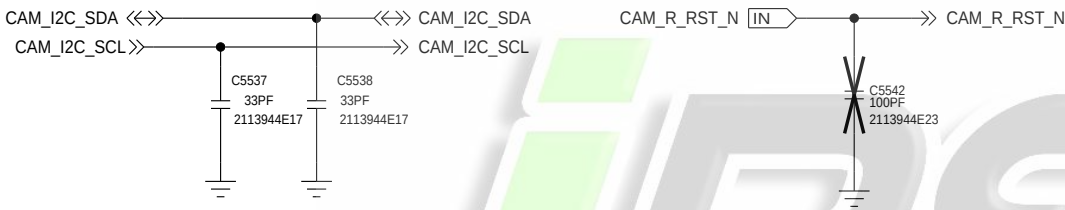
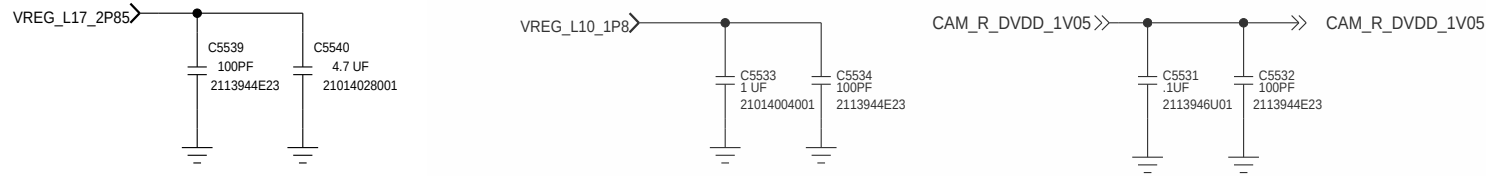
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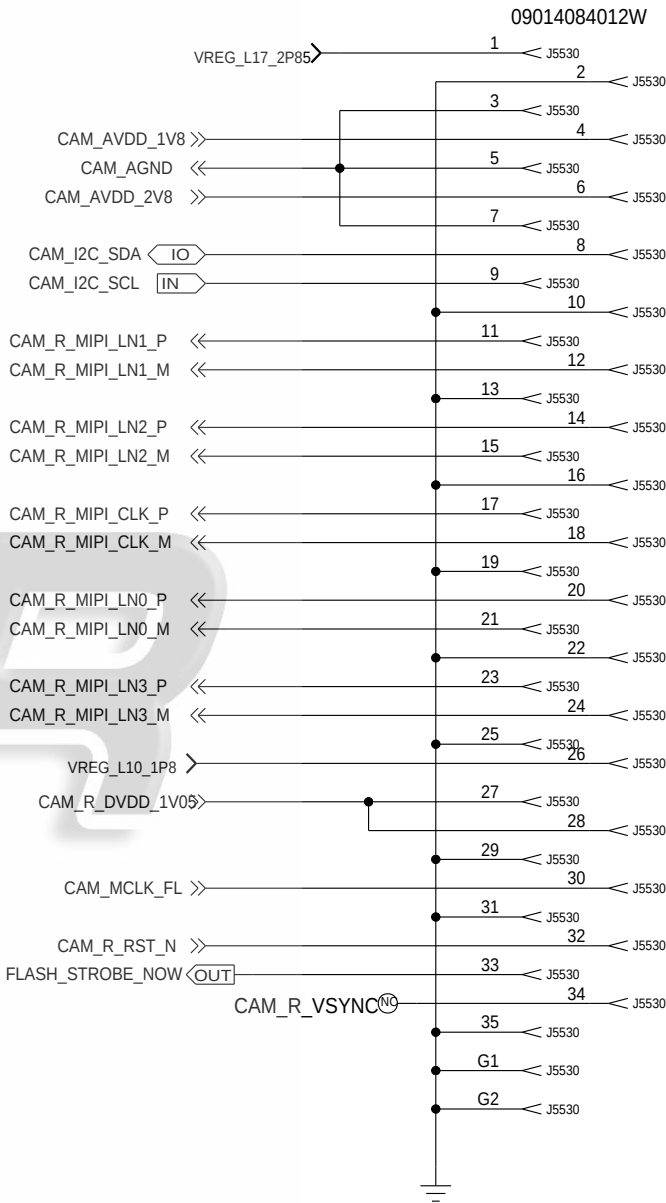
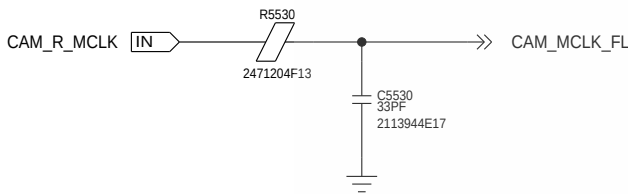
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		CAD FILE NAME:	PAGE 34 OF 64

CAM: REAR

REF:5530-5549



EMI/ESD FILTER



APPROVALS

ENGINEER: JOHN KRAMER

LAST MODIFIED BY: AZIZ NOORI
Tue Nov 15 13:45:54 2016

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MOTOROLA
MOBILE DEVICES

BLOCK TITLE:

multimedia

PROJECT NAME
houdini_09.cpm

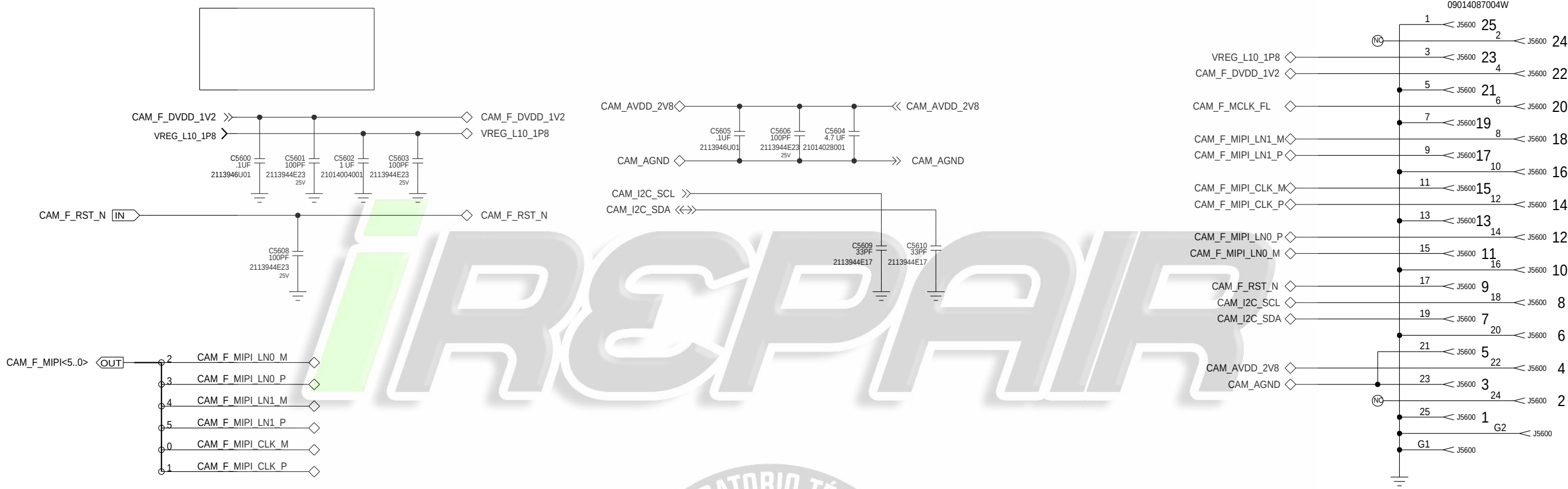
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CAD FILE NAME:

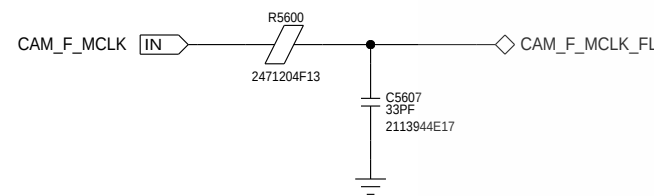
CAM: FRONT

REF 5600-5620

FF CAMERA CONNECTOR FRONT CAM I2C ADDR:



EMI/ESD FILTER



APPROVALS

ENGINEER: JOHN KRAMER

LAST MODIFIED BY: NATE CONNELL
Tue Nov 15 13:45:55 2016

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MOBILE DEVICES

BLOCK TITLE:

multimedia

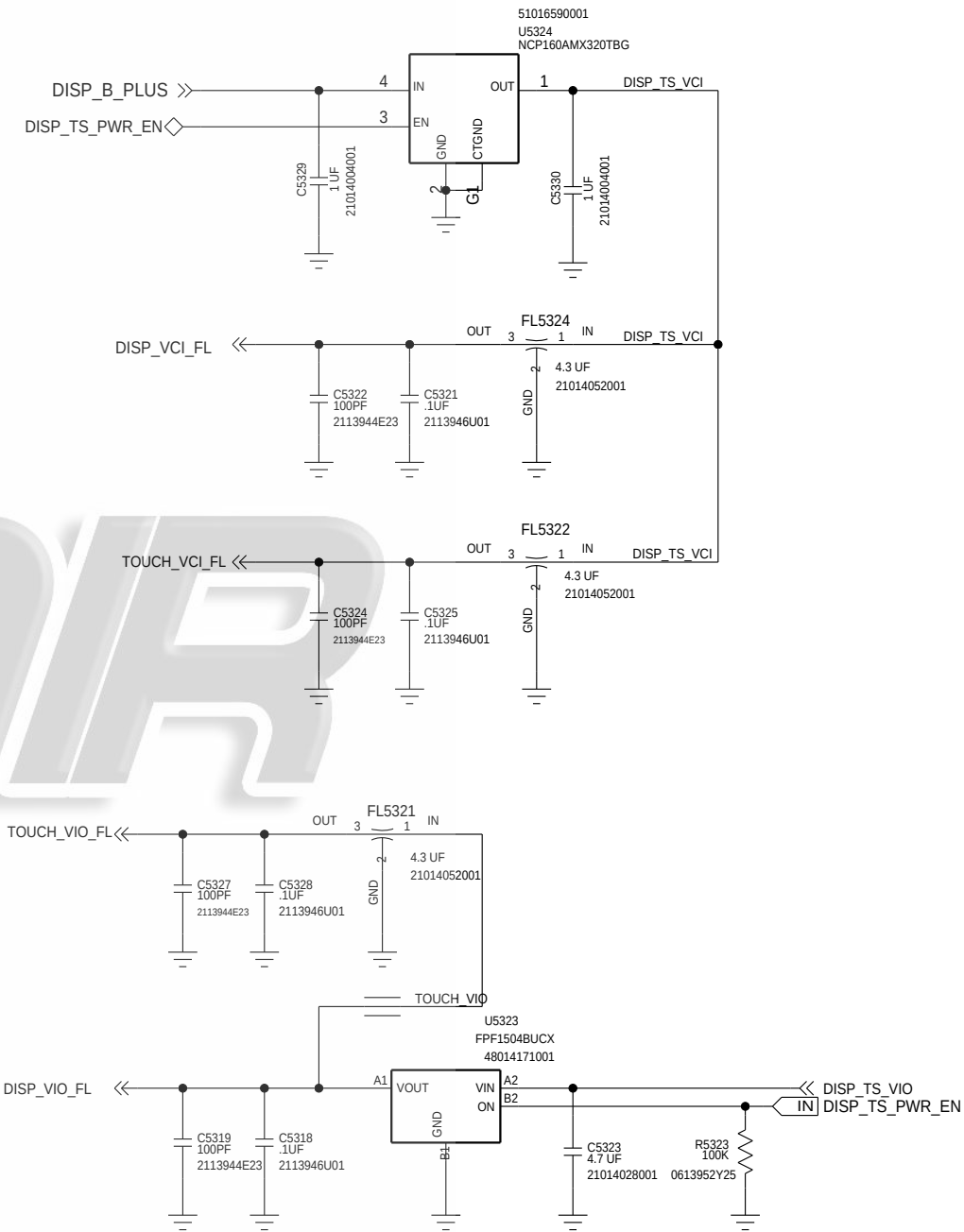
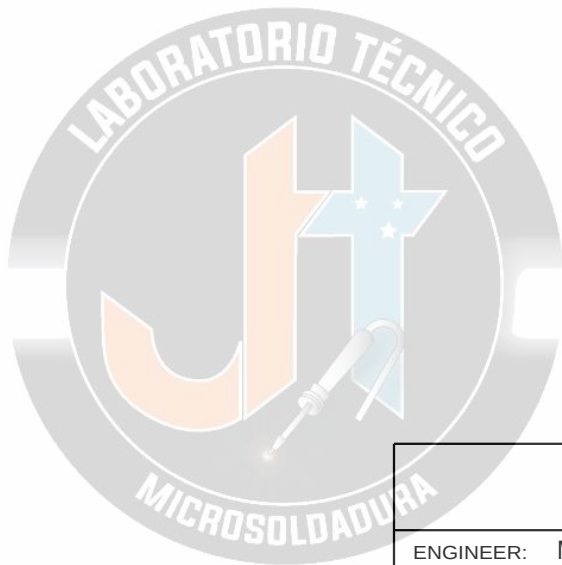
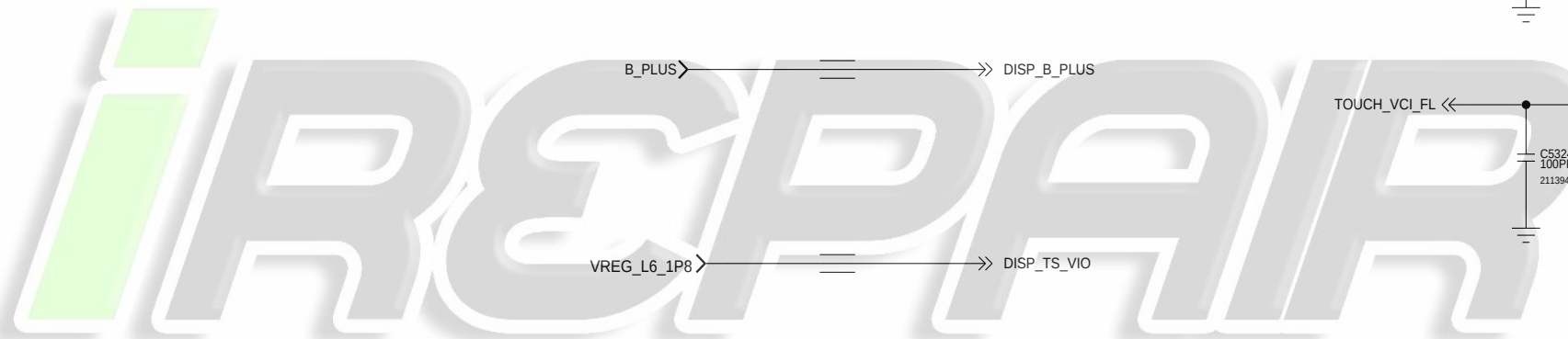
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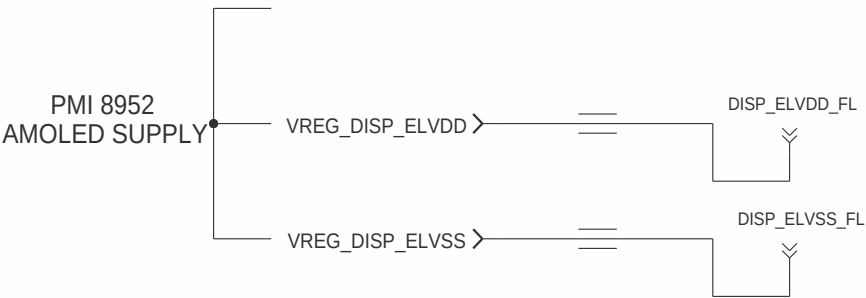
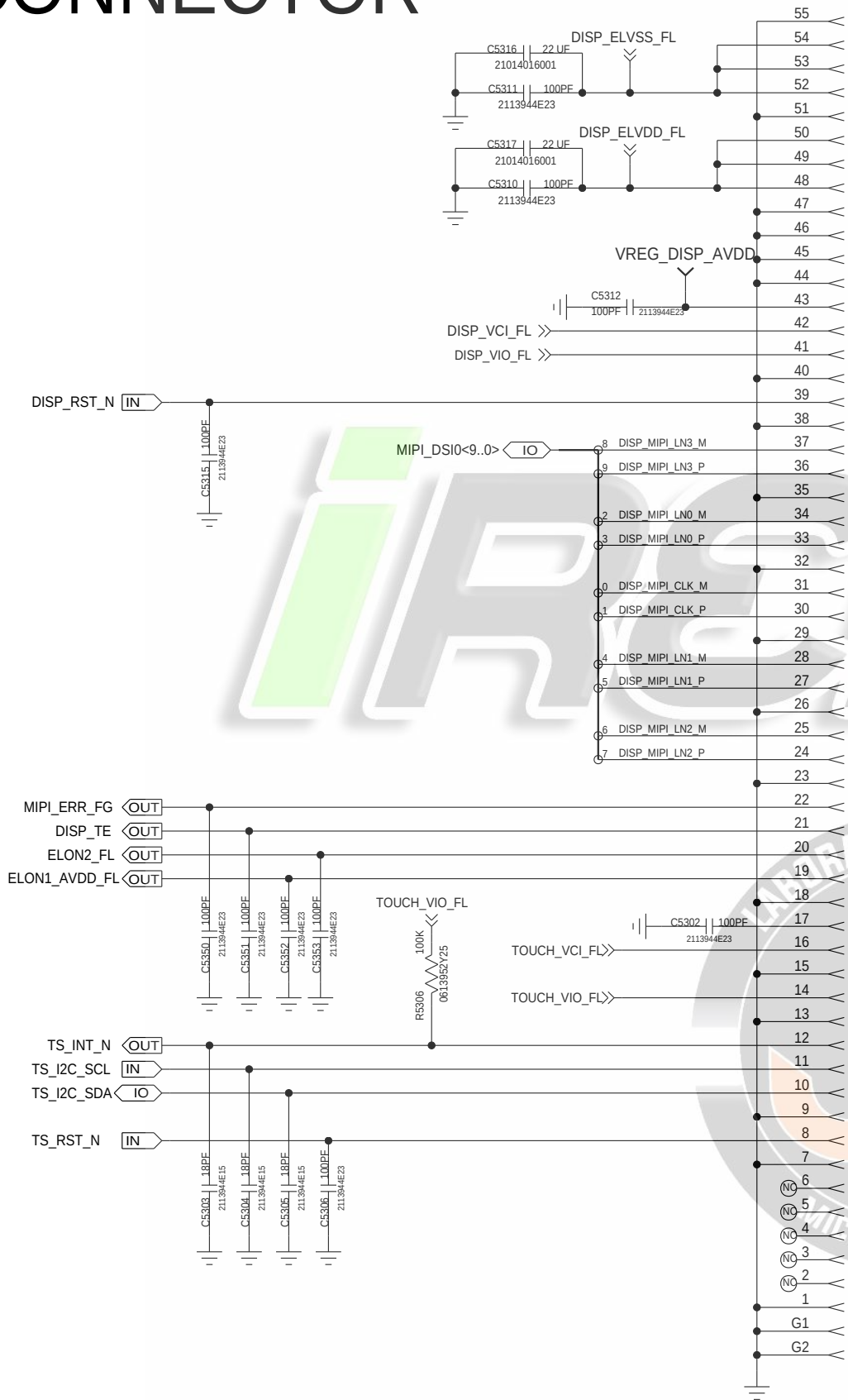
PAGE 36 OF 64

DISP: POWER



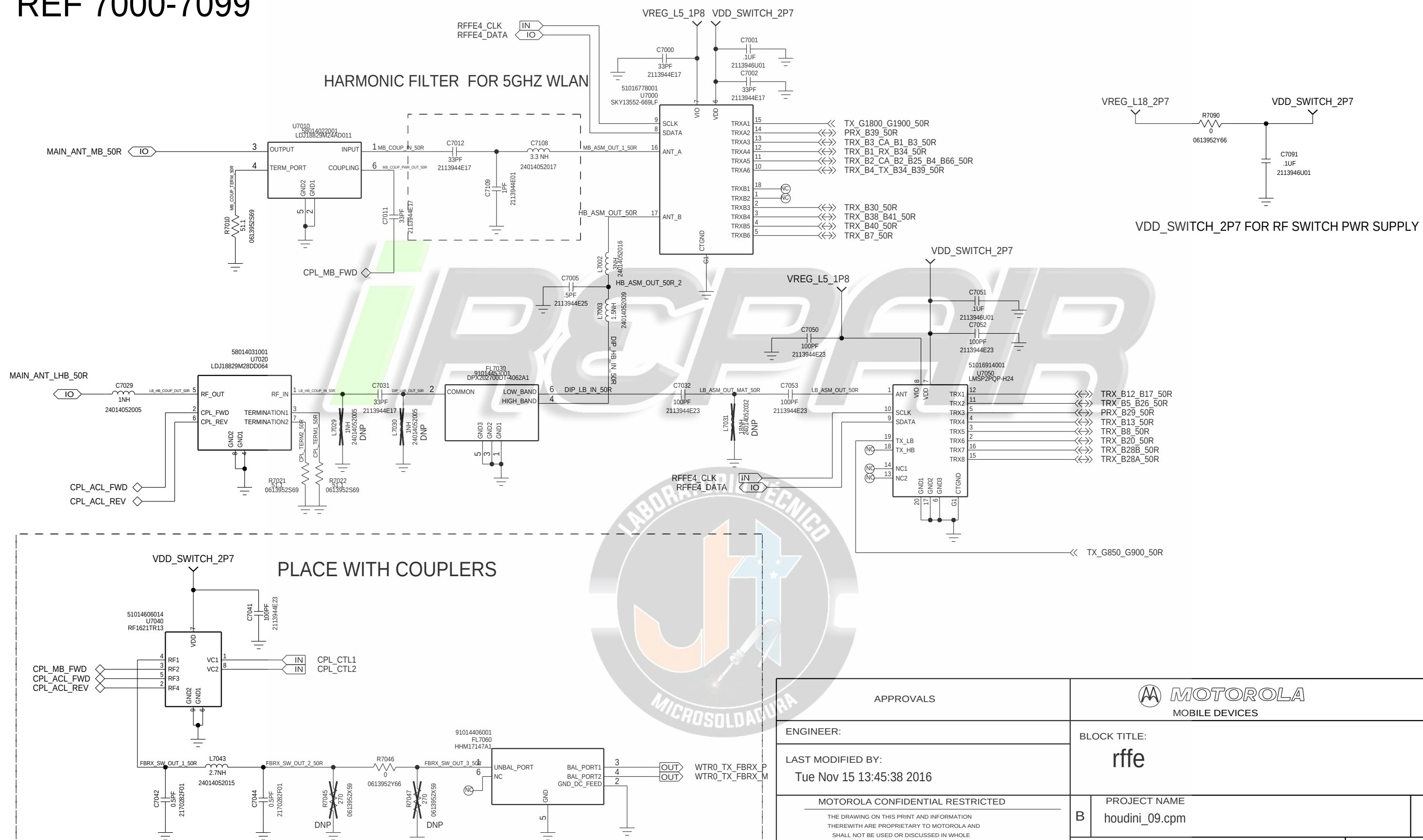
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		CAD FILE NAME: PAGE 37 OF 64	

DISP: CONNECTOR



APPROVALS		MOTOROLA MOBILE DEVICES	
ENGINEER: MMI		BLOCK TITLE: multimedia	
LAST MODIFIED BY: MMI Tue Nov 15 13:45:56 2016		PROJECT NAME houdini_09.cpm	REV 01
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RFFE:TRX ASM
REF 7000-7099



APPROVALS

ENGINEER:

LAST MODIFIED BY:

Tue Nov 15 13:45:38 2016

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BLOCK TITLE:

rffe

	PROJECT NAME
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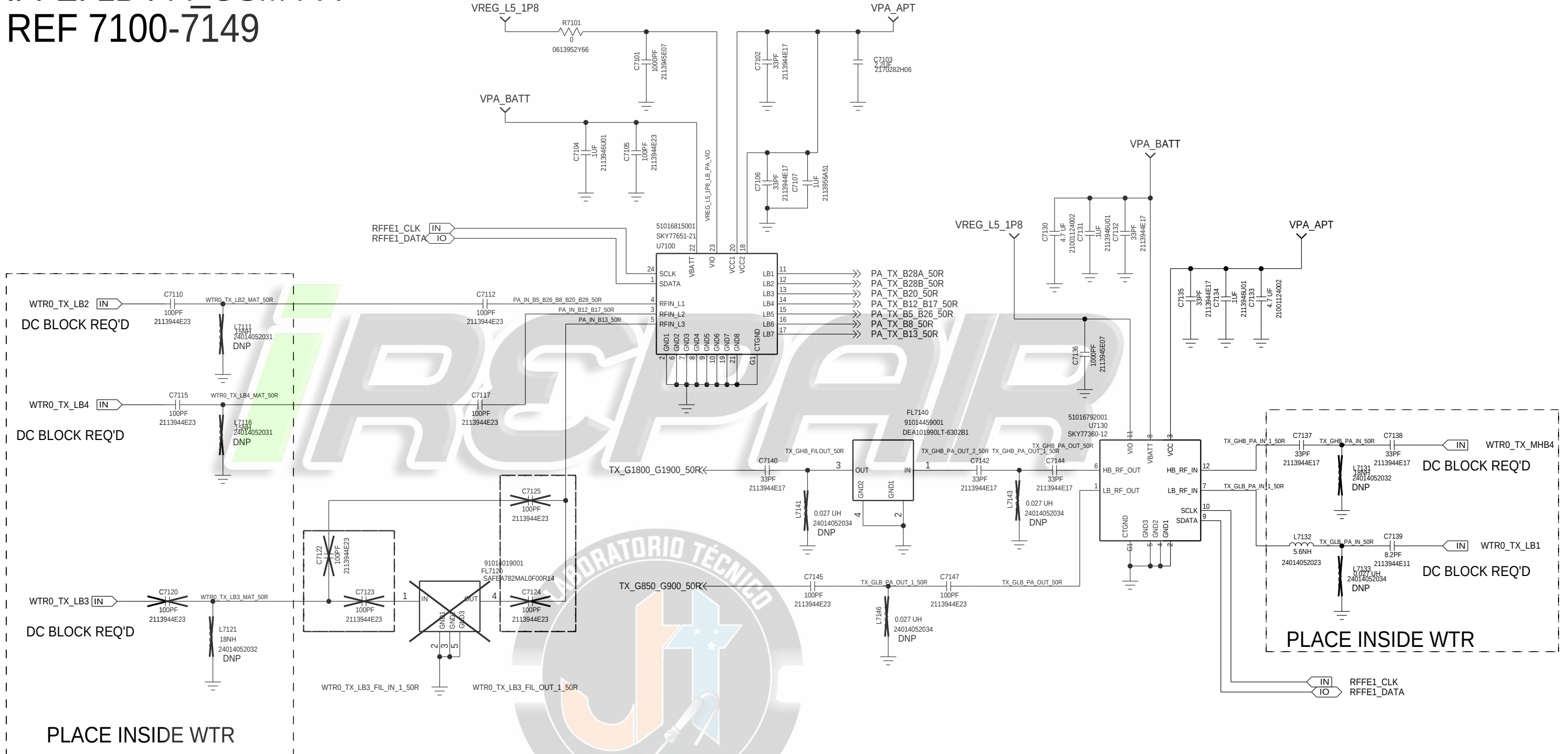
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CAD FILE NAME:

REV
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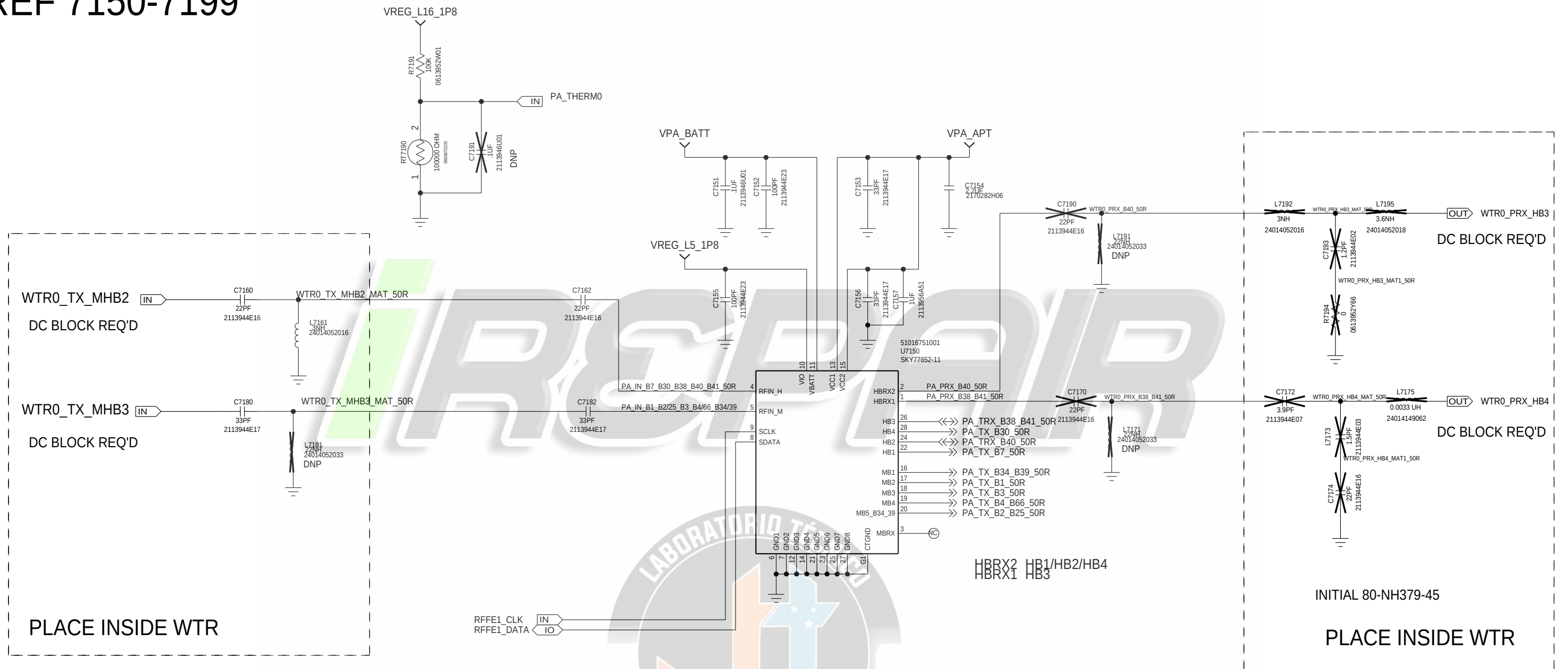
PAGE 45 OF 64

RFFE: LB PA_GSM PA
REF 7100-7149



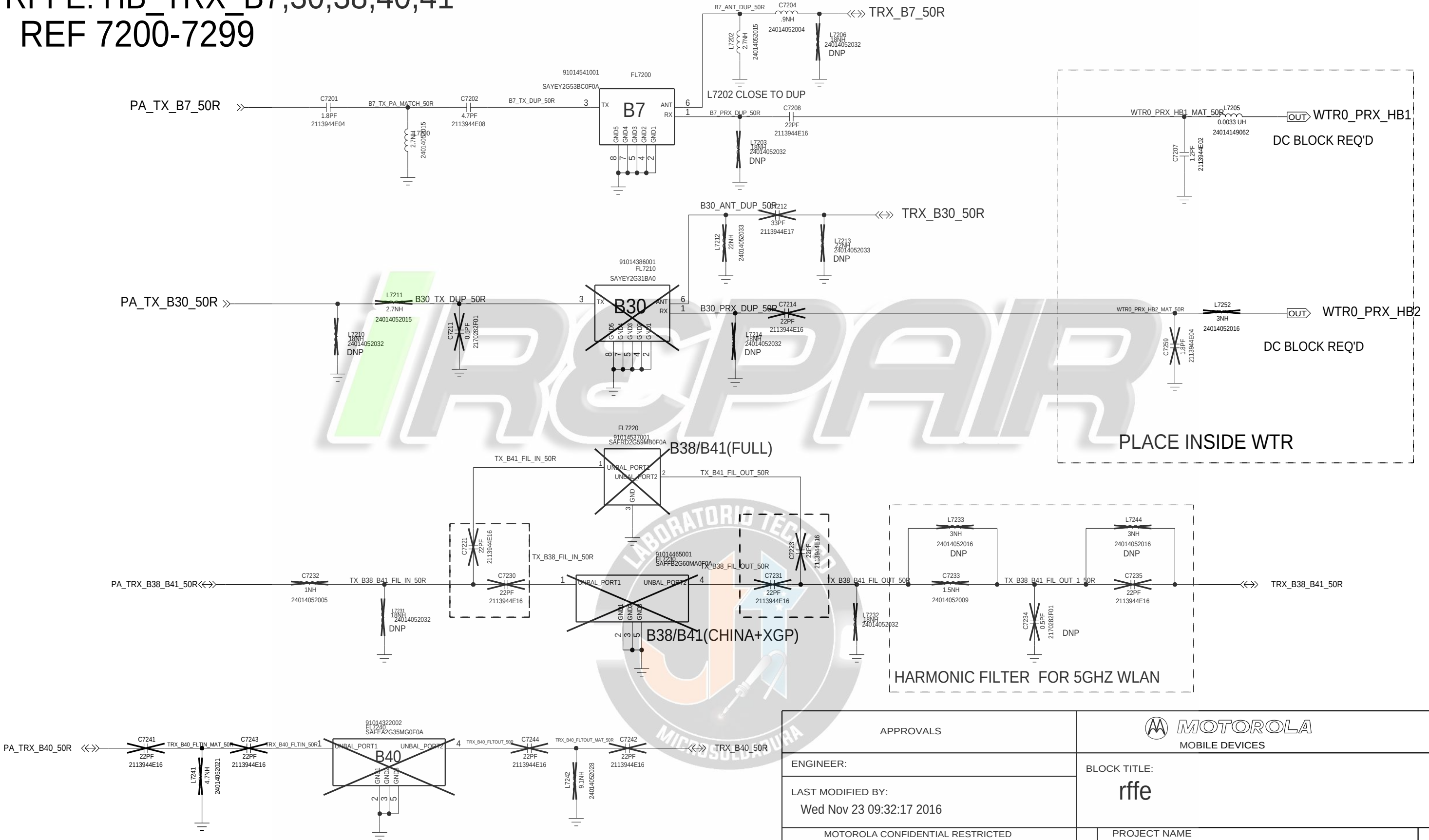
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		CAD FILE NAME:	PAGE 46 OF 64

RFFE: MB/HB PA_SKY77652-11
REF 7150-7199



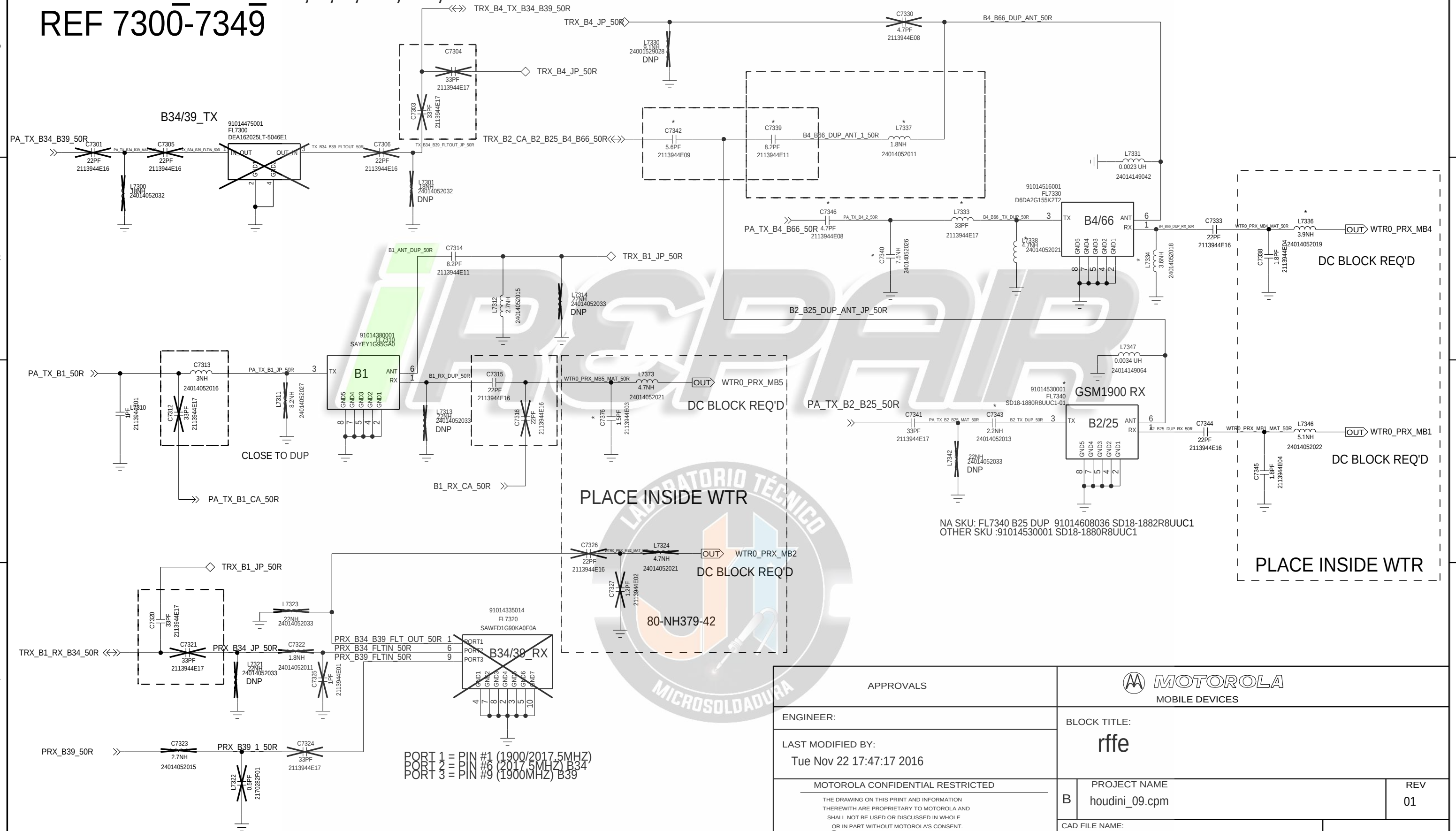
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		CAD FILE NAME:	
		PAGE 47 OF 64	
		REV 01	

RFFE: HB_TRX_B7,30,38,40,41
REF 7200-7299



APPROVALS		 MOTOROLA MOBILE DEVICES	
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LAST MODIFIED BY: Wed Nov 23 09:32:17 2016			
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		CAD FILE NAME:	PAGE 48 OF 64

RFFE : MB_TRX_B1,2,4,34,39,G1900
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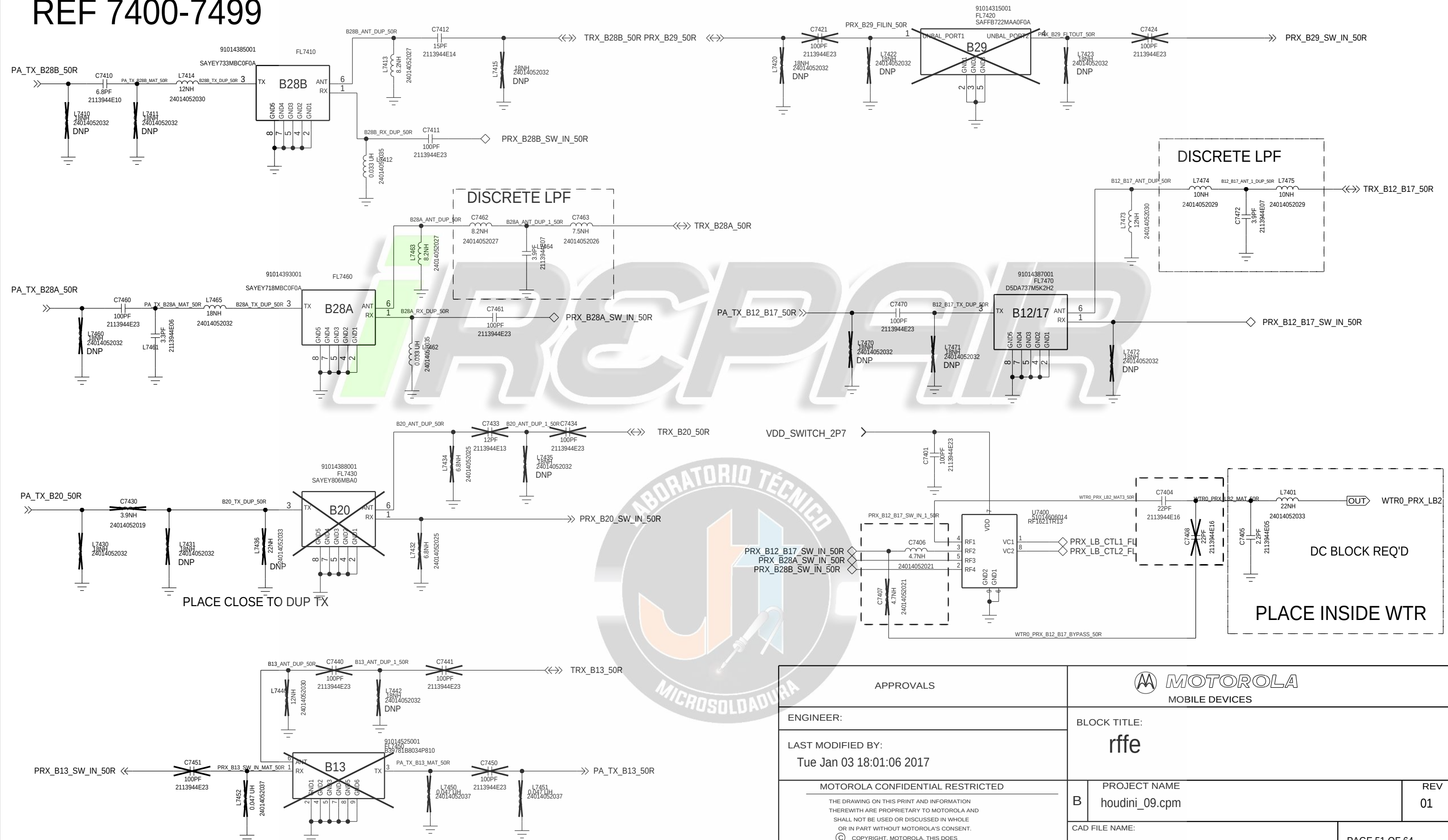


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		CAD FILE NAME:	PAGE 49 OF 64

D

A

RFFE: LB_TRX_B12,13,17,20,28,29
REF 7400-7499



APPROVALS		 MOTOROLA MOBILE DEVICES	
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		CAD FILE NAME:	PAGE 51 OF 64

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2

CBA

RFFE: DRX ASM_HB_B7,30,38,40,41

REF 7600-7699

D

C

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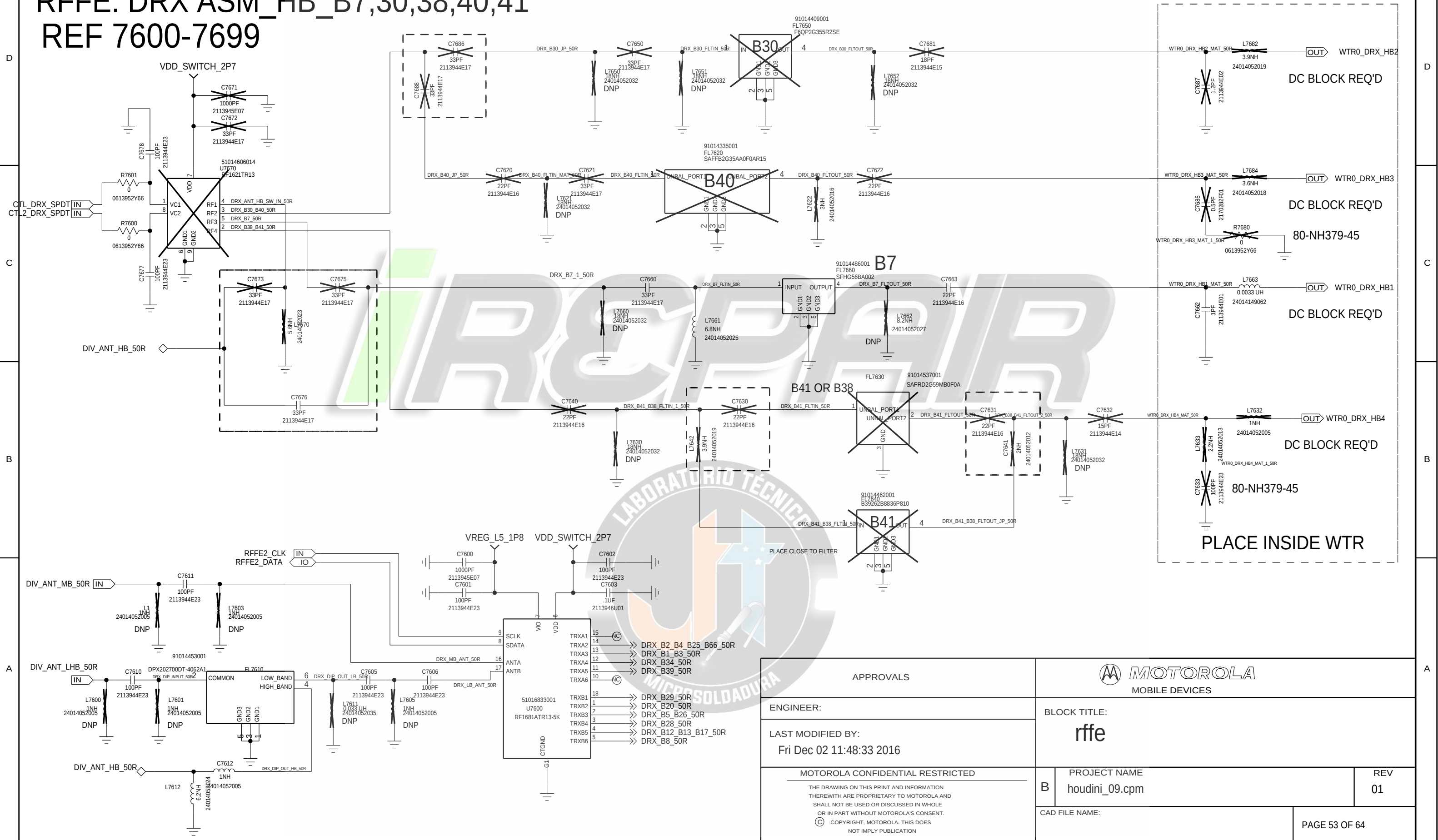
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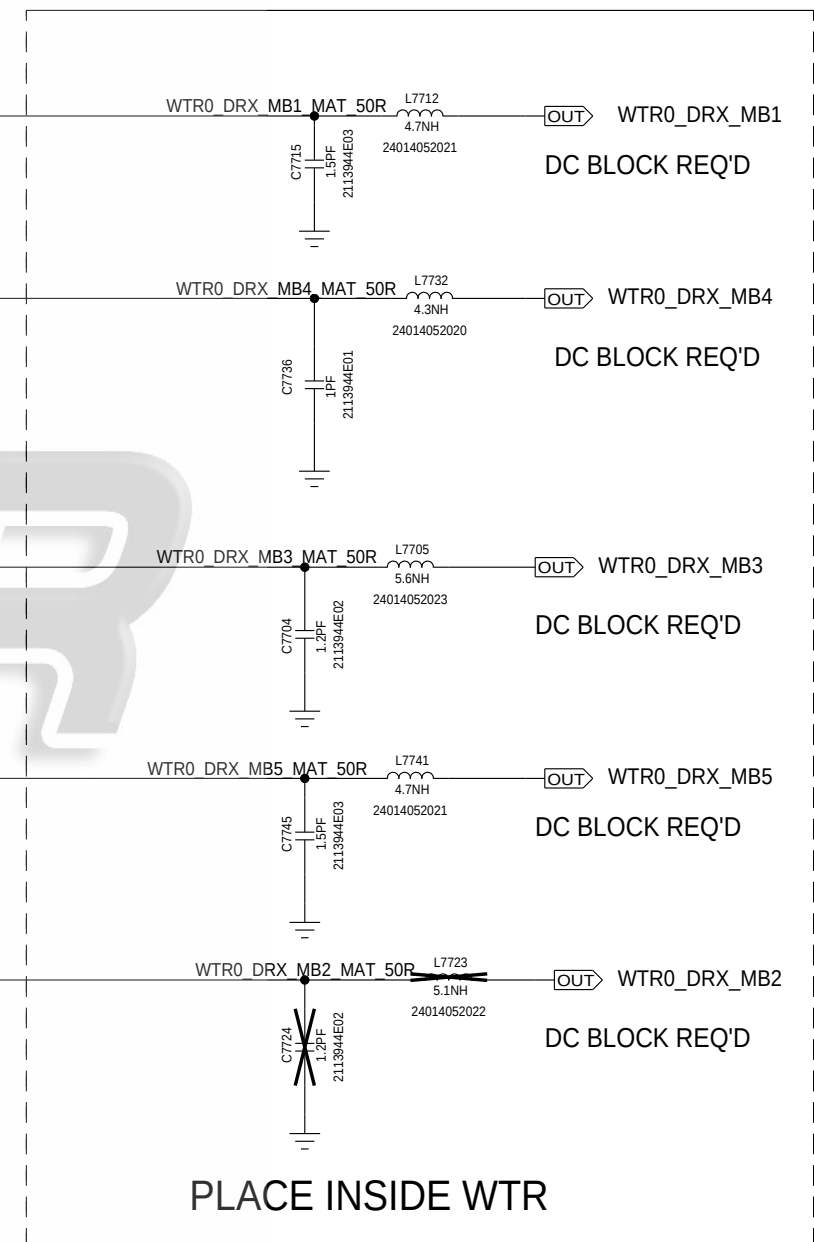
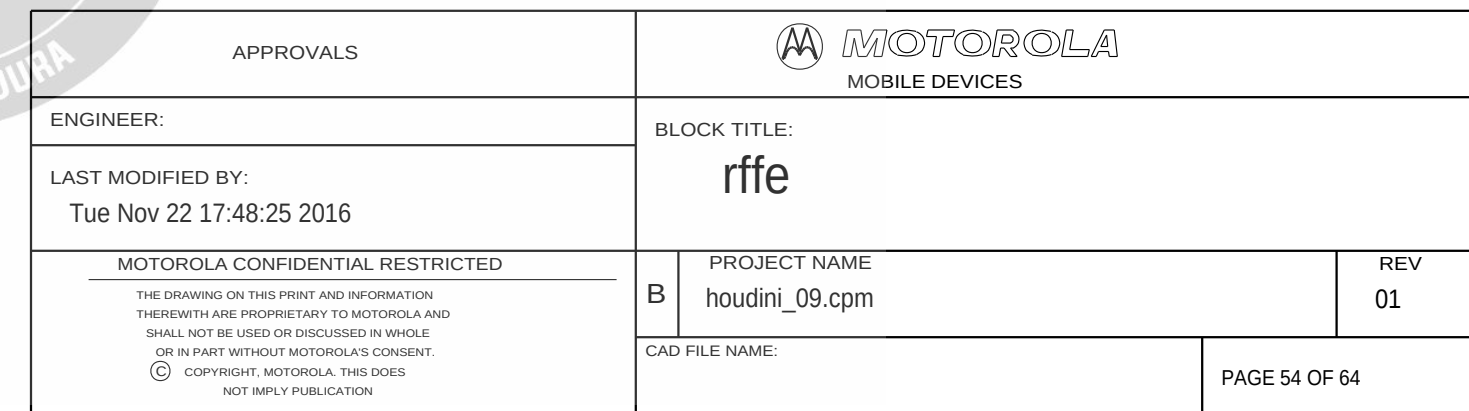
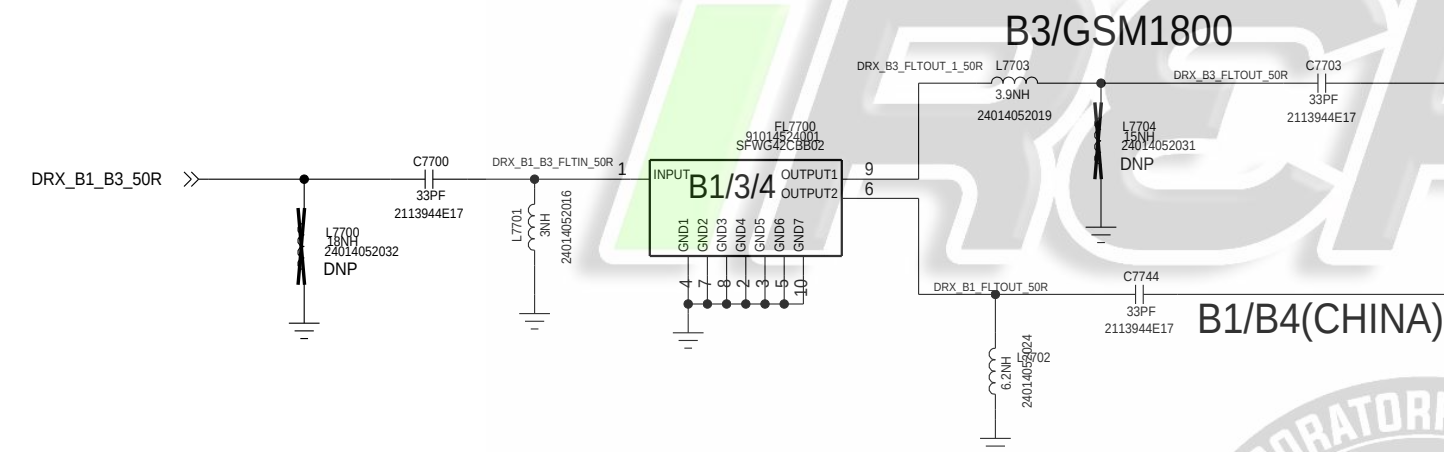
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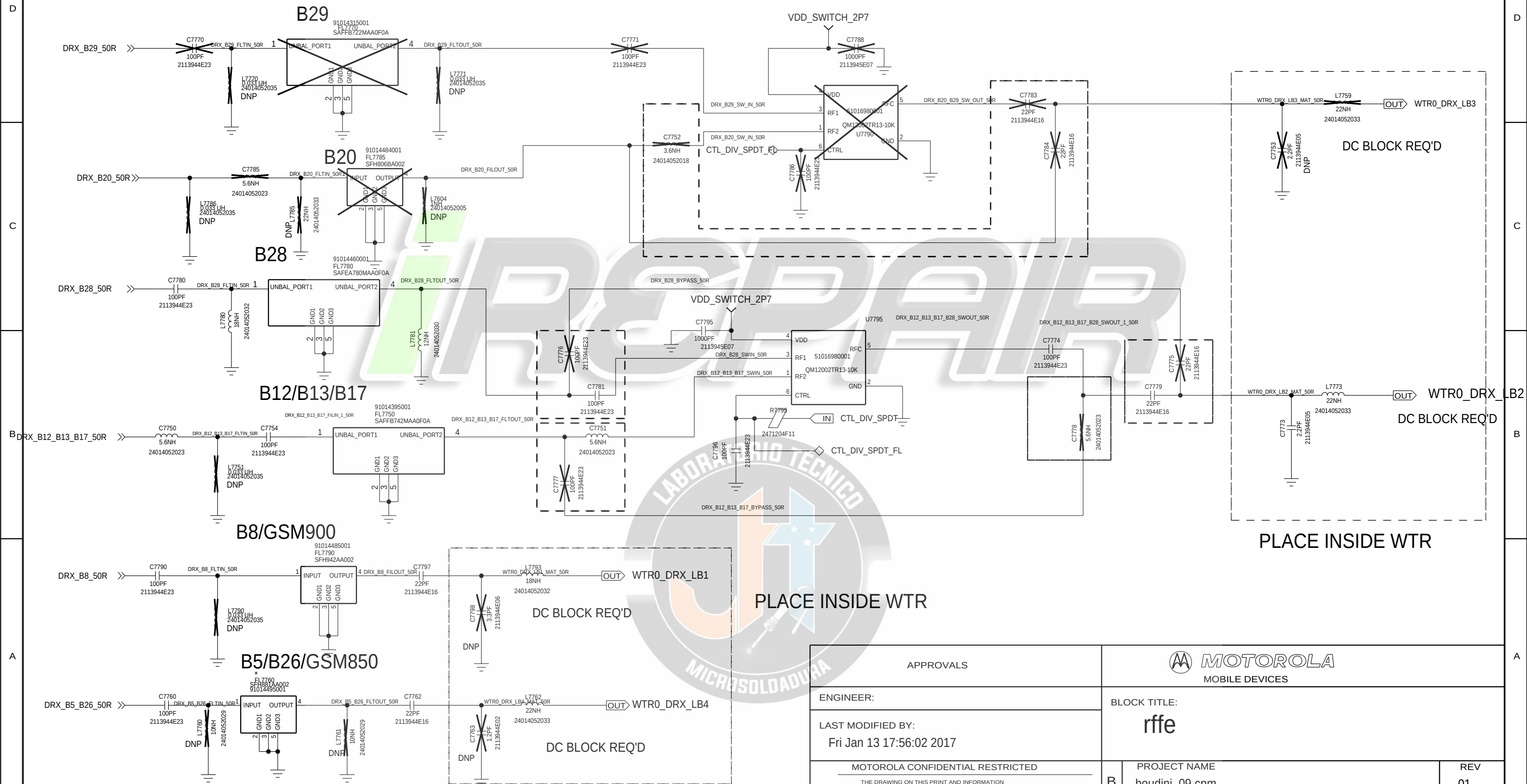


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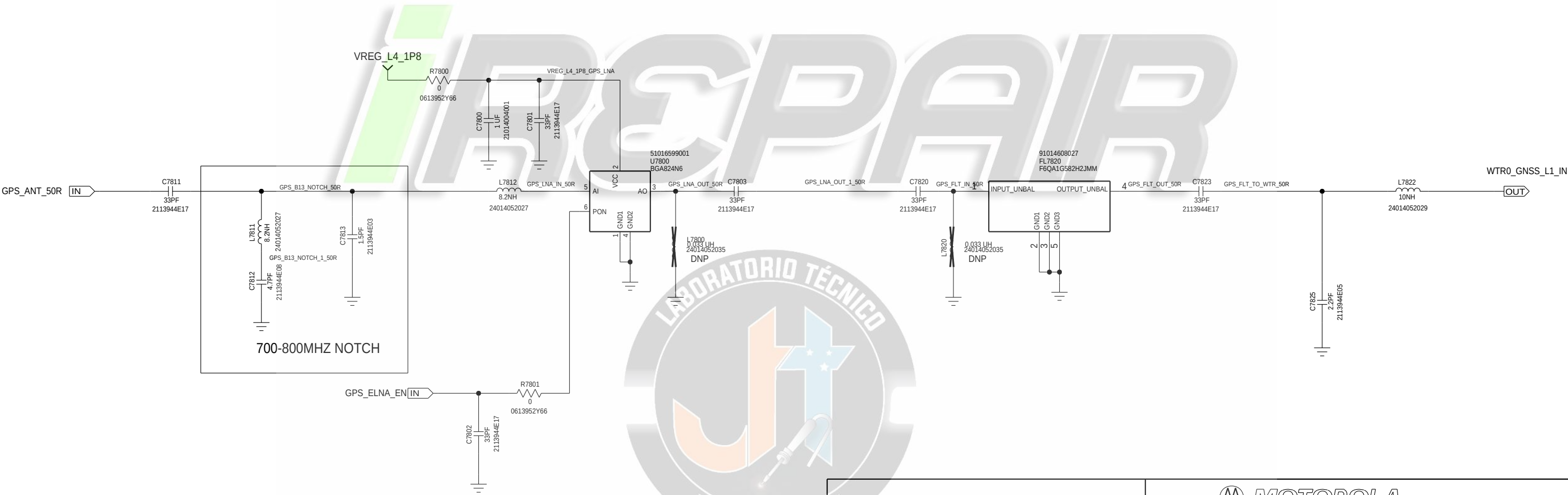
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OTHER SKU B5 91014495001 SFH881AA002

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		PAGE 55 OF 64	

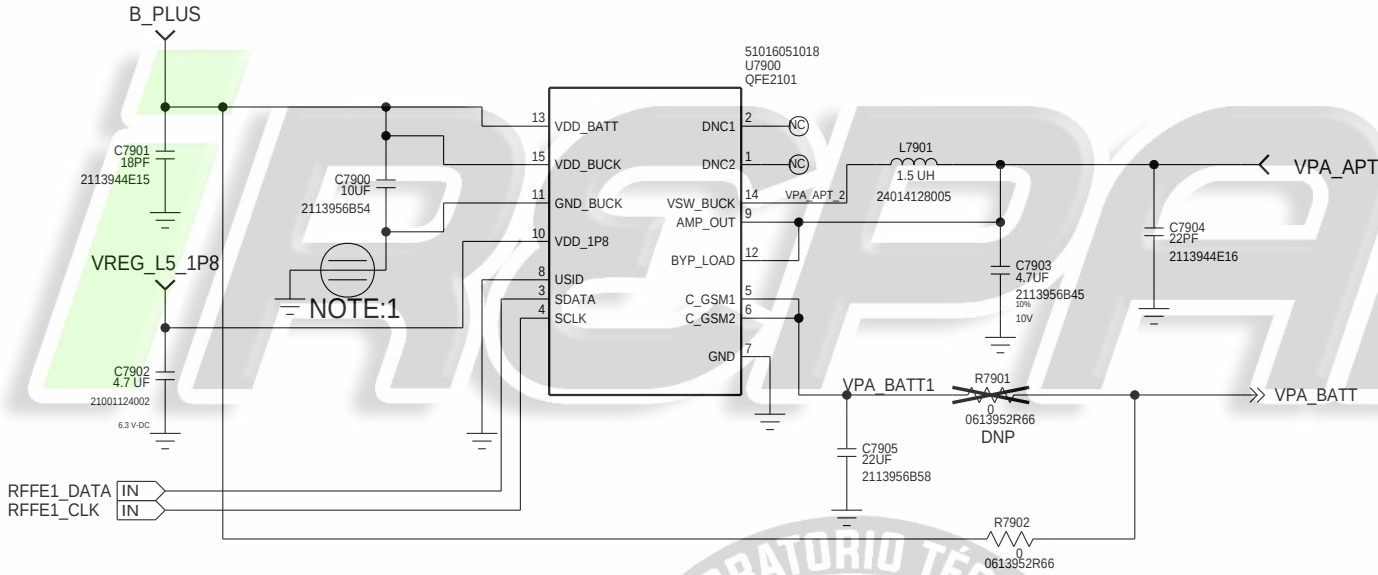
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		REV 01	
		CAD FILE NAME:	
		PAGE 56 OF 64	

RFFE: QFE2101
REF7900-7949

DCDC

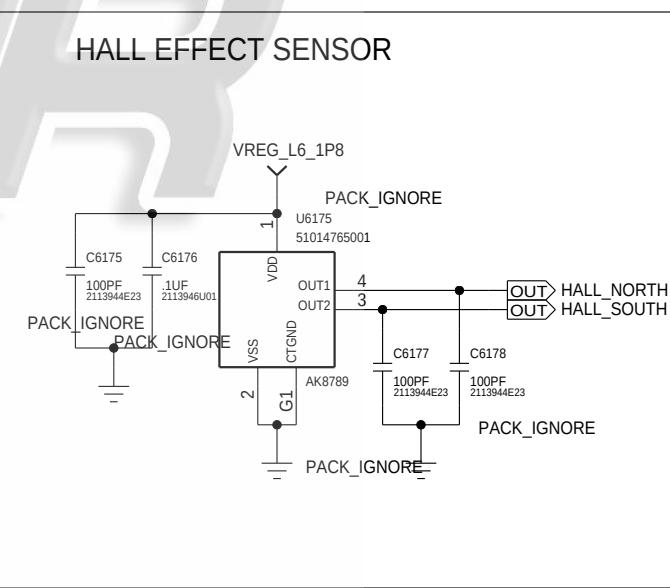
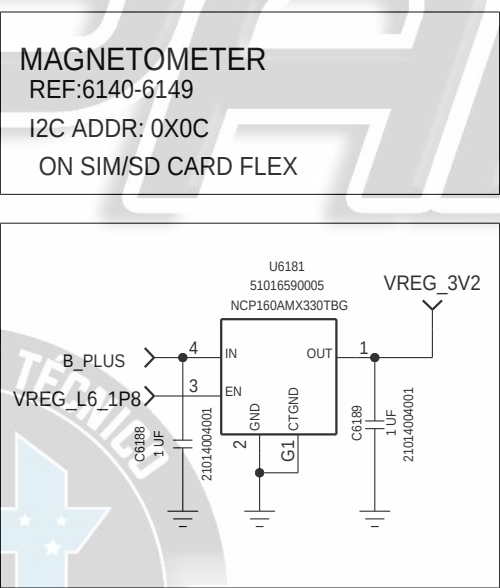
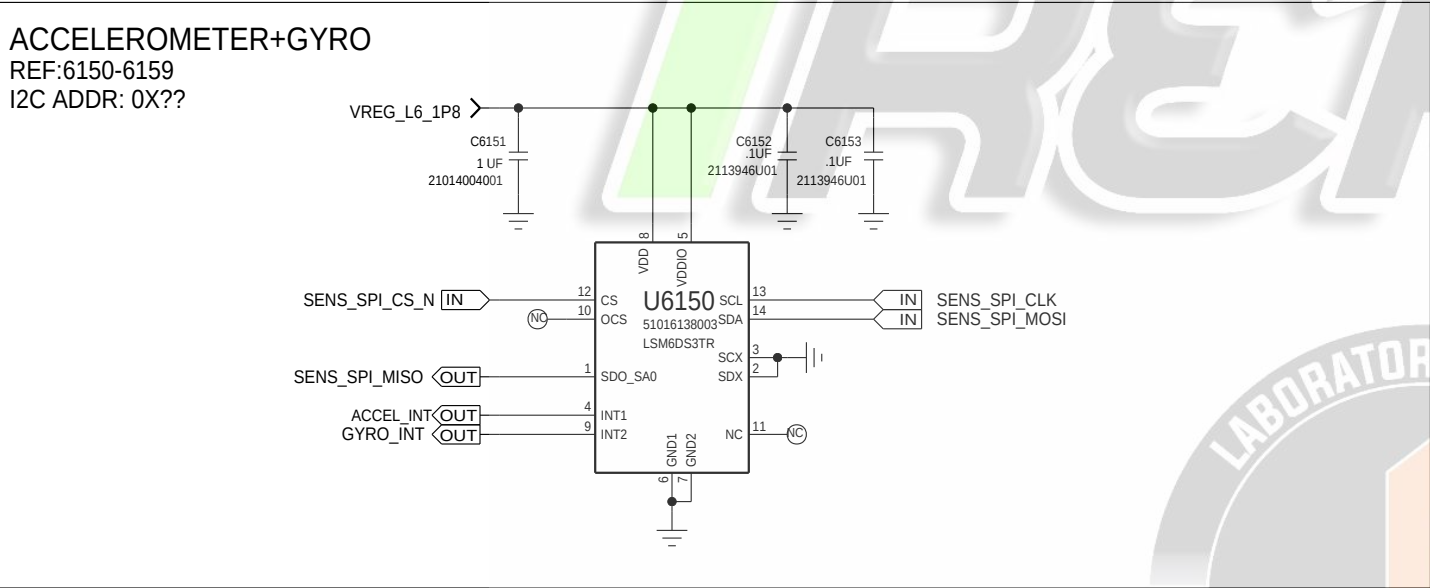
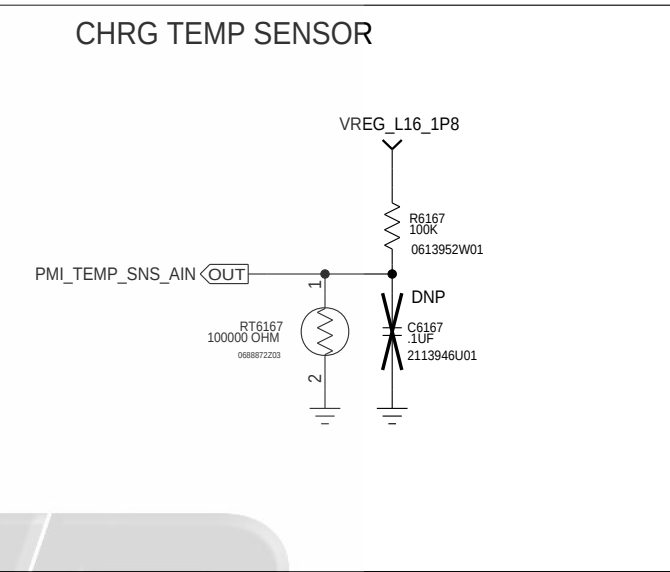
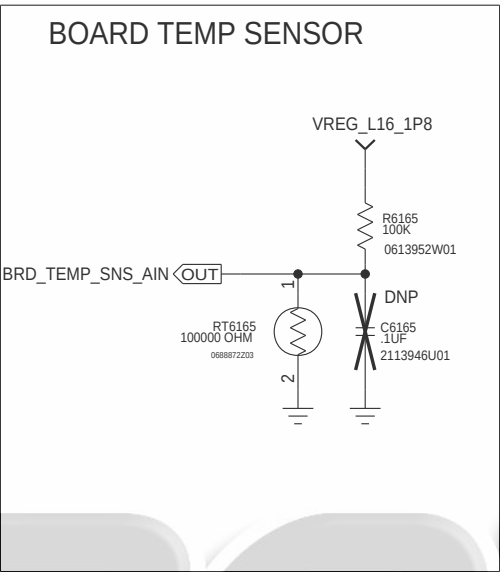
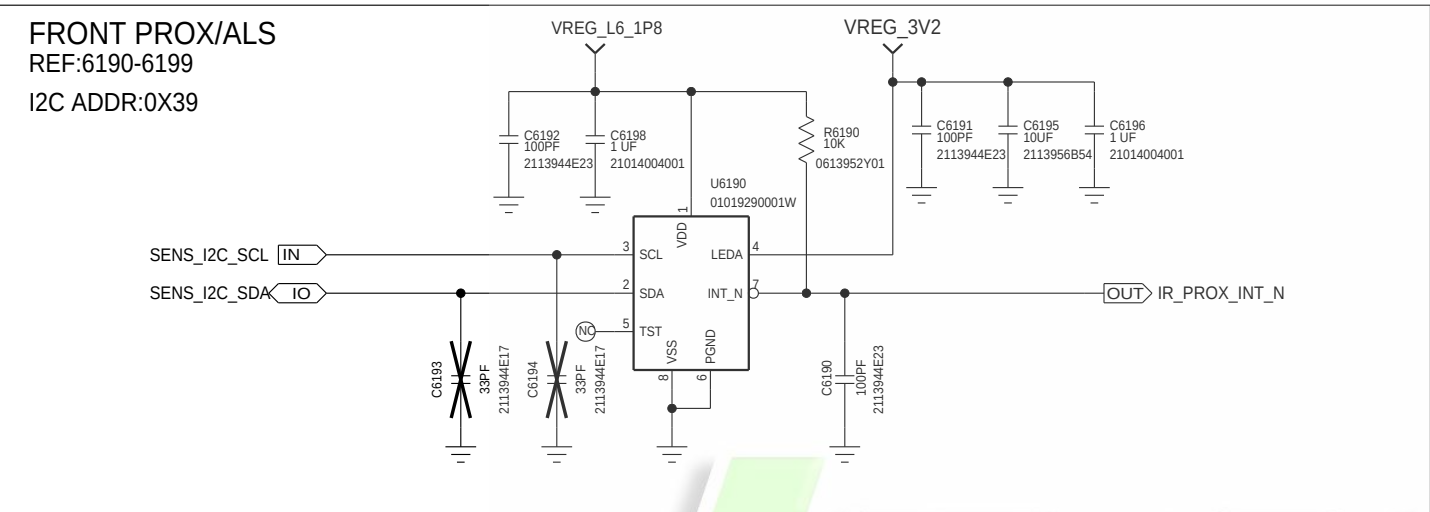


DO NOT VIA DIRECTLY TO GROUND PLANE. KEEP ISOLATED ON SURFACE LAYER;
BYPASS OPTION FOR VPA_VBATT
DNP APPROPRIATELY

APPROVALS		 MOTOROLA MOBILE DEVICES	
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		REV 01	
		CAD FILE NAME:	
		PAGE 57 OF 64	

XMCS: SENSORS

REF: 6100-6199

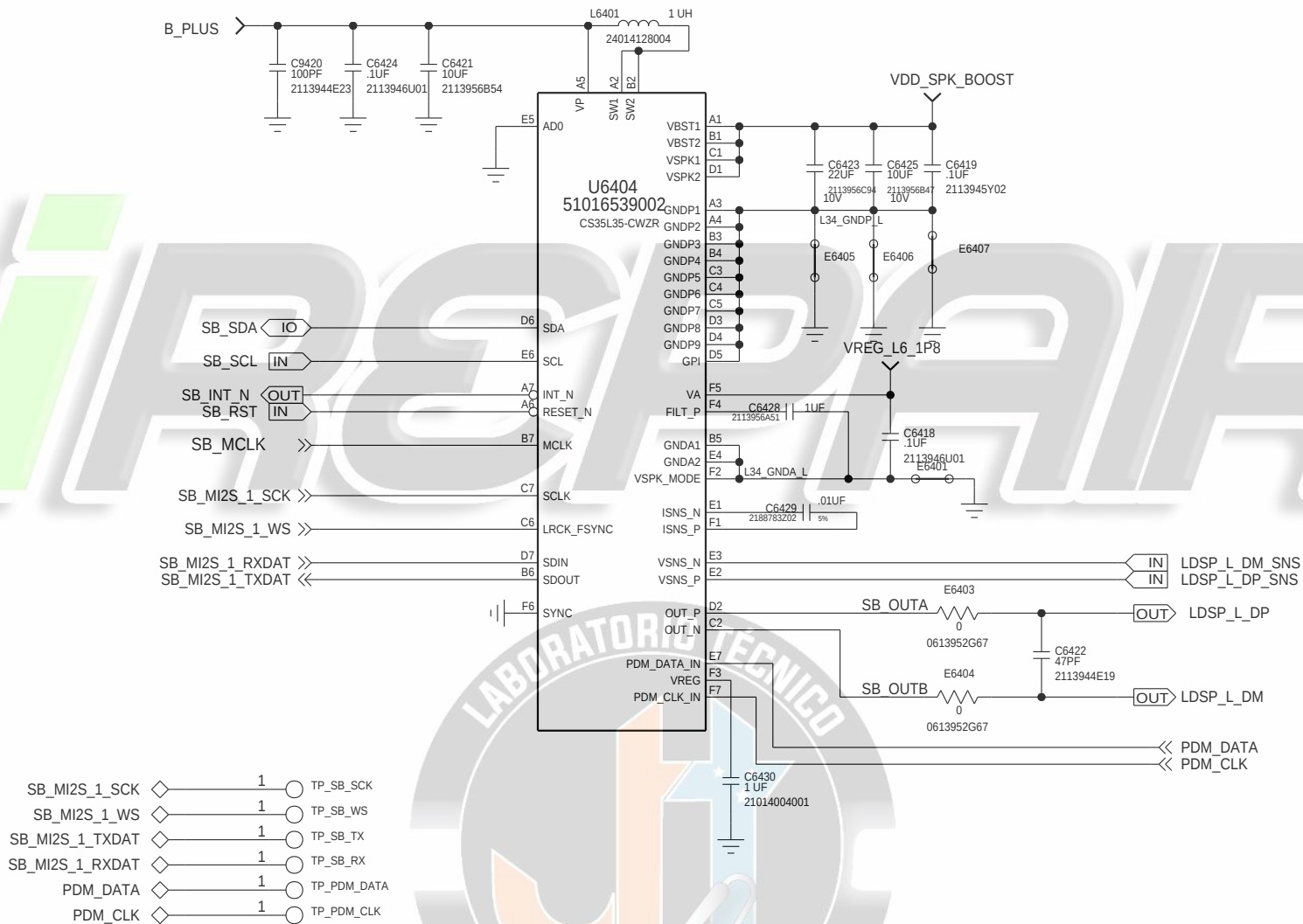


JOHN GORSICA/NATE CONNELL APPROVALS		MOTOROLA MOBILE DEVICES	
ENGINEER: HUNG PHAM		BLOCK TITLE:	
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		PAGE 58 OF 64	

XMCS: SMART BOOST

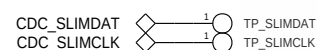
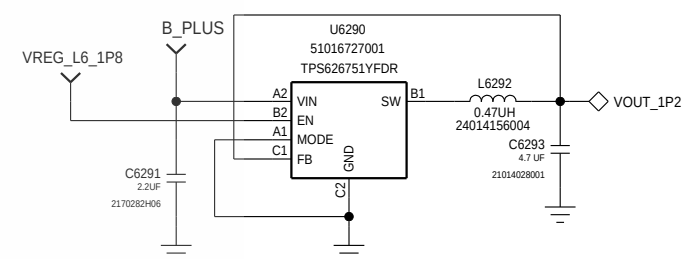
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*** BOOST IC I2C ADD. 0X80



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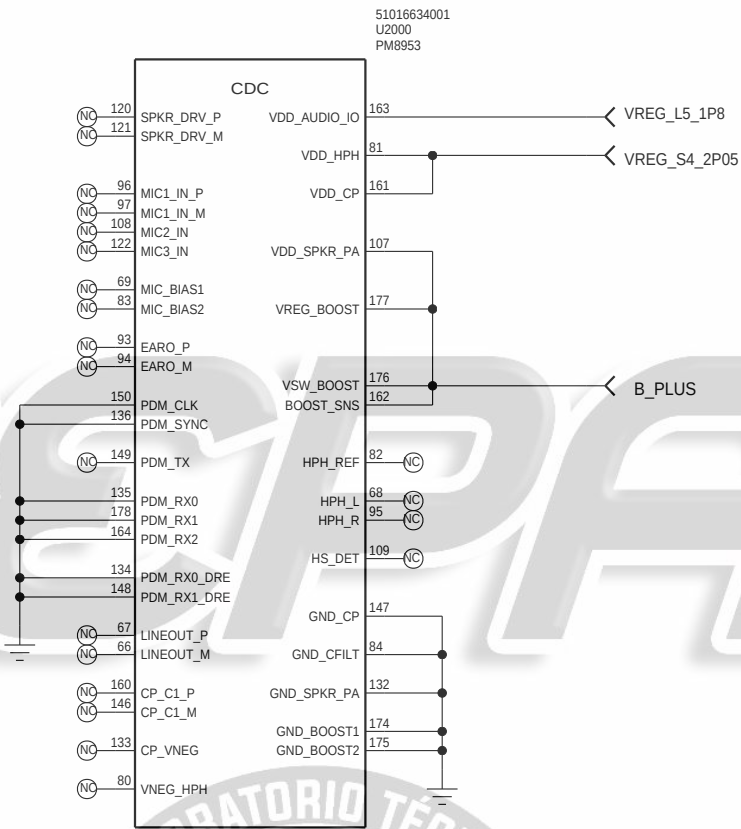
REF 6250-6299



APPROVALS		 MOTOROLA MOBILE DEVICES		
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XMCS: PMIC CODEC

REF 2500-2579



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		REV 01	
		CAD FILE NAME:	
		PAGE 61 OF 64	

REF 3000-3099

VREG_L4_1P8

VREG_L1_1P0

WTR_VDD_RF2

WTR_VDD_RF1_TVCO

WTR_VDD_RF1_TSIG

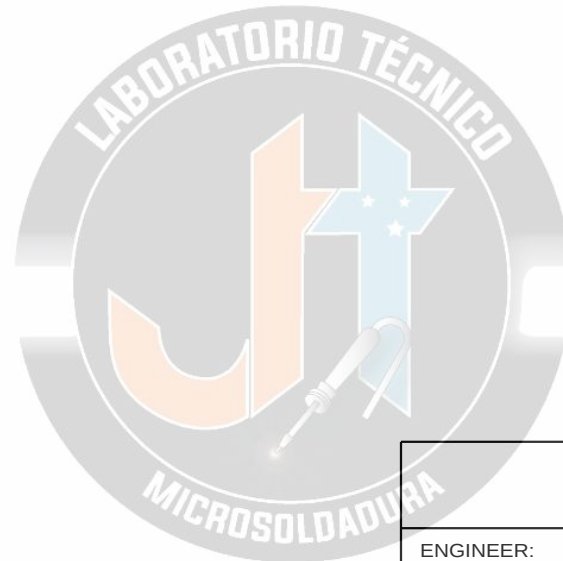
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WTR_VDD_RF1_RX1

WTR_VDD_RF1_RX2

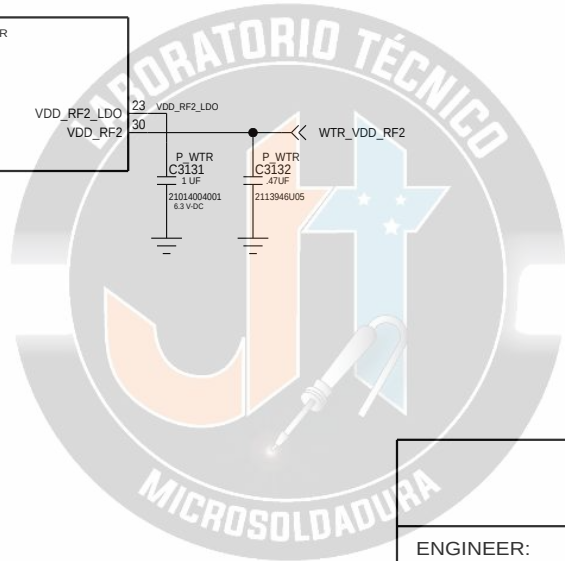
C5001
10 uF

P_WTR
21014029001
10 V-DC



APPROVALS		 MOTOROLA MOBILE DEVICES	
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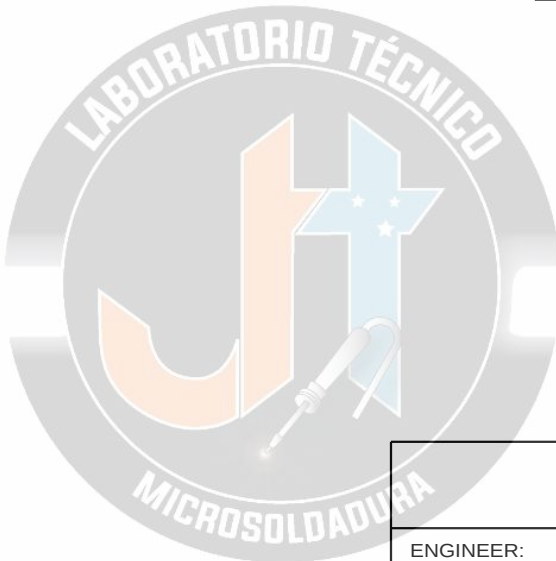
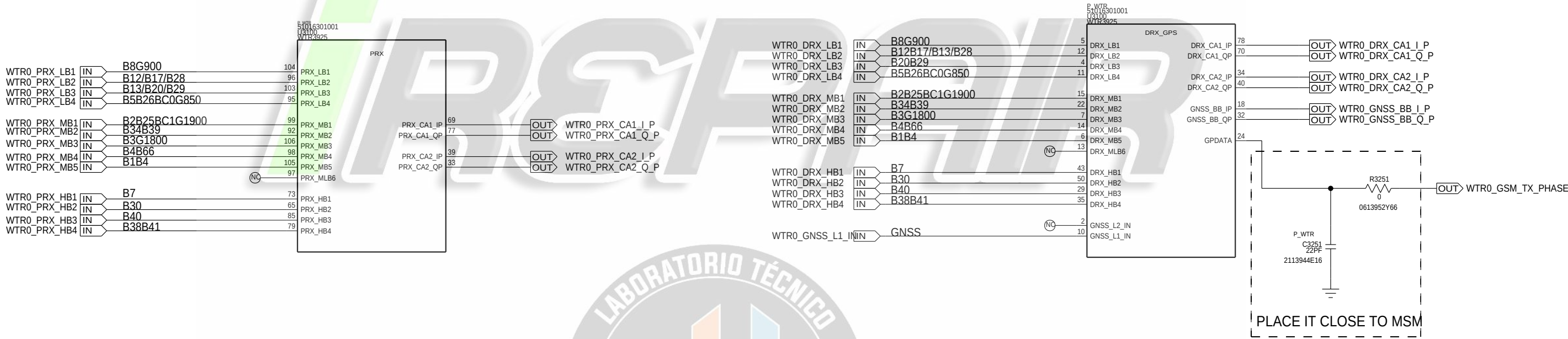
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WTR3925: PRX/DRX/GPS

REF 3200-3299



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